Cloud Data Integration for Developers

Student Guide

Version: CDI-AUG24-DEV-202409



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Cloud Data Integration for Developers

Version: CDI-AUG24-DEV-202409

September 2024

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Document Conventions

This guide uses the following formatting conventions:

If you see…	It means…	Example
>	Indicates a sub menu to navigate to.	Click Repository > Connect. In this example, you should click the Repository menu or button and choose Connect.
boldfaced text	Indicates text you need to type or enter.	Click the Rename button and name the new source definition **S_EMPLOYEE**.
UPPERCASE	Database tables and column names are shown in all UPPERCASE.	T_ITEM_SUMMARY
italicized text	Indicates a variable you must replace with specific information.	Connect to the Repository using the assigned *login_id*.
Note:	The following paragraph provides additional facts.	**Note:** You can select multiple objects to import by using the Ctrl key.
Tip:	The following paragraph provides suggested uses or a Velocity best practice.	**Tip:** The m_ prefix for a mapping name is…

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- Documentation and Knowledge Base
- Global Customer Support
- Professional Certification

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https://www.informatica.com/services-and-training/support-services/contact-us.html

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Class Introduction

- Instructor
  - Role and Informatica
  - Background
  - Contact Details
- Attendee
  - Name
  - Company
  - Role with respect to the product
  - Expectations from the course

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Modules Overview

- Module 1: Informatica Cloud Overview
- Module 2: Introduction to Data Integration Service
- Module 3: Synchronization Task
- Module 4: Data Transfer Task
- Module 5: Basics of Cloud Mapping Designer
- Module 6: IICS Transformations
- Module 7: Joiner and Lookup Transformations
- Module 8: Mapplets
- Module 9: Mapping Parameterization
- Module 10: Mapping Task

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Modules Overview (continued)

- Module 11: Replication Task
- Module 12: Masking Task
- Module 13: Taskflows
- Module 14: IICS REST APIs

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Module Objectives

After completing this module, you will be able to:

- Describe IICS as an iPaaS solution
- Explain IICS for ETL
- Define the key terminologies used in IICS
- Explore the IICS architecture
- Discuss the IICS Interface
- IICS services for Data Integration

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Topic

Introduction to IICS

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Informatica Intelligent Cloud Services Overview

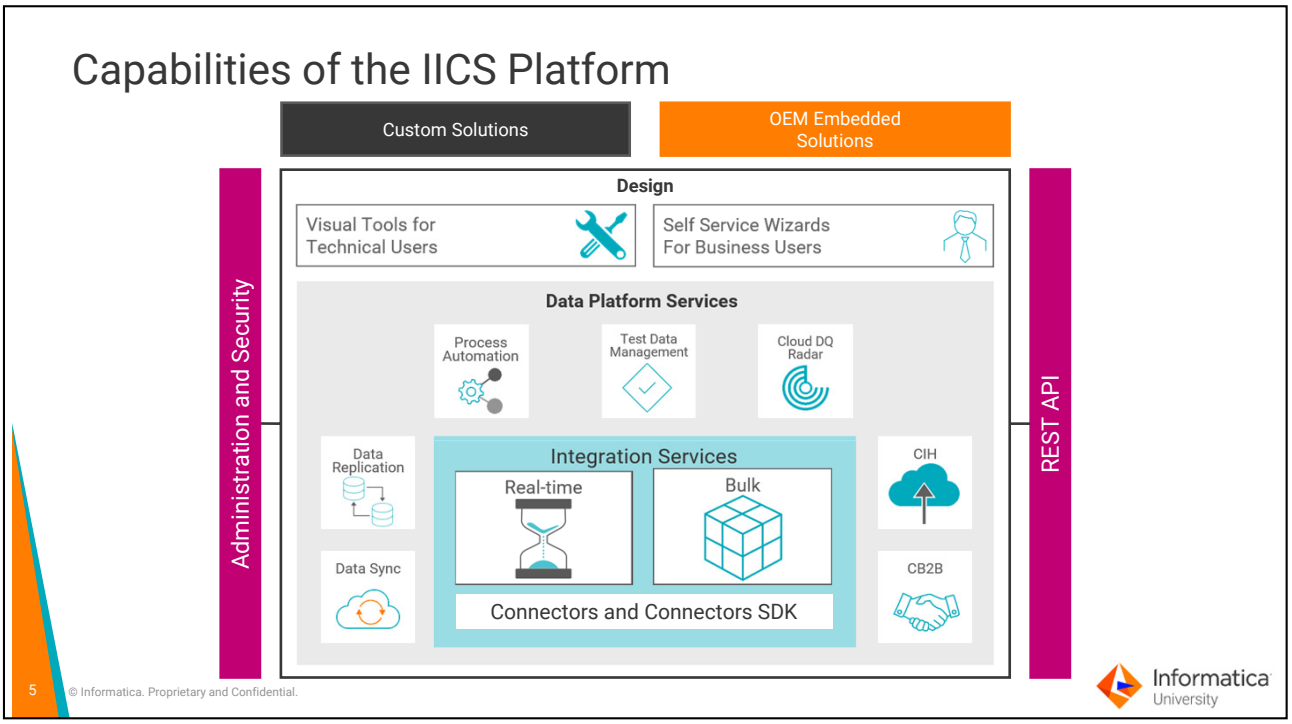
- Informatica Intelligent Cloud Services (IICS) is a cloud-based data integration platform that provides a variety of features such as data integration, application integration between cloud and local applications.
- IICS is a next-generation iPaaS solution that you can use to exchange data between applications or business partners.
- It allows you to integrate, synchronize, and relate all data, applications, and processes that reside on-premise or in cloud.
- With IICS, you can easily perform complex business processes, such as:
  - Import or migrate data between data sources
  - Perform object synchronization to create a copy of data on multiple applications
  - Design process integration to connect multiple processes
  - Replicate data between data sources for backup and archiving purposes

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IICS stands for Informatica Intelligent Cloud Services. It is a next-generation iPaaS solution that allows you to exchange data between applications or to exchange data externally with business partners. You can use IICS to integrate, synchronize, and relate all data, applications, and processes that reside on-premise or in your Cloud environment.
You can use IICS to perform the following business processes:
Importing or Migrating Data – Businesses often need to import data from an external system or migrate data from one contact management system to another
Object Synchronization – Many organizations use IICS to synchronize copies of data in multiple systems. For example, you have “Account” and “Contact” data in Salesforce, and the same information is in an Accounting system, you can use IICS to synchronize such information in each of these systems.
Process Integrations – Process integrations help you connect one process with another and update both systems with appropriate information. For example, you have a Salesforce system where you track all sales-related activities such as opportunities. When an opportunity closes, you can update that information in your order management system.
Replication and Archiving – This involves taking a backup of your data regularly or at scheduled intervals. You can also replicate the data to an on-premise database to run analytics.

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The IICS platform provides real-time integration service and bulk integration service. It also provides connectors to connect to the data sources, and an SDK toolkit to create custom connectors. The core integration platform services include:

- Data Synchronization
- Data Replication
- Process automation
- Test data management
- Cloud DQ Radar
- Cloud Integration Hub (CIH)
- Cloud B2B

The platform holds all these services together with the help of visual tools for technical users and self-service wizards for business users. The platform also encompasses data governance and ensures that administration and security are seamless across the network. You can expose the dataset using REST API and utilize reusable codes from Informatica Market Place. The Informatica Marketplace is an open platform to host solutions that support all phases of the Data Integration lifecycle. Finally, the IICS platform allows you to use OEM Embedded Solutions.

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Benefits of Using the IICS Platform

- Some of the benefits of using the IICS platform are:
  - Supports cloud, on-premise, and hybrid data management systems
  - Enterprise-class reliability and performance
  - End-to-end data management and governance
  - Supports both traditional data and big data platforms
  - Easy integration with on-premise and Cloud-based applications
  - Supports all major platforms

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There are many benefits to use IICS platform that will give you a strategic advantage over your competitors.

IICS platform supports Cloud, on-premise, and hybrid data management systems. It also has an enterprise-class reliability and performance, with end-to-end data management and governance. It supports both traditional data and big data platforms. For example, IICS supports traditional data systems such as relational databases and data warehouses, as well as big data platforms such as Amazon Web Services and Hadoop.

You can easily integrate IICS with on-premise and Cloud-based applications. An important benefit of IICS is that it supports all major platforms such as Salesforce, Workday, Tableau, Microsoft Azure Blob, and so on.

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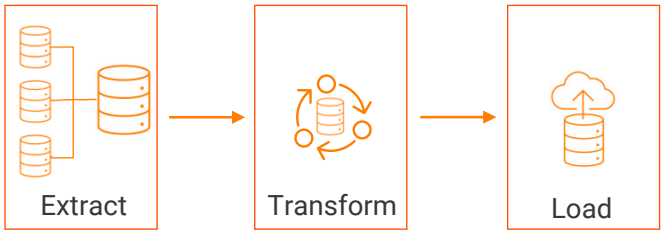
IICS for ETL

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ETL Overview

- ETL stands for Extract, Transform, and Load data.
- ETL is a data integration process that merges data from multiple data sources into a single data store.
- The data is then loaded onto the data warehouse or other target system.
- The ETL process is explained in the following image:



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- Extraction of data is the first phase of Extract, transform and Load.
- Data is collected from one or many data sources. It is then seized in the temporary storage, where the next two steps are carried out.
- In the transformation phase, data is processed to make its values and structure conform consistently with its intended use case.
- The load phase moves the transformed data into the target system.

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Cloud ETL

- Cloud ETL extracts structured and unstructured data from any data source type.
- Both on-premises and cloud data warehouses may contain the data.
- It then combines and transforms the data.
- It loads the data into a central area so that it can be accessed whenever necessary.
- IICS is Informatica's data integration tool for cloud.
- Cloud ETL is used to make high-volume data readily available for analysts, engineers and decision makers.

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- IICS provides option to users for running ETL codes in the cloud rather than integration services on premise.
- Data integration can be done for vast amount of data from multiple sources in less time than any other ETL tool.

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Use Cases of Informatica Cloud

Data Synchronization	Cloud application to cloud application, On-premise application to cloud application, On-premise application to on-premise application.
System Maintenance Tasks	Performs Create, Update and Delete operations.
Data Replication	Copy datasets, Backup or Warehouse data.
Mappings	Process to extract, transform and load data from multiple sources to target.
Monitoring	The activity log is used for successful and unsuccessful jobs.

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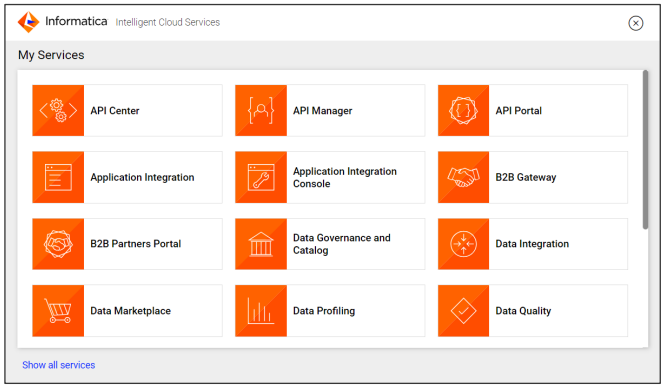
IICS Interface

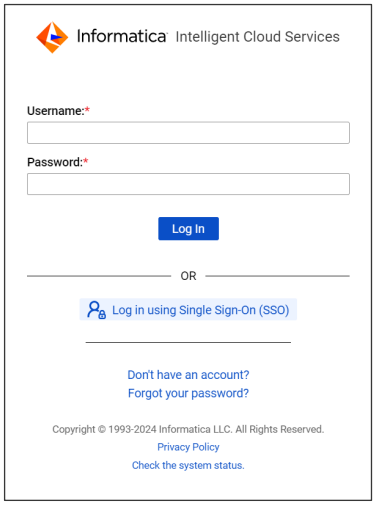
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IICS Interface

- To access the IICS interface use the following URL
  https://dm-us.informaticacloud.com/identity-service/home
- After you login, you will be directed to My Services page.





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IICS Services for Data Integration

- The list of services on the My Services page changes according to your user privileges and Org license.
- The services used more often in this course are mentioned below:

Service	Description
Administrator	Administrator provides organization management capabilities across Informatica Intelligent Cloud Services.
Data Integration	Use Data Integration to create data integration projects and run data integration tasks
Data Ingestion and Replication	Use the IICS Data Ingestion and replication to ingest and replicate data, at scale from database, application, file, and streaming data sources and transfer the data to selected cloud targets and messaging systems

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IICS Key Terminologies

Term	Definition
Source	Location from where you retrieve data
Target	Location to which you move the data
Task	Specifies the following Data Integration Task types: • Synchronization Task • Replication Task • Mapping Task • Masking Task
Connection	Provides the information that IICS needs to connect to an on-premise or Cloud application
Org	An IICS organization, generally one per company

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Source: A source is the location from where you retrieve data. For example, if you load data from a CSV file into Salesforce, your source is the CSV file.
Target: A target is the location to which you move the data. If you load the data from a flat file into Salesforce, your target is the object in Salesforce where you load the data.
Task: A task specifies the Data Integration job, including the source and target, mappings, and other advanced options. Synchronization, Replication, Mapping, Data Ingestion and Replication, and Masking are examples of the types of tasks.
Connection: A connection provides the information that IICS needs to connect to the data sources and databases.
Org: An Org is an IICS organization. Generally, you have one Org per company where all users login and create objects. However, in some cases a company may have a production and a sandbox Org. Large companies may have separate business units that require separate production Orgs.

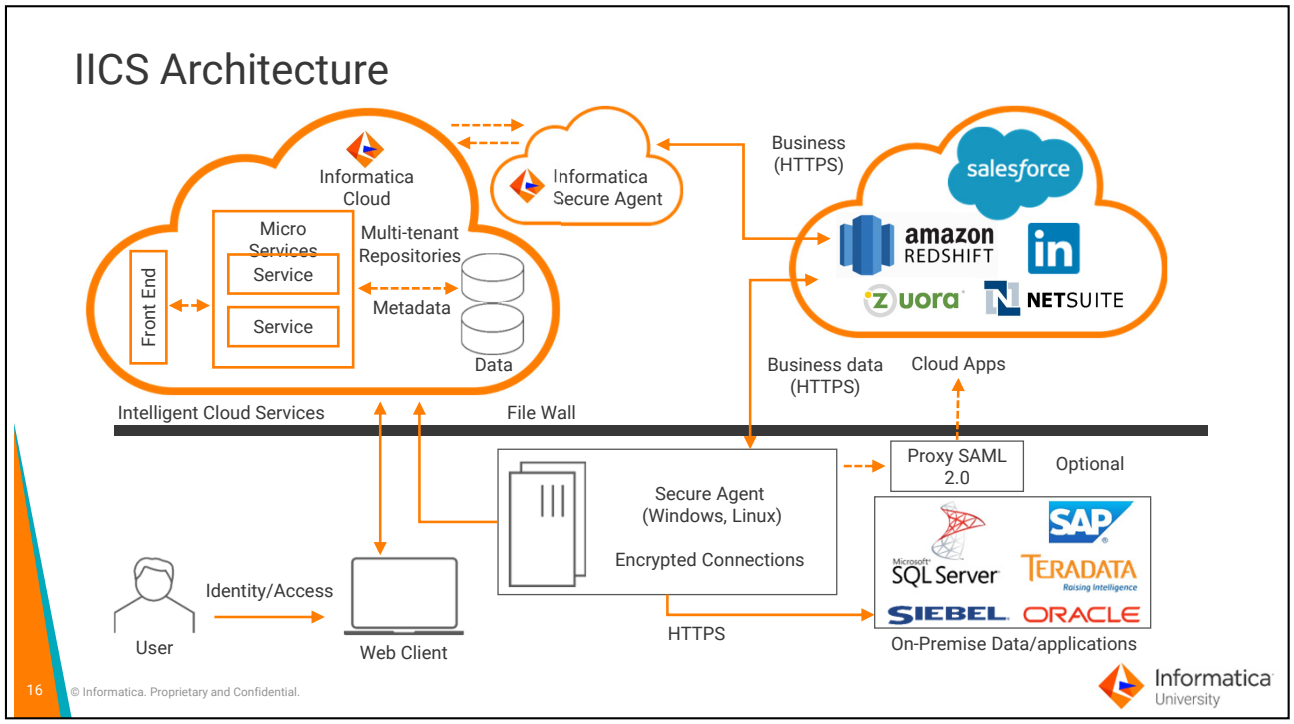
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Informatica Cloud Architecture

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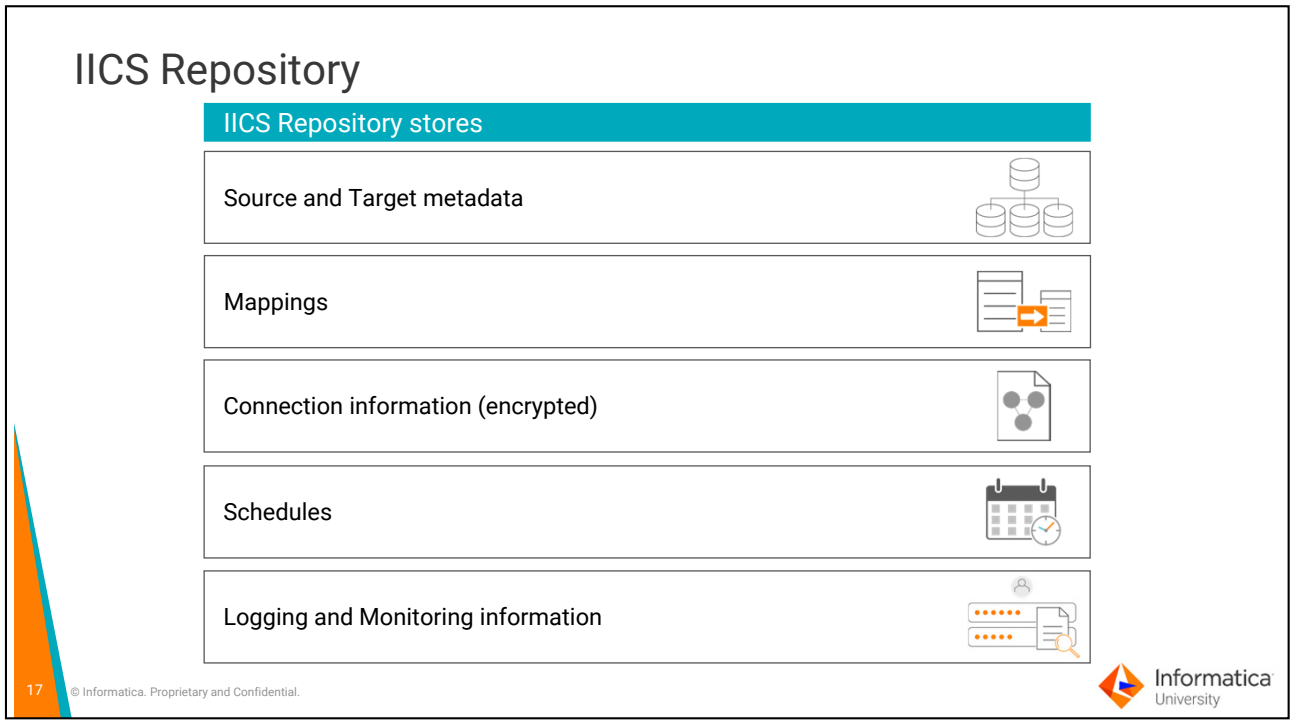
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You have already seen that IICS is an iPaaS-based application. This means that you can access it from any machine with an internet access and a web browser application. The application has multiple wizards that guide to build and test your Data Integration tasks. You can also use the application to schedule jobs and monitor the progress and result of the jobs.

When you access the application, your web browser connects to the Informatica Cloud Services through a secure HTTP. The Informatica Cloud Services includes the IICS repository that stores various information about the task.

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As you create, schedule, and run tasks, information is written to the IICS repository. This repository stores source and target metadata, mappings, connection information, schedules, and logging and monitoring information.

- **Source and Target metadata**: The repository stores metadata for each source and target object. This includes field names, data type, precision, and other information about the source and target object.
- **Mappings**: When you create a Data Integration task, the repository stores mappings and transformation rules.
- **Connection information**: The repository stores information that enables you to connect to specific source and target systems. The repository stores this information in an encrypted format.
- **Schedules**: You can configure the tasks to run automatically with the use of various scheduling options. The repository stores information regarding these schedules.
- **Logging and monitoring information**: The repository stores the results of all jobs. You can log in to IICS and view status details.

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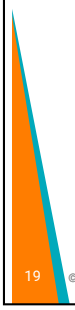
Knowledge Check

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Informatica Cloud is an iPaaS solution. What does iPaaS stand for?

A. Integration Platform as a Service
B. Infrastructure Platform as a Service
C. Information Platform as a Service
D. Intermediate Platform as a Service



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Answer: a

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ETL stands for-

a. Extraction, transformation and loading
b. Extract, transactions and load
c. Enable, transactions and load
d. All the mentioned

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Answer: a

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Lab Activity

Lab 1-1: Navigating the IICS interface

In this lab, you will perform the following:

- Navigate to various services in IICS

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Module Summary

This module showed you how to:

- Describe IICS as an iPaaS solution
- Explain IICS for ETL
- Define the key terminologies used in IICS
- Explore the IICS architecture
- Discuss the IICS Interface
- IICS services for Data Integration

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Module 2

Introduction to Data Integration Service

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Module Objectives

After completing this module, you will be able to:

- Describe the Data Integration (DI) Service
- List the Cloud Data Integration assets and components
- Explain the pre-requisite for using Data Integration Service
- Discuss types of runtime environments
- Define Connections

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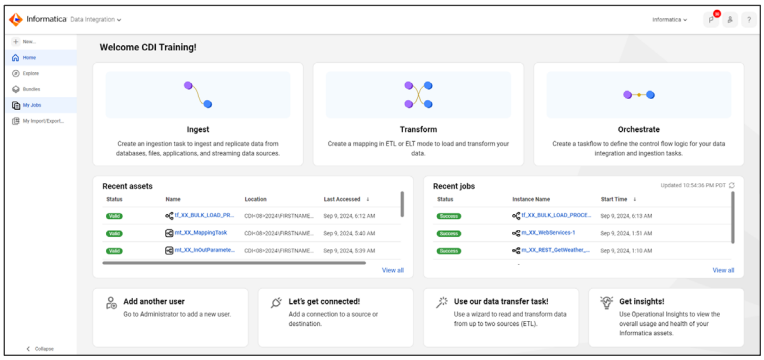
Introduction to Data Integration Service

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Data Integration Service Overview

- Data Integration is an on-demand subscription service of IICS.
- It provides a complete platform for cloud integration and data management.
- You can use the Data Integration service to build and run complex integrations using pre-built functions and templates.



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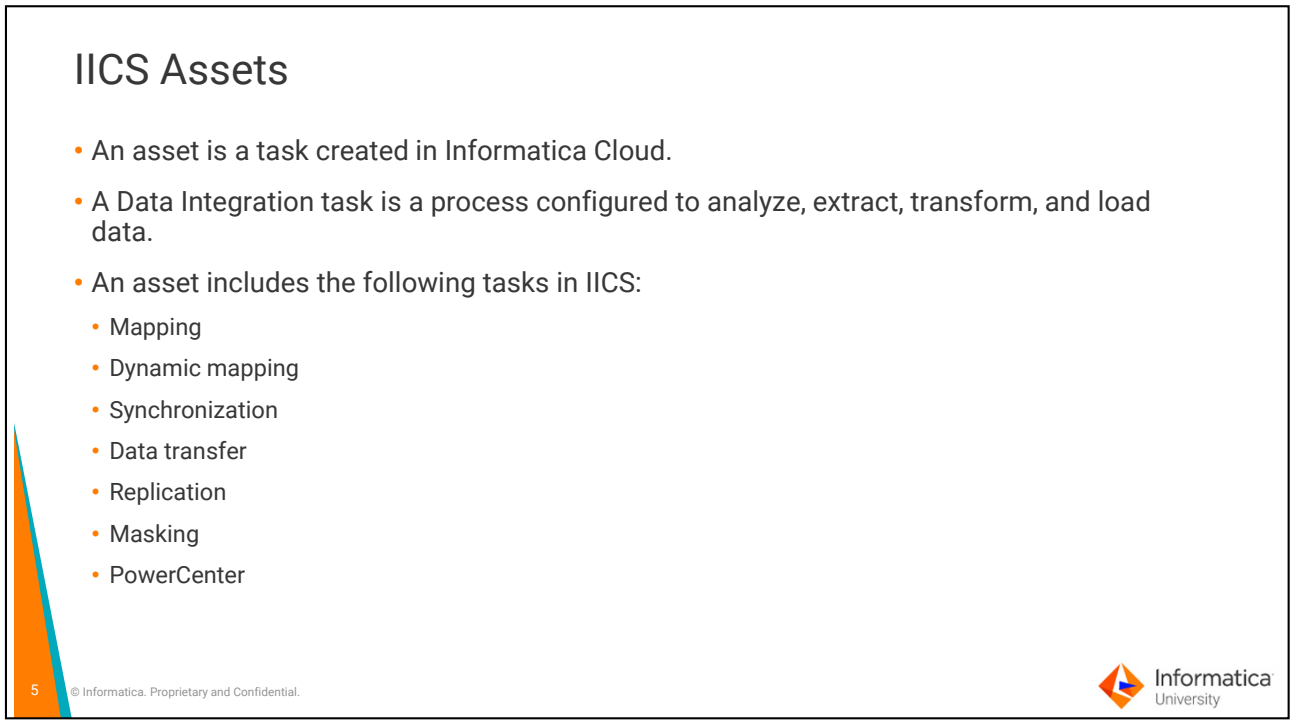
To access the service, you need to select the Data Integration service from the **My Services** page of IICS. The Home page provides a guided and unified experience for Data Integration users and Data Ingestion and Replication users. The Home page is customized for each user in an organization Primary task panels: Use the following panels to load and transform your data:

- Ingest. Create Data Ingestion and Replication tasks to ingest and replicate your data.
- Transform. Create data transfer tasks or mappings to load and transform your data.
- Orchestrate. Create taskflows to define the control flow logic for your tasks..
- **Runtime Environments**: Shows the status of runtime environments available in your organization.
- **Recent Jobs:** Provides a list of jobs ran in the last 24 hours.
- **Recent Assets**: Lists the assets that are most recently modified.

You can also access the following pages from the navigation bar of the service Home page:

- **New**: Used to create new data integration assets.
- **Home**: Used to return to the Home page.
- **Explore**: Used to create and manage projects and assets.
- **Bundles**: Used to create and manage bundles of assets to share with your organization or other organizations.
- **My Jobs**: Used to view the status of your mapping, task, and taskflow instances.
- **My Import/Export Logs**: Used to view the status of your imports and exports.

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- Mapping: Use to process data based on the data flow logic defined in a mapping or Visio template.
- Dynamic mapping: Use to run multiple jobs with different parameters based on the data flow logic defined in the same mapping.
- Synchronization: Use to load data and integrate applications, databases, and files. Includes add-on functionality such as mapplets.
- Data transfer: Use to move data from a source to a target. Optionally, sort and filter data before loading it to the target.
- Replication: Use to replicate data from Salesforce or database sources to database or file targets. You might replicate data to archive the data, perform offline reporting, or consolidate and manage data.
- Masking: Use to replace source data in sensitive columns with realistic test data for non-production environments. Masking rules define the logic to replace the sensitive data. Assign masking rules to the columns you need to mask.
- PowerCenter: Use to import a PowerCenter workflow and run it as a Data Integration PowerCenter task.

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IICS Components

- Components are assets that support mappings and tasks.
- You can create the following components to support tasks in IICS:
  - Business services
  - File listeners
  - Fixed-width file formats
  - Hierarchical schemas
  - Intelligent structure models
  - Mapplets
  - Saved queries
  - Shared sequence
  - User-defined functions

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Supporting Services

- Use the following IICS services to successfully create integration projects in IICS:

Data Integration	Administrator
Required to manage data integration projects, create assets such as mappings, tasks, and taskflows, and run tasks and taskflows.	Perform administrative tasks such as user management, configuring runtime environments, creating connections, managing licenses, and schedules.

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Topic

Getting Started with Data Integration Service

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Pre-requisites to Use Data Integration Service

Before you can start creating assets in , the following steps must be performed:

1. Configure a runtime environment for your IICS Org
2. Create connections to use in assets
3. Create a project to organize assets in folders

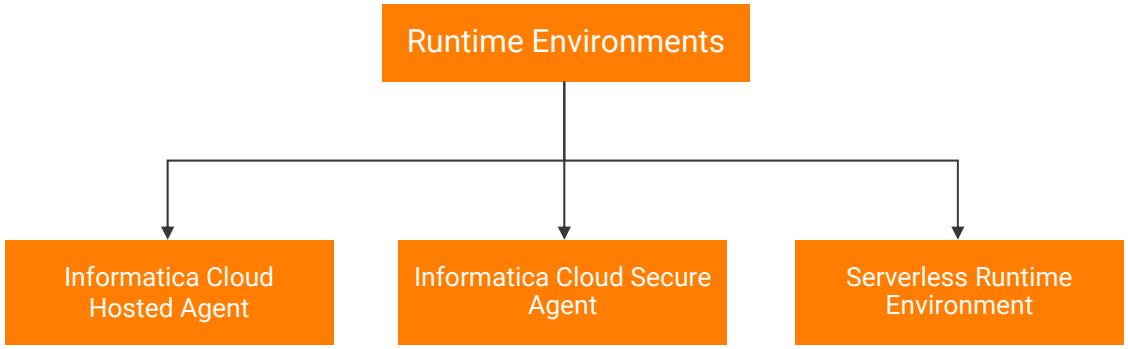
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IICS Runtime Environment

- A runtime environment is the execution platform that runs assets such as tasks and taskflows in IICS.
- You must have at least one runtime environment in each organization so that users in the organization can run tasks.



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The projects, assets and folders within them can be managed on the Explore page. The Explore page is a Informatica Intelligent Cloud Services feature that is available for most services. If you have multiple services enabled in your Org, the Explore page may show projects, folders, and materials for all of them.

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Informatica Cloud Hosted Agent

- You can use Informatica Cloud Hosted Agent to run tasks that use certain connectors.
- Used to run synchronization, mapping, and replication tasks.
- Uses certain connectors such as Amazon S3, Google Analytics, Marketo, Microsoft Azure Blob Storage, and Workday.
- Hosted Agent can process limited volumes of data.
- You cannot add, delete, or configure a Hosted Agent.

Note	You must have the **Cloud Runtime** license enabled in your Org to use Cloud Hosted Agent to run the tasks.

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The Hosted Agent is used to run synchronization, mapping, and replication tasks in IICS that uses certain connectors as source or target.

As, the Hosted Agent runtime environment is managed by Informatica Cloud Data Integration, so, you cannot add, delete, or configure a Hosted Agent.

Note: To use a hosted agent for a connection, you must check the list of connectors supported by hosted agent from the IICS help section.

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Informatica Cloud Secure Agent

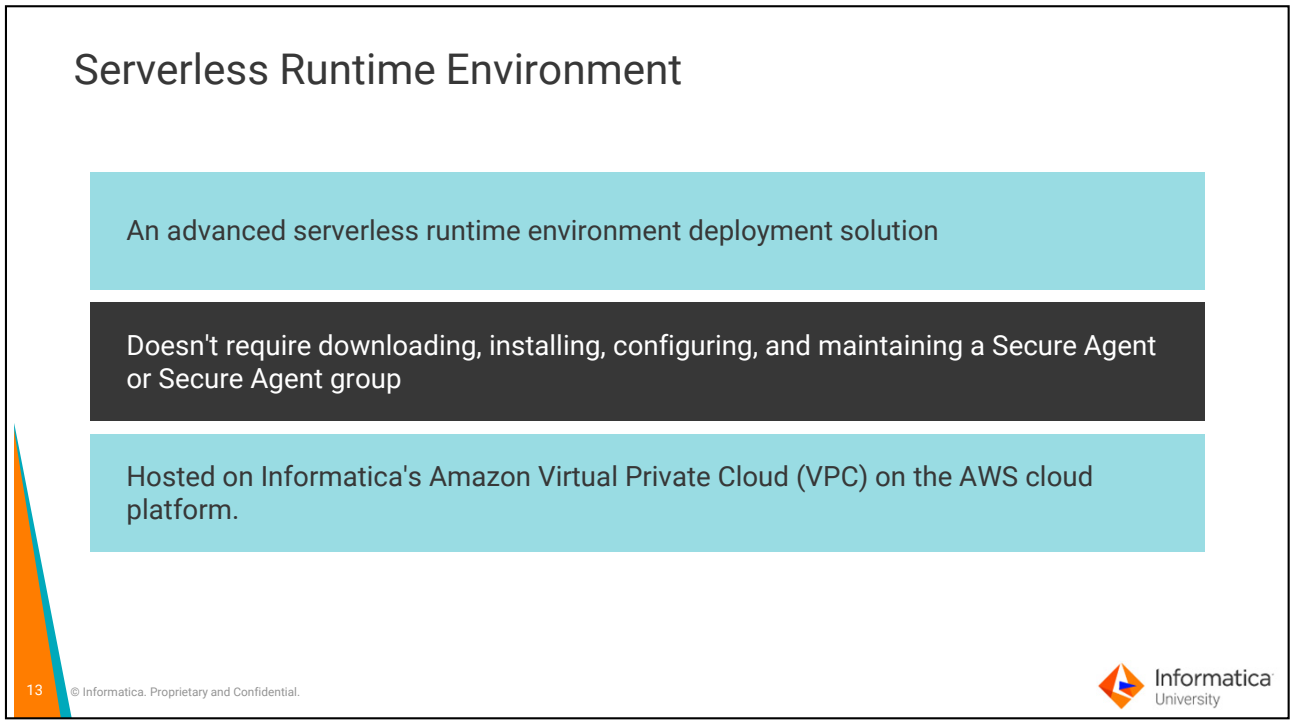
It is a Lightweight and self-upgrading program that runs tasks and enables secure communication between your organization and IICS

- Allows you to access all your local resources that reside behind your firewall.
- Available for Windows and Linux platforms.
- Can have only one agent per machine.
- It is linked to the IICS Org that you install it from.
- Uses pluggable microservices for data processing.
- Your application data never gets staged or run through Informatica Cloud servers.

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A serverless runtime environment is an advanced serverless deployment solution that doesn't require downloading, installing, configuring, and maintaining a Secure Agent or Secure Agent group. You can use a serverless runtime environment in the same way that you use a runtime environment when you configure a connection or some types of tasks in Data Integration.

A serverless runtime environment is hosted in Informatica's Amazon Virtual Private Cloud (VPC) on the AWS cloud platform. The serverless runtime environment creates a cross-account elastic network interface (ENI) to connect to your cloud environment.

To use a serverless runtime environment, your cloud environment must be on the AWS cloud platform and your VPC must have default tenancy. A serverless runtime environment cannot connect to a VPC with dedicated instance tenancy.

Note: To use a serverless runtime environment, your organization must have the serverless environment license. Additionally, users that require access to the Serverless Environments configuration page must have the PRIVILEGES.VIEW_AGENT_GROUP permission.

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IICS Connections

- A connection is an Informatica Cloud object.
- Provides access to data in the cloud and on-premise applications, databases, and flat-files.
- Specifies the location of sources, lookup objects, and targets included in a task.
- Use connectors to create connections in IICS.
  - IICS provides the following two types of connectors:

Native Connectors	Add-on Connectors
• Pre-installed in an IICS Org. • Does not require any additional setup. Examples- • SQL Server • Flat File • Oracle • FTP/SFTP	• Created by Informatica or its partner. • Need to add the connector to your IICS Org. Examples- • Amazon Redshift • Workday • Amazon Web Services S3 • Microsoft Azure Blob Storage V3

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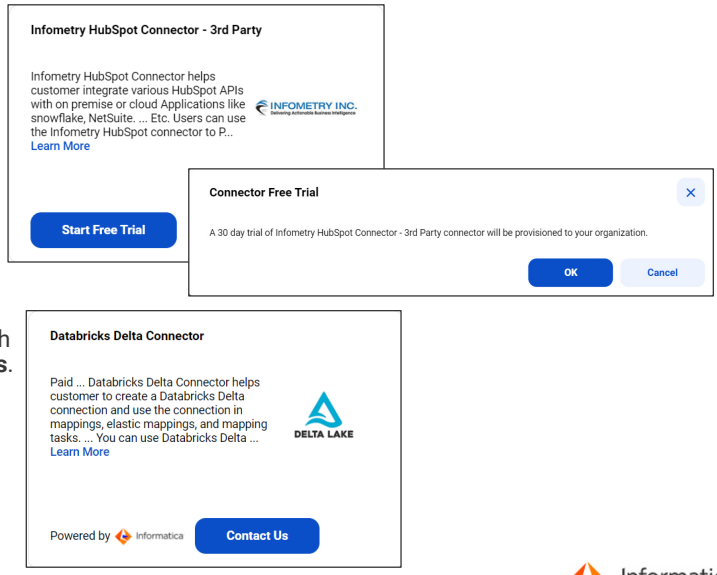
A connection is an Informatica Cloud object that provides access to data in the cloud and on-premise applications, platforms, databases, and flat files. Connections specify the location of sources, lookup objects, and targets included in a task.

You can create connections using connectors. You can create a connection for any connector that is pre-installed in IICS. You can also create a connection by installing an add-on connector created by Informatica or an Informatica partner. Add-on connectors provide connectivity for connection types that are not pre-installed in IICS.

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Steps to Enable Add-On Connectors

1. In Administrator service, select **Add-On Connectors**.
2. Perform either of the following steps:
   a. To start a free trial for a connector, click **Start Free Trial** for that connector.
   b. Click **OK** to confirm.
   c. To buy a license for a connector with an expired free trial, click **Contact Us**. After which, an Informatica representative will contact you.



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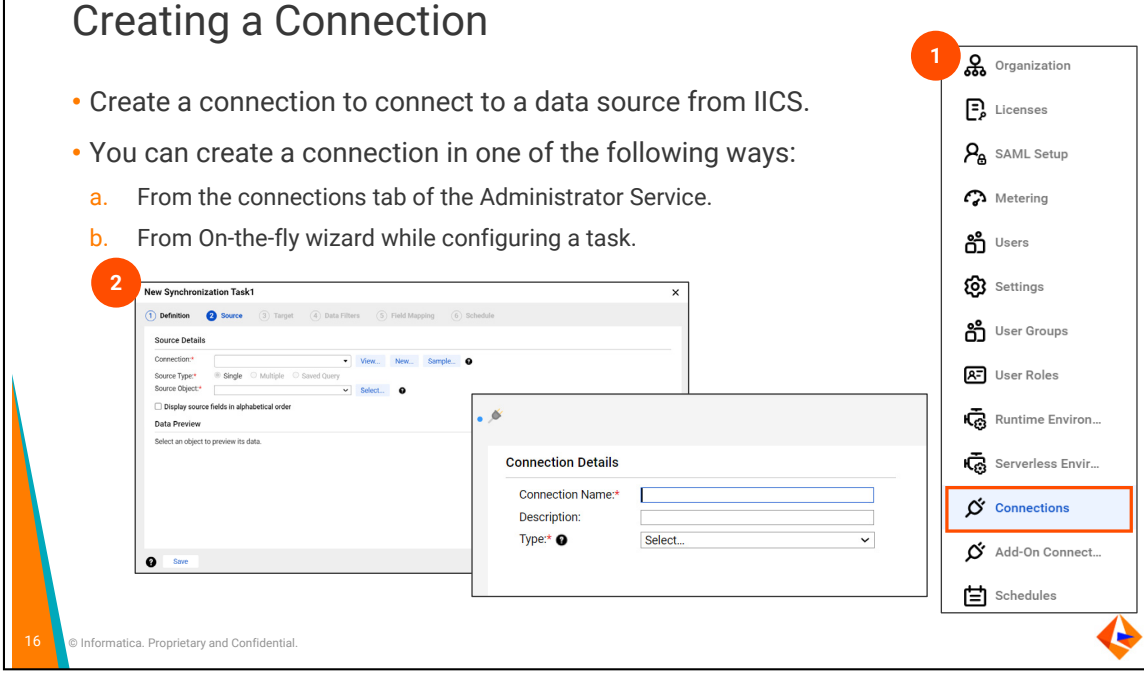
1. After the connector is enabled, the connection type becomes available to your organization and sub-organizations.
2. To use a connector in sub-organization, you must enable the connector in the parent organization. As, add-on connectors cannot be installed in a sub-organization.
3. You can disable the connector license for the sub-organization, if a sub-organization does not use a connector that is available to the parent organization.

To use the connection in IICS, refer the online help section to review the documentation for the connector's properties.

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Creating a Connection

- Create a connection to connect to a data source from IICS.
- You can create a connection in one of the following ways:
  a. From the connections tab of the Administrator Service.
  b. From On-the-fly wizard while configuring a task.



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To connect to a data source from IICS, you must create a connection.

In IICS, you can configure connections in one of the following ways:
1. On the Administrator's service **Connections** page.
2. In a wizard when you configure a mapping or task.

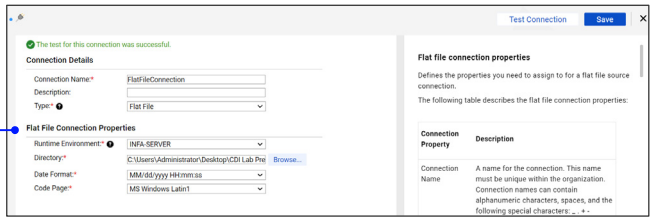
The connections page lists all the connections configured in the organization. When you configure a connection, the connection becomes available for use within the organization. If you use sub-organizations and you want a connection to be available to multiple sub-organizations, create the connection in each sub-organization.

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Configuring Connection

- Configure following properties to create a connection:
  - Connection Name (Must be unique within the Org)
  - Description
  - Type
  - Connection Specific properties
- Once the connection is configured, test and save the connection configuration to use it in IICS.

Connection Specific properties



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To configure a connection, you must enter following connection details:

- **Connection Name**: Name of the connection. Each connection name must be unique within the organization.
- **Description**: Optional. Description of the connection.
- **Type**: Type of the data source to connect to.
- **Connection specific properties**: It changes according to the type of connection. For example, the properties for flat-file connection changes from properties for salesforce connection.

Note: Refer the Data Integration Connections document or IICS connection properties in the help section to review a type's connection properties.

For most of the connection types (except for Software-as-a-Service applications), you need to specify the runtime environment for the connection. So, you must have a Secure Agent running in your IICS Org before creating any connection.

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Project and Folders

- Create projects and project folders on the **Explore** page.
  - The Explore page is available for most of the IICS services, where you can create assets.
  - Example: Data Integration service
- A Project can contain multiple folders to organize the assets used in the project.
- To create a project, in the Explore page, click **New Project**.
  - To create a folder within a project, select the folder name and click **New Folder**.

Important:

- You cannot create folders within folders.
- You cannot use the following characters on Explore page:
  # ? ' | { } " ^ & [ ] / \

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The projects, assets and folders within them can be managed on the Explore page. The Explore page is a Informatica Intelligent Cloud Services feature that is available for most services. If you have multiple services enabled in your Org, the Explore page may show projects, folders, and materials for all of them.

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Module Summary

This module showed you how to:

- Describe the Data Integration (DI) Service
- List the Cloud Data Integration assets and components
- Explain the pre-requisite for using Data Integration Service
- Discuss types of runtime environments
- Define Connections

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Module 3
Synchronization Task

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Module Objectives

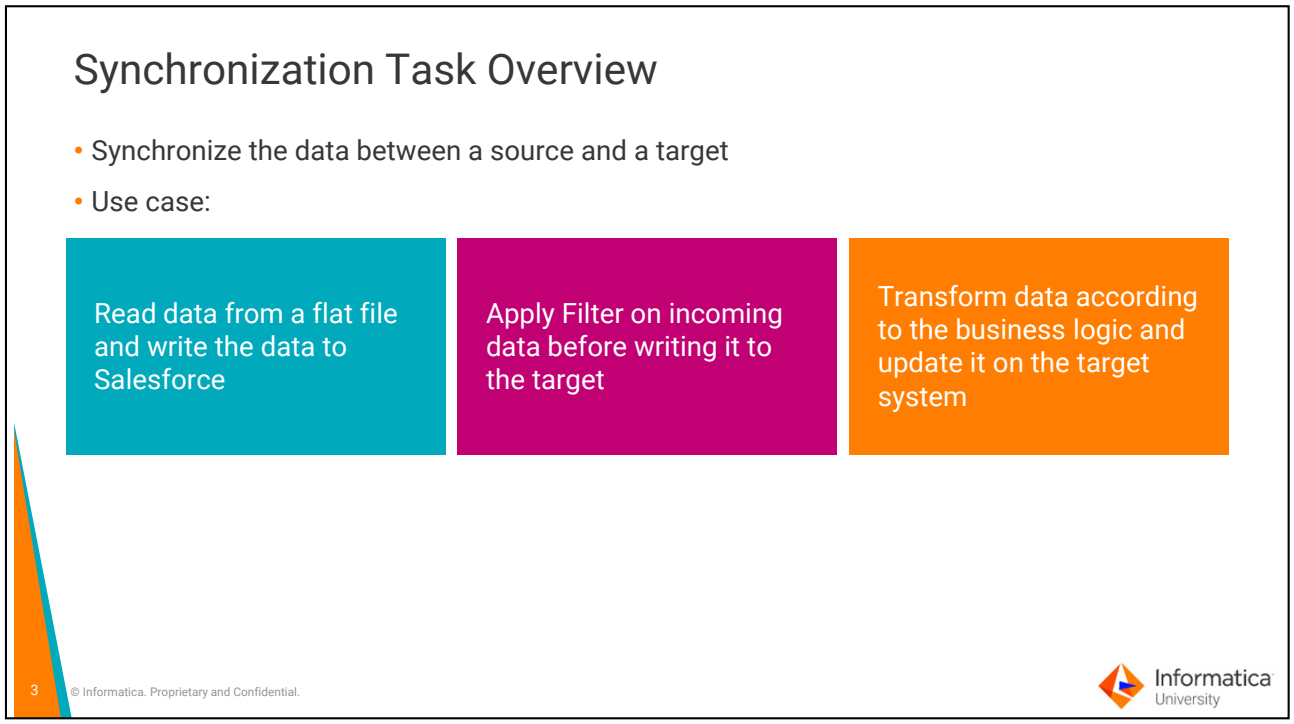
After completing this module, you will be able to:

- Define Synchronization Task
- Discuss Synchronization Task Wizard
- Explain Saved Queries
- Create a Synchronization Task
- Identify status of a Synchronization Task

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A Synchronization Task allows you to synchronize the data between a source and a target. In IICS, the supported source and target types include Database connection, Flat File connection, and Salesforce connection.

Some examples of business scenarios where you can use a synchronization task are:
- Reading data from a Flat File and writing the data to a Salesforce account
- Applying Filter on incoming data before writing it to the target
- Transforming the data according to the business logic and updating it on the target system

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Synchronization Task Wizard

- Six step wizard that allows user to synchronize data between database, flat-file, and Salesforce as source or target.
- Allows use of expressions to transform the data as per business logic or use data filters to filter data before writing it to targets.
- The following six steps are part of synchronization task wizard:

Definition Step	Source Step	Target Step
Data Filters Step	Field Mapping Step	Schedule Step

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Topic

Definition Step

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Step 1 – Definition Step

- Specify the Task Name, Location, Description, and the Task Operation
  - Task Name specifies the name of the synchronization task
  - Location specifies the location where you want to save the task
  - Description specifies a brief note about the task
  - Task Operation specifies the operation that the synchronization task performs
- Select one of the following task operations:
  - Insert
  - Update
  - Upsert
  - Delete

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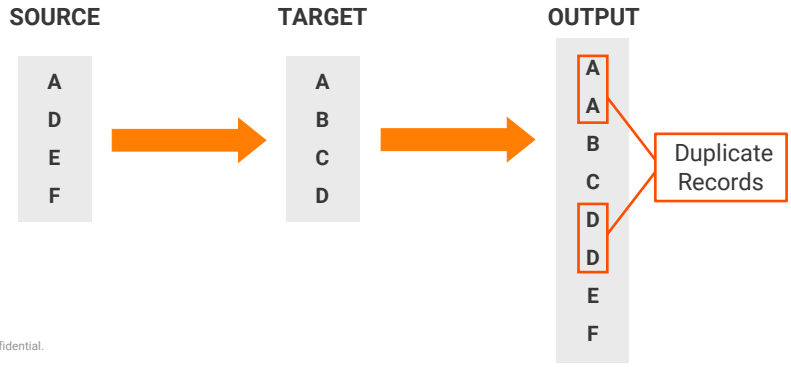
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Step 1 – Definition Step (continued)

Insert Operation

- Inserts all source rows into a target
- Ideal Use: one-time or initial data load
- Risk of duplicate records in the target



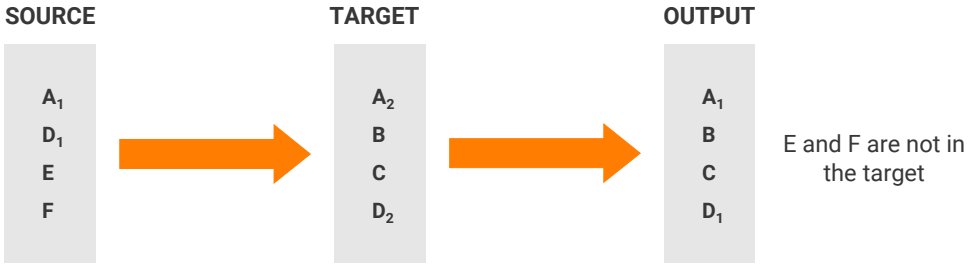
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Step 1 – Definition Step (continued)

Update Operation

- Updates rows in the target that exist in the source
- Does not update source rows that do not exist in the target



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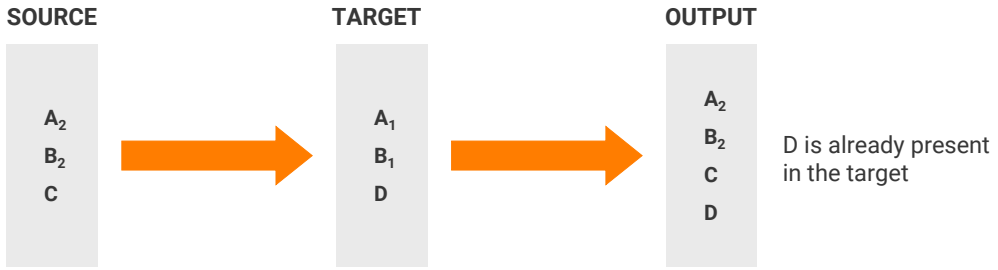
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Step 1 – Definition Step (continued)

Upsert Operation

- Updates existing records and insert new records in the target
- Does not support flat file target



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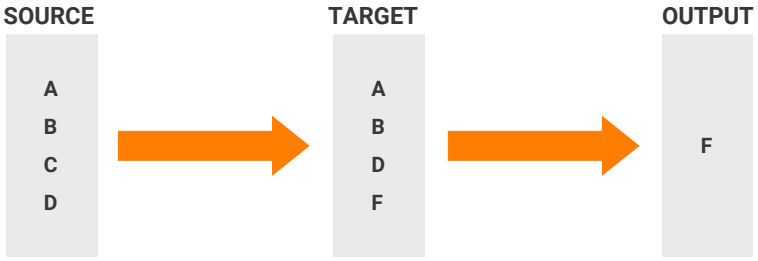
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Step 1 – Definition Step (continued)

Delete Operation

- Deletes all rows from target that exists in the source



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Topic

Source and Target Steps

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Step 2 – Source Step

- The source for a synchronization task can be:

Single Flat File | Database Table | Salesforce Object | Saved Query

- Single object
- perform operation on a single source object
- Multiple object
- configure multiple database tables or Salesforce objects as the source
- Saved Query
- an IICS component used to run SQL statements against a database

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Saved Queries

- A saved query is an IICS component used to run SQL statements against a database.
- You can use a saved query component in following IICS assets:
  - as source object in a synchronization task
  - as a query in an SQL transformation.
- The SQL statement for a saved query must be a SELECT statement.
  - Data Integration uses the SQL statement to retrieve source column information.
  - You can edit the data type, precision, or scale of each column before you save the saved query.
- Perform the following steps to use a saved query in a synchronization task:
  - Create the saved query component.
  - Select the saved query as source object in synchronization task.

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Step 3 – Target Step

- The target for a synchronization task can be:

Single Flat File | Database Table | Salesforce Object

- Target connections that you can use also depend on the task operation you select
  - Example: If you select the Upsert operation, you cannot use a flat file target connection
- Target Object can be an existing object, or you can create a new target at runtime

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Topic

Data Filters Step

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Step 4 – Data Filters Step

- Limits the data retrieval from the source
- You can apply two types of data filters – simple and advanced
- Use a simple data filter when all the filter conditions can be joined together using the AND operator
- Use advanced data filter when your source type is a flat file

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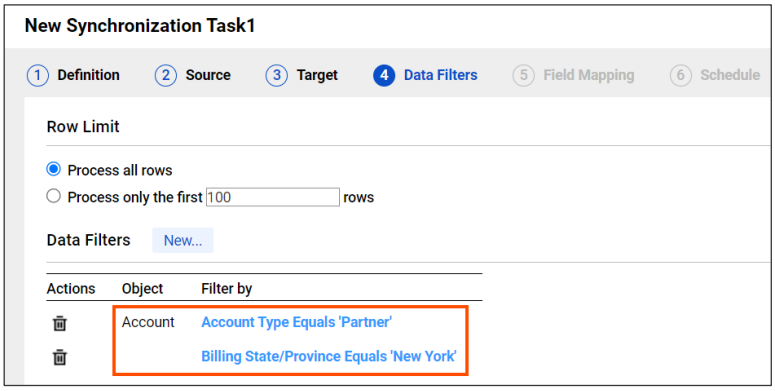
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Step 4 – Data Filters Step (continued)

Simple Filters – Example

- Source: Salesforce Accounts
- Scenario: Only load Partner accounts in New York



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Step 4 – Data Filters Step (continued)

Advanced Filters

- Use an advanced data filter:
  - when the filter conditions are complex
  - when the connection is a flat file connection
  - when it is necessary to use the OR operator in the data filter
- In an advanced filter, one expression contains all the filter conditions

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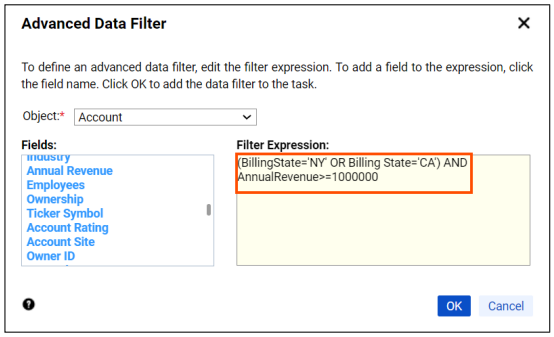
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Step 4 – Data Filters Step (continued)

Advanced Filters – Example

- Source: Salesforce Accounts
- Scenario: Only load accounts where Billing State is either New York or California, and Annual Revenue is greater than or equal to 1,000,000
- In an advanced filter, one expression contains all the filter conditions



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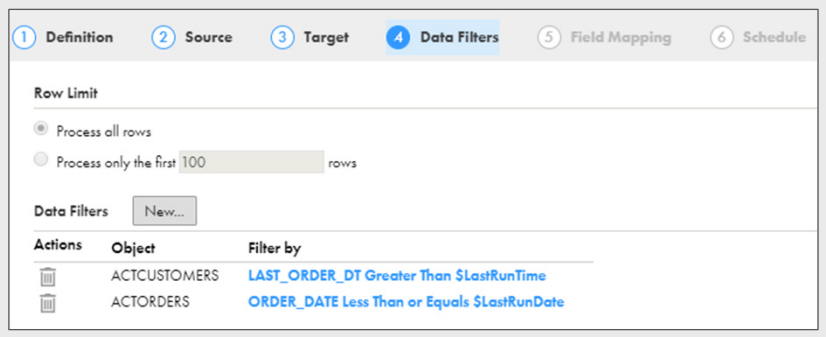
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Step 4 – Data Filters Step (continued)

Data Filter Variables

- You can use System Variables to filter new or modified records
- IICS provides access to the following System Variables:
- $LastRunDate
- $LastRunTime
- $ErrorFileName
- $SuccessFileName



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Topic

Field Mapping and Schedule Step

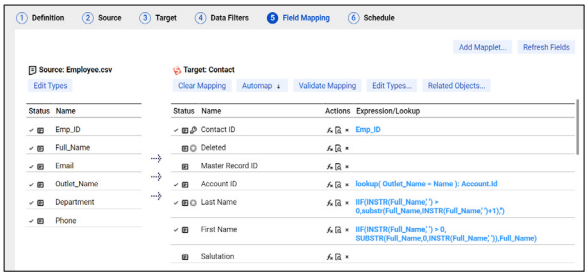
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Step 5 – Field Mapping Step

Automatch and Clear Mapping

- Define the mapping between the source fields and the target fields.
- The Automatch feature matches the source field with the target field in the following ways:
  - the Exact Field name option matches fields with the exact same name
  - the Smart Match option matches fields with similar names
- Use the Clear Mapping option to clear or rematch all the field mappings.



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Step 5 – Field Mapping Step (continued)

Field Properties

- The following icons identify the field properties:

Primary Key	External ID	Non-null
		
Indicates that field is primary key for object	Indicates that field is external ID for object (Salesforce.com only)	Indicates that field cannot contain a null value

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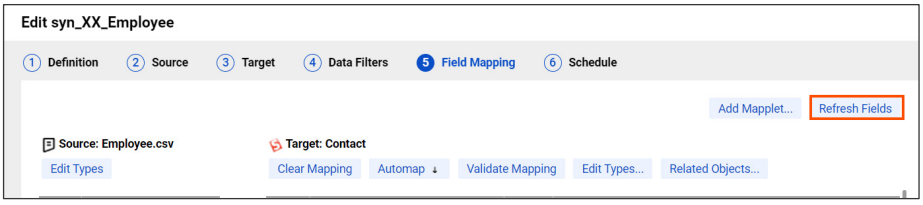
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Step 5 – Field Mapping Step (continued)

Refresh Fields

- When you create a synchronization task, the Data Integration service stores field metadata for all the source and target fields
- When you change the source or target of an existing task, you must use the Refresh Fields option



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Step 5 – Field Mapping Step (continued)

Edit Types and Validate Mapping

- Map the fields with compatible data types
- To configure or edit the field data types, IICS provides the Edit type option
- Edit type option is not available for all target types
- After you map the source and target fields, you must always validate the mapping

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Step 5 – Field Mapping Step (continued)

Field Expression

- Transforms source data before loading to target.
- You can use field expressions in the following scenarios:
  - mapping multiple source fields to a single target field
  - converting data values (to/from date and string)
  - performing data clean-up (trimming leading or trailing blank spaces and removing unnecessary characters)

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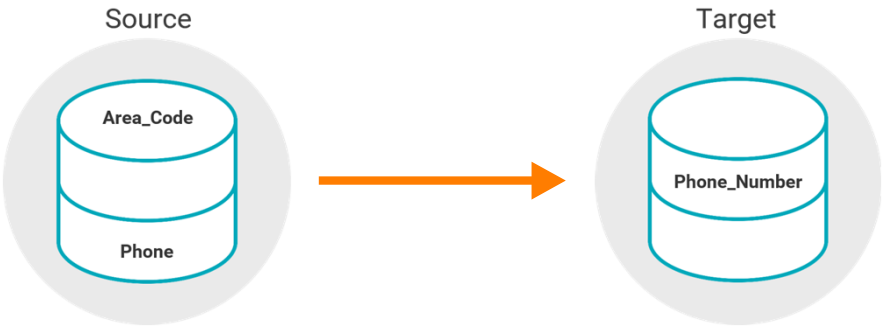
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Step 5 – Field Mapping Step (continued)

Field Expression – Example

- When you drag and drop two source fields onto a single target field, IICS automatically writes the field expression



Expression: concat (Area_Code, Phone)

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Step 5 – Field Mapping Step (continued)

Field Lookup

- Use the Field Lookup option to look up some fields for integration purposes
- Field Lookup allows you to retrieve information from any lookup connection
- Field lookup retrieves the data based on a lookup condition

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Step 5 – Field Mapping Step (continued)

Field Lookup – Example

Name	Address	City	State	zip
DHL	34 West Street	Mckinney		75070

	A	B	C	D	E	
1	Zip	lat	long	City	State	County
32899	75063	32.767268	-96.777626	IRVING	TX	DALLAS
32900	75065	33.131089	-97.008563	LAKE DALLAS	TX	DENTON
32901	75067	33.047871	-97.050897	LEWISVILLE	TX	DENTON
32902	75068	33.17679	-96.958293	LITTLE ELM	TX	DENTON
32903	75069	33.141438	-96.588295	MC KINNEY	TX	COLLIN
32904	75070	33.230381	-96.627018	MC KINNEY	TX	COLLIN
32905	75074	33.109044	-96.578819	PLANO	TX	COLLIN
32906	75075	33.162417	-96.71546	PLANO	TX	COLLIN

Lookup return value

Name	Address	City	State	zip
DHL	34 West Street	Mckinney	TX	75070

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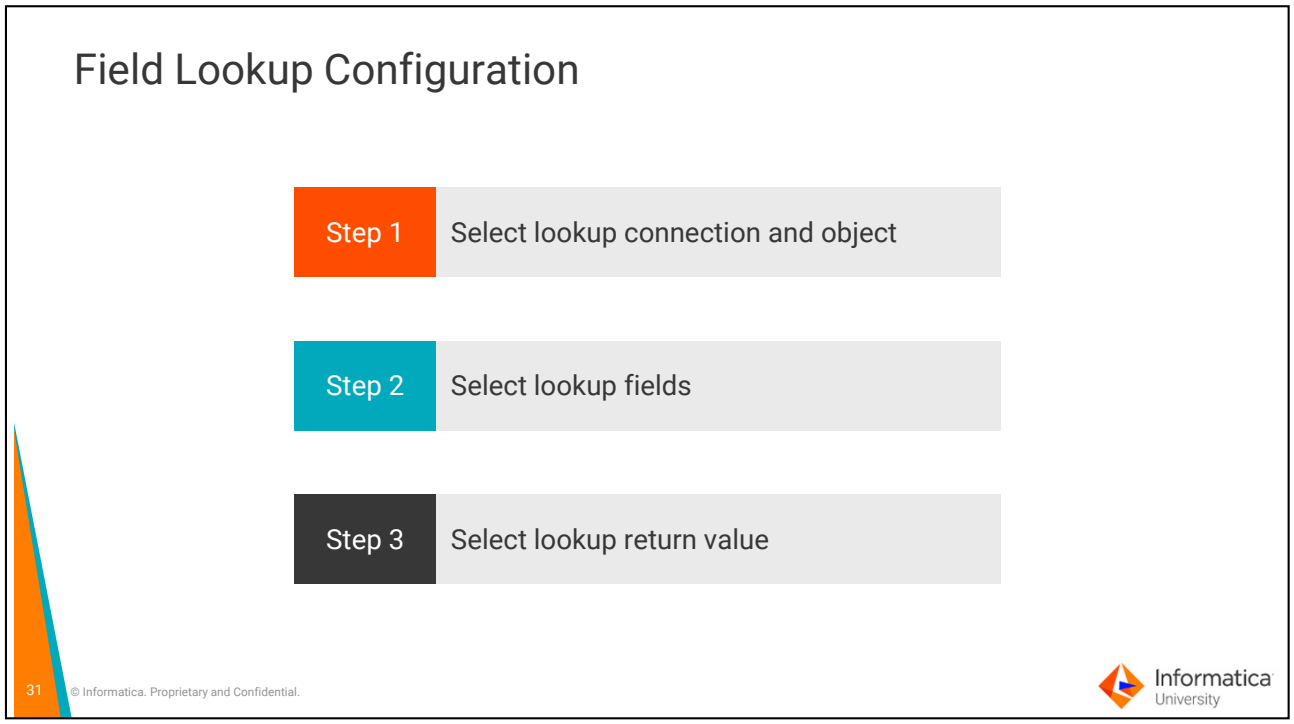
Rules and Guidelines for Lookup

- If the lookup is on a flat file:
  - must be comma-delimited
  - make sure Secure Agent has access to flat file directory (Windows OS)
- Minimize number of lookups per task
- Source field and lookup field in the lookup condition must have compatible data types

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Configuring a Field lookup is a three-step process.

Step 1: You must select a connection and an object to perform the lookup. The lookup connection does not need to be the same as the source or the target connection for the task.

Step 2: You must specify the lookup to compare.

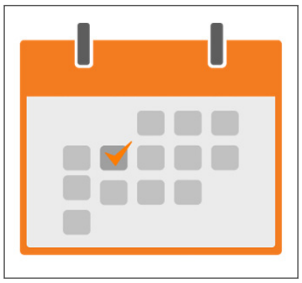
Step 3: You must select the return value. You get the return value when the lookup finds a match in the lookup object.

When you configure a field lookup, you must also configure how the Data Integration service handles multiple matching return values. The Data Integration service can randomly choose a matching value or return an error. When the lookup returns an error, the Data Integration service writes the row to the error rows file.

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Step 6 – Schedule Step

- On the Schedule page of the synchronization task wizard, you can specify to run a synchronization task manually or schedule it to run at a specific time or interval.
- Email notifications allow you to monitor the status of the task.



Allows you to run tasks at a specific time or at regular intervals

Create a schedule from the Administrator service or from the Schedule step of the synchronization task wizard

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Topic

Knowledge Check

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Which of the following operations does not support flat file target?

A. Insert
B. Update
C. Delete
D. Upsert

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Answer: d

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Which of the following source and target types are prohibited from usage in synchronization tasks?

A. Database
B. Flat file
C. Snowflake
D. Salesforce

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Answer: c

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Lab Activity

Lab 3-1 Creating a Salesforce Connection

In this lab, you will perform the following:

- Create a Salesforce connection

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Lab Activity

Lab 3-2 Creating a Synchronization Task

In this lab, you will perform the following:

- Create and configure a synchronization task
- Run the task and validate the results in Salesforce

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Lab Activity

Lab 3-3 Using Filter, Expression, and Lookup in a Synchronization Task

In this lab, you will perform the following:

- Create data filter
- Create field expressions
- Use a lookup to relate outlet name and account name

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Lab Activity

Lab 3-4 Using Saved Query in a Synchronization Task

In this lab, you will perform the following:

- Create a saved query
- Use a saved query in the Synchronization task

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Lab Activity

Lab 3-5 Using Upsert Operation in Synchronization Task

In this lab, you will perform the following:

- Create a Synchronization task using the Upsert operation
- Map fields using Salesforce External ID
- Run the task and validate the results in Salesforce

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Module Summary

This module showed you how to:

- Define Synchronization Task
- Discuss Synchronization Task Wizard
- Explain Saved Queries
- Create a Synchronization Task
- Identify status of a Synchronization Task

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Module Objectives

After completing this module, you will be able to:

- Define Data Transfer Task
- Describe Data Transfer Task wizard
- Create a Data Transfer Task
- Configure Data Transfer Task

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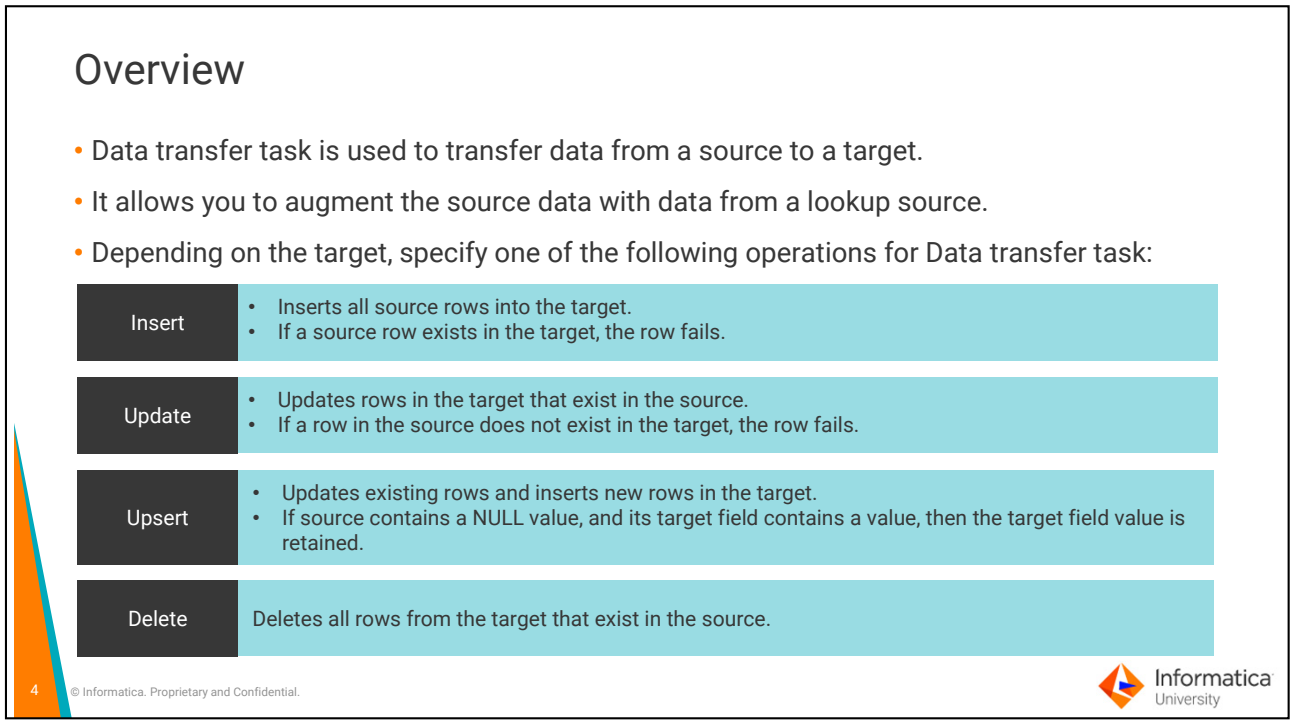
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Topic

Data Transfer Task

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You can use a data transfer task to transfer data from a source to a target. For example, you might use a data transfer task to transfer data from an on-premise database to a cloud data warehouse. When you configure a data transfer task, you must specify one of the following task operations based on your selected target:

- Insert
- Update
- Upsert
- Delete

Depending upon the source connection, you can sort and filter the data before loading it to the target. In case of multiple filter or sort conditions, the task applies the conditions in the listed order.

To see if a data transfer task applies to the connectors you are using, see the help for the relevant connectors.

Note: To create and run data transfer tasks, you need the Data Integration Advanced license.

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Topic

Data Transfer Task Wizard

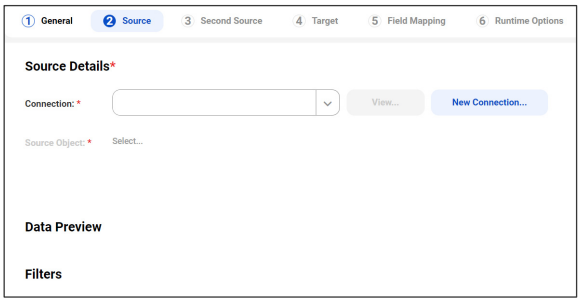
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Source Step

You can select a single source to transfer data from.

- The source connection you choose will determine the formatting and additional options that you can set up for the source.
  - For example, for a flat file source, you can configure formatting options such as the formatting type.
  - For Salesforce sources, you can configure advanced options such as row limit, and bulk query.



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Source Filters

- Apply filter conditions to the source data you send to the target to filter it.
- You can apply the following types of filter conditions:

Simple	• Select the source field and configure the operator and value to use in the filter. • The task evaluates the filter conditions using the AND logical operator to join the conditions.
Advanced	• Create a filter expression using the expression editor. You enter one expression that contains all filters. • You can use source fields and built-in functions in the expression.

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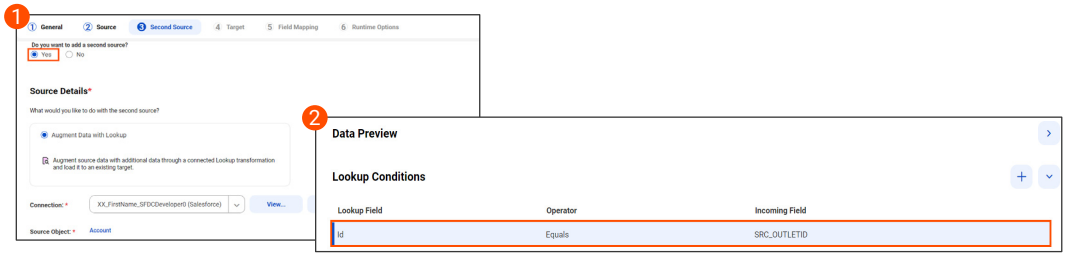
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Second Source Step

You can add a second source to be used as a lookup source when configuring a data transfer operation.

- The task queries the lookup source based on the lookup condition that you specify and returns the result of the lookup to the target.
- You can preview the data in the lookup source.
- To preview rows in alphabetical order, select Display fields in alphabetical order.



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- Configure the lookup source on the Second Source page.
- When you want to add a related value or values from the lookup source to the source data, choose a second source.
- For example, the source is an orders table that contains a customer ID field.
- You might retrieve the customer name and address from the lookup source so that you can include them in the target object.
- The task returns all fields from the lookup source.
- You must set up one or more lookup conditions when you choose a second source to utilize as a lookup source.
- Second source filters- You can apply filter conditions to filter the combined data.

You can configure the Simple and Advanced types of filters:

- **Simple**: To configure a filter condition, select a source field and configure the operator and value to use in the filter.
- **Advanced**: Create a filter expression using the expression editor. You enter one expression that contains all filters.

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Lookup Condition

You must set up one or more lookup conditions when you choose a second source to utilize as a lookup source.

- When a lookup source returns values, it is determined by a lookup condition.
- Example for Lookup:
  - You might configure the following lookup condition when the original source contains the CustID field.
  - When the customer IDs match, you want to return values from the lookup source, which contains the CustomerID field:

Lookup Field	Operator	Incoming Field
CustomerID	Equals	SRC_CustID

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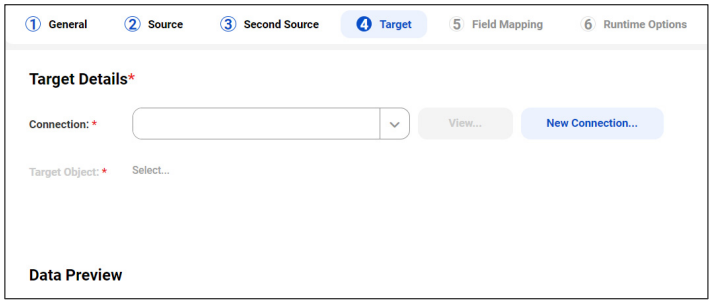
- When you configure a lookup condition, you compare the value of one or more fields from the original source with values in the lookup source.
- A lookup condition consists of an operator, an incoming field from the original source, and two fields from the lookup source.
- You can use the following operators in a lookup condition:
  - Equals
  - Not Equals
  - Less Than
  - Less Than or Equals
  - Greater Than
  - Greater Than or Equals

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Target Step

A single object may be the target of a data transfer task.

- Select a target object or create a new target object at run time.
- The task operations that you can select depend on the target connection that you use.
- Select Display fields in alphabetical order to preview rows in that order.



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Field Mapping

To map source fields to target fields in a data transfer task, configure field mapping.

- Configure field mapping on the Field Mapping page of the data transfer task wizard.
- You can configure the following field mapping options:

Options	• Configure which fields to display. • Choose one of the following display choices by clicking Options:	• Show All • Show Mapped • Show Unmapped
Automap	• Fields with the same or a similar name are linked automatically through data integration. • Click Automap and select from the following mapping options:	• Exact Field Name • Smart Map • Undo Automap
Actions	• Additional field link options. Provides the following options:	• Map Selected • Unmap Selected • Clear Mapping

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- You must map at least one source field to a target field.
- Click the name of the mapped field to configure a field expression after mapping a field.
- You can include fields and built-in functions in the expression but not user-defined functions.

Field data types: Each field in the source and target is given a data type when you build a data transfer task using Data Integration.

- When you add a field to a target that you create at run time, you select the data type.

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Runtime Options Step

Configure runtime options on the Runtime Options page.

- A data transfer task can be executed either manually or on a schedule.
- Select **Run on a schedule**, choose the schedule, and then click OK to run a data transfer task on a schedule.
- Click **Save**.

Note: You must create the schedule in Administrator before you can select it in the task.

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Configuring a Data Transfer Task

To configure a data transfer task, perform the following steps:

1. Define the data transfer task.
2. Configure the source.
3. Optionally, configure a second source to use as a lookup source.
4. Configure the target.
5. Configure field mappings.
6. Configure runtime options.

- You can click Save at any moment while completing the data transfer task wizard to save your work.
- Use the Validation panel to validate the task.

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Topic

Knowledge Check

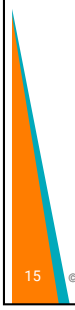
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To run a data transfer task on a schedule, edit the task in the data transfer task wizard to associate the task with a schedule.

a. True

b. False



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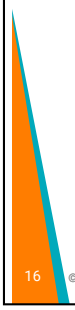
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Answer: a

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Which of the following operations updates all rows in the target that also exist in the source and inserts all new source rows into the target.

a. Update
b. Insert
c. Upsert
d. Delete



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Answer: c

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Lab Activity

Lab 4-1 Creating a Data Transfer Task

In this lab, you will perform the following:

- Create a Data transfer task
- Configure the two sources
- Configure the runtime options

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Module Summary

This module showed you how to:

- Define Data Transfer Task
- Describe Data Transfer Task wizard
- Create a Data Transfer Task
- Configure Data Transfer Task

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Module 5

Basics of Cloud Mapping Designer

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Module Objectives

After completing this module, you will be able to:

- Discuss Cloud Mapping Designer
- List the Mapping Designer Terminologies
- Discuss Mapping Data Preview
- Define Data Flow Run Order
- Discuss Field rules
- Copy Transformations
- Customize Mapping Preview

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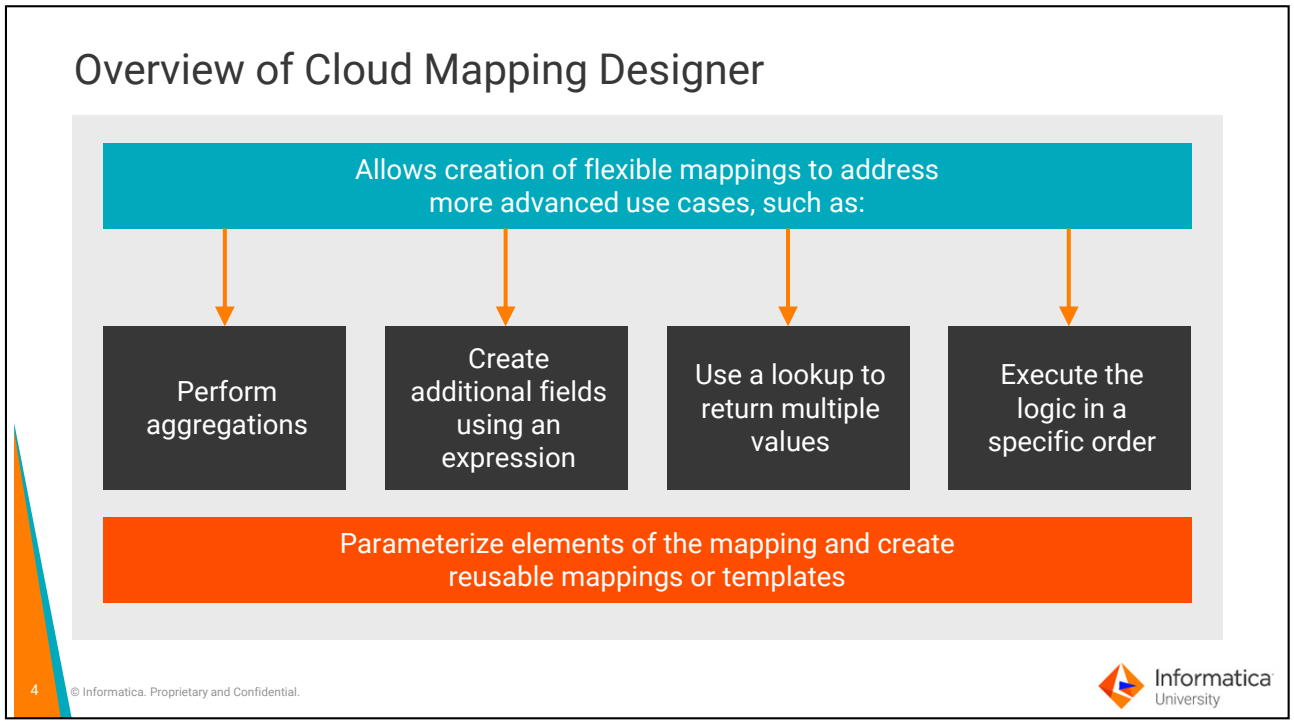
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Topic

Cloud Mapping Designer

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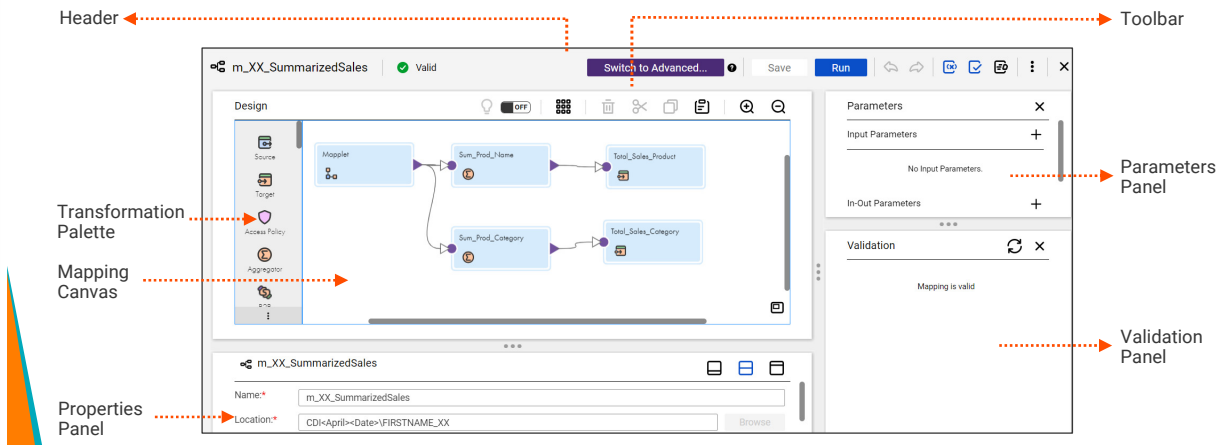
The Cloud Mapping Designer is a simple web-based interface that allows you to create end-to-end mappings. You can use the Cloud Mapping Designer to handle specific use cases that cannot be handled using the Synchronization task. The Cloud Mapping Designer allows you to perform transformation logic on the data. For example, you can perform aggregations, create additional fields using an expression, or use a lookup to return multiple values. The Cloud Mapping Designer can also execute the logic in a specific order. For example, you can filter data using the return value from a lookup.

The Cloud Mapping Designer allows you to create a mapping that includes multiple sources or targets. It also allows you to join data from heterogeneous sources. For example, you can join the data from a Sequel Server database table with the data from Salesforce. You can also write data to multiple targets.

The Cloud Mapping Designer also enables you to parameterize elements of the mapping, which allows you to create reusable mappings or templates. You can parameterize all aspects of the mapping, including sources and targets, filter criteria, lookup conditions, and so on.

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Cloud Mapping Designer



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Properties panel displays configuration options for the mapping or selected transformation. Different options display based on the transformation type.

The header section displays the following information, buttons, and icons:

- Mapping name and Mapping status. The status can be either Valid or Invalid.
- Save button. And Run buttons to save and run the mapping respectively.
- Actions icon to create and save a mapping task based on the mapping.

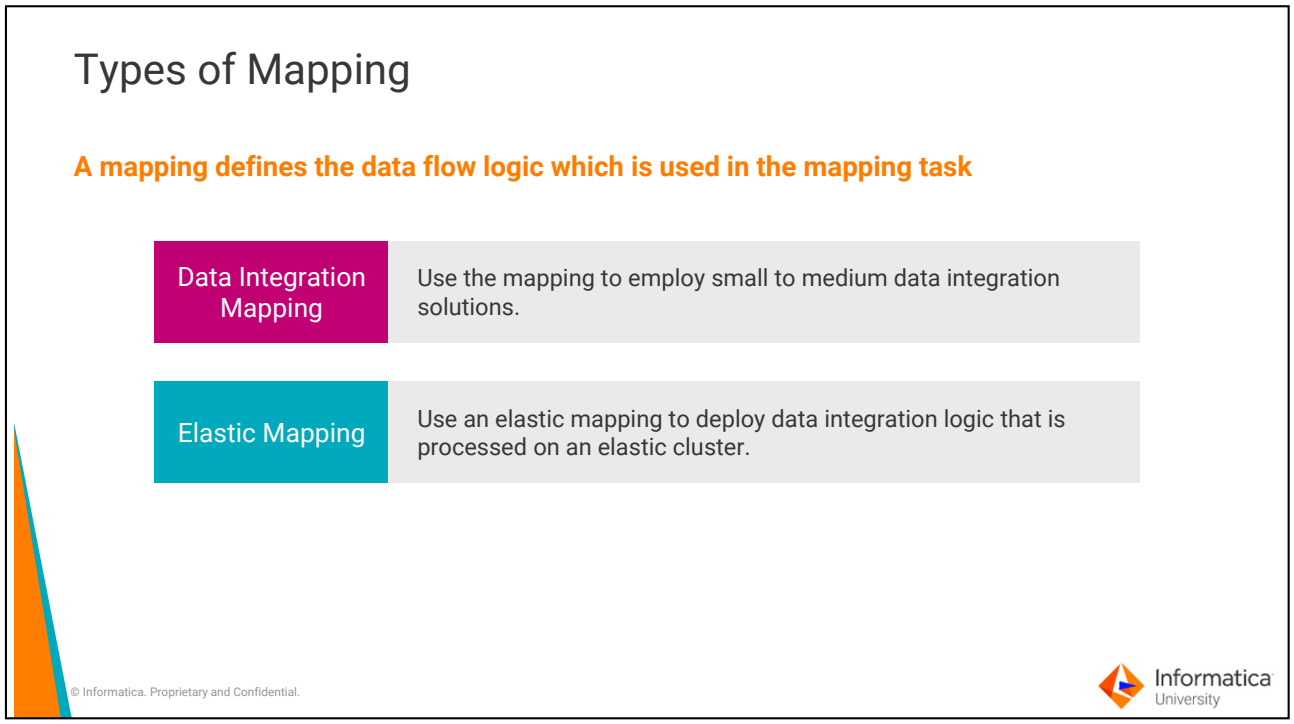
Transformation palette lists the transformations that you can use in the mapping. You use the Mapping canvas to configure a mapping. When you create a mapping, a Source transformation and a Target transformation gets added by default onto the canvas to configure. The Parameters panel lists the parameters in the mapping. You can create, edit, and delete parameters, and see where the mapping uses parameters in this section.

Validation panel lists the transformations in the mapping and displays details about mapping errors.

Toolbar displays the following icons and buttons:

- Recommendation toggle turns the CLAIRE™ recommendations on or off.
- Arrange All icon arranges the transformations in the mapping.
- Delete icon to delete a transformation.
- You can also Cut, copy, paste, zoom in, and zoom out the mapping objects using their respective icons.

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- To configure mappings, use data Integration mapping designer.
- You can use transformations such as filter transformation, expression transformation in your mappings
- You can use components such as mapplets, business services in the mappings.

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Topic

Mapping Templates

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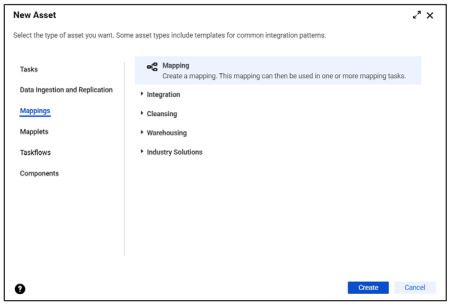
Mapping Templates

Instead of building a mapping from scratch, you can use a Mapping template.

- Mapping templates are divided into the categories:

Integration | Cleansing | Warehousing

- You can use a mapping template as is or you can reconfigure the mapping.



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- You create a mapping using a copy of the mapping template when you choose a mapping template in the New Asset dialogue box.
- The mapping contains pre-populated transformations.
- To view the function of a transformation, its configuration, and the parameters used, click on any of the transformations in the mapping.

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Mapping Designer Terminologies

Term	Description
Canvas	Area where you can create a mapping
Shapes	Represent sources, targets, and data transformations
Mapping	Defines end-to-end integration that you create using Mapping Designer
Parameterized Mapping	Mapping that contains one or more parameters
Mapping Task	Allows you to configure a mapping
Deploy	Apply changes made to a mapping across all dependent mapping configuration tasks

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Canvas refers to the area where you create a mapping.

Shapes represents sources, targets, and data transformations. You can drag and drop shapes on to the canvas.

Mapping defines the end-to-end integration that you create using the Mapping Designer.

Parameterized Mapping refers to a mapping that contains one or more parameters.

Mapping Task is an automatically generated wizard that allows you to configure a mapping, including schedules and parameter value assignments.

When you make changes to a mapping, you can promote the changes to any dependent mapping task by deploying them

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Mapping Lifecycle

Design a Mapping

- Must contain at least one source and one target
- Optionally create parameters for connections, objects, and transformations

Run and Test the Mapping

- Verify that the mapping works
- Enter values for parameters

Create Mapping Tasks

- Run on schedule or as part of task flow
- Enter values for parameters

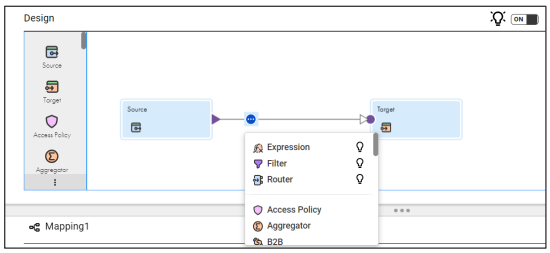
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CLAIRE Recommendations

- Enable Informatica's AI engine, CLAIRE for transformation recommendations during mapping design.
- Click the Add Transformation icon to add transformations to a mapping directly on the mapping canvas.
- The Add Transformation icon appears when you hover over the link between transformations or when you select an unconnected transformation.



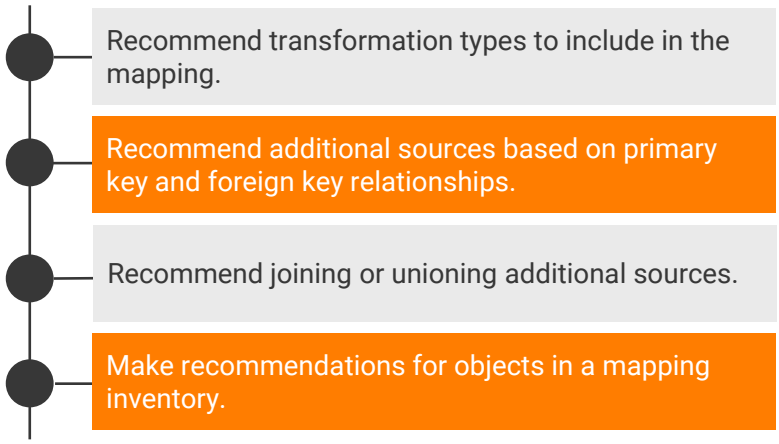
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CLAIRE Recommendations (continued)

- During the mapping design process, CLAIRE can offer the following kinds of recommendations:



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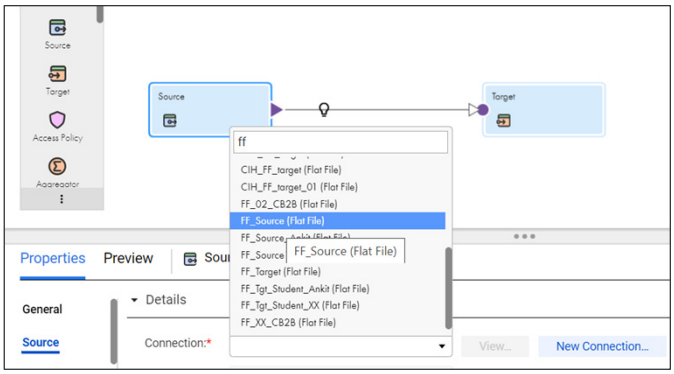
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Selection List Search in Transformation

To recommend changes to the data flow, CLAIRE consults design metadata as well as your mapping's present flow.

- Enter a search string in the selection list to search for a particular connection



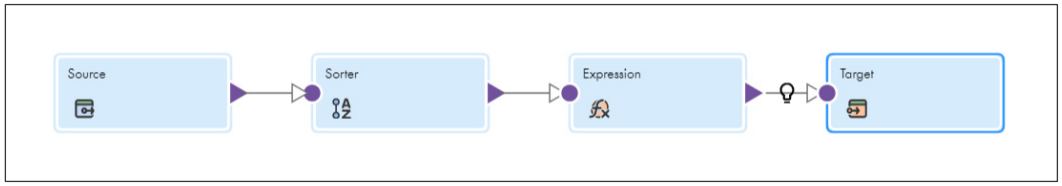
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Mapping

- Defines the flow of data from the source to the target
- Add transformation shapes to perform data transformation tasks
- Define rules for fields that are part of the transformations
- Links visually represent the flow of data in a mapping



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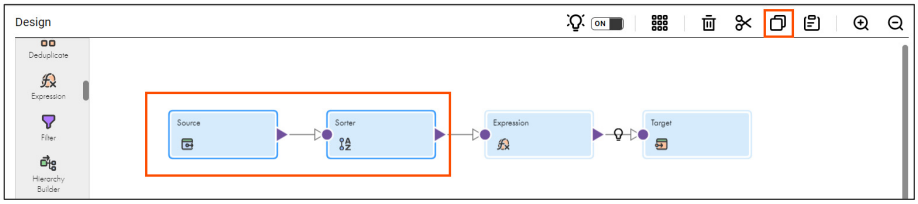
A mapping is an asset that you create in the Cloud Mapping Designer. It defines the flow of data from the source to the target. To create a mapping, you must add transformations to the mapping to perform various data transformation tasks. The links in the mapping visually represent the flow of data in the mapping.

- You can also configure parameters in the mapping to enable additional flexibility for the mapping. Parameters act as placeholders for information that you define in the mapping task.
- You can also use various components such as mapplets, business services, hierarchical schema, and so on in mappings. Components are assets that support mappings. The use of a Component in a mapping can be mandatory or optional depending upon the transformation. For example, you must use a business services asset in a mapping that includes a Web Service transformation. However, a saved query component is useful when you want to reuse a custom query in multiple mappings, but it is not mandatory.
- To use mappings and components, your organization must have the appropriate licenses. Apart from parameters and components, you can also define rules for fields that come with the transformations. The rules are based on defined criteria and allows users to create flexible mappings. For example, you can choose to include or exclude fields of a particular data type to remain valid even when you add new fields to the source.

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Copying Transformations

- Using Mapping Designer, you can copy and paste multiple transformations between the following open assets:
  - Between mappings
  - Between elastic mappings
  - Between mapplets
  - From a mapping or elastic mapping to a mapplet
- To copy one or more transformations, you must select the transformation(s) and click the Copy icon.



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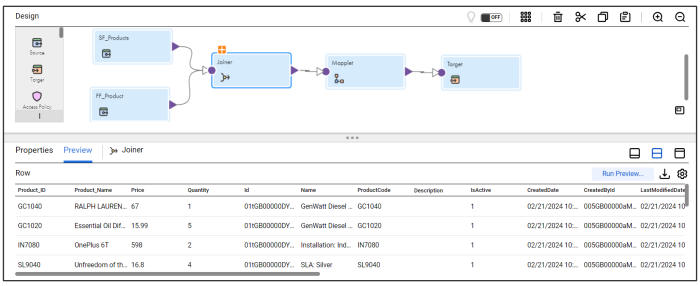
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Mapping Data Preview

- Preview data for individual transformations to test the mapping logic
- Stores preview results on the Secure Agent machine in the following directory:

<Secure Agent installation directory>/apps/Data_Integration_Server/data/cache/preview

- Data preview is not supported for:
  - Web Services transformation
  - Sequence Generator transformation
  - Target transformation
  - Data Masking transformation
  - Hierarchy Builder transformation
  - Velocity transformation



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- When you design a mapping, you can preview data for individual transformations to test the mapping logic. Data Integration displays the mapping results for the selected and the upstream transformations. To preview mapping data, your organization must have the appropriate license. You can preview data for a transformation on the Preview panel of the transformation. Data Integration also stores preview results in CSV files on the Secure Agent machine in the following directory: <Secure Agent installation directory>/apps/Data_Integration_Server/data/cache/preview.

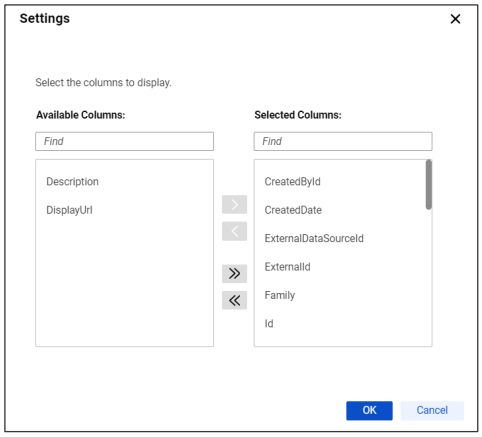
- When you run a preview job, Data Integration creates a temporary mapping task that contains a virtual target immediately downstream of the selected transformation. Data Integration discards the temporary task after the preview job completes and displays the data that is transformed by the selected transformation. You can preview data for a transformation if there are no mapping validation errors in the selected or upstream transformation.

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Customizing Preview Results

To recommend changes to the data flow, CLAIRE consults design metadata as well as your mapping's present flow.

- To customize the preview results, you can:
  - Choose the columns to display in the Preview panel
  - Re-order the columns
- You can customize the preview results from the Settings option.



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- When you run the data preview for a mapping transformation, you can select which columns to display or reorder the columns in the Preview panel of the mapping designer. You can customize the mapping preview from the Settings option in the Preview panel.

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Topic

Data Flow Run Order

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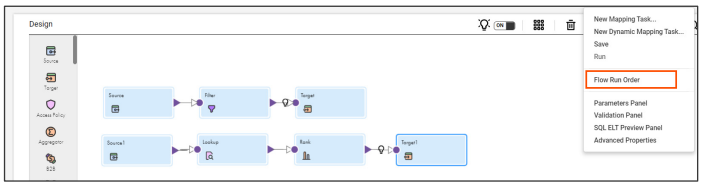
Data Flow Run Order

- Specify the order in which to run the individual data flows in a mapping.

  Example: When inserting, removing, or updating tables with primary or foreign key restrictions, you might want to specify the flow execution order.

- To configure the data flow run order in the Mapping Designer:

  1. Click Actions
  2. Select Flow Run Order
  3. Select a data flow and set the order
  4. Click Save





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- When you create a mapping that has multiple data flows, you can specify the order in which the data flows must be run. A flow includes all connected sources, targets, and transformations in a mapping.
- You can have multiple flows in a mapping. You can specify the data flow run order when you want Data Integration to load the targets in the mapping in a particular order. For example, you might want to specify the flow run order when inserting, deleting, or updating tables with primary or foreign key constraints. You can specify the flow run order for data flows with any target type.

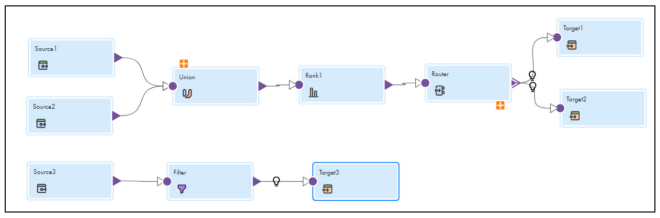
To configure the order in which the Data Integration runs the data flows in a mapping, perform the following steps in the Mapping Designer:

- Click Actions and select Flow Run Order.
- In the Flow Run Order dialog box, select a data flow, and use the arrows to move it up or down.
- Click Save.

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Data Flow Run Order (continued)

- You cannot alter the load order of the targets in a flow if there are numerous targets.
- The following image shows a mapping with two data flows:
  - In this example, there are two pipelines in the top flow and one in the bottom flow.
  - When you configure the flow run order, you cannot configure the run order of the pipelines within a data flow.
  - The following image shows the flow run order for the mapping:
  - Prior to performing the second flow, Data Integration executes the top flow and loads target3.
  - When Data Integration runs the second flow, it loads Target1 and Target2 concurrently.





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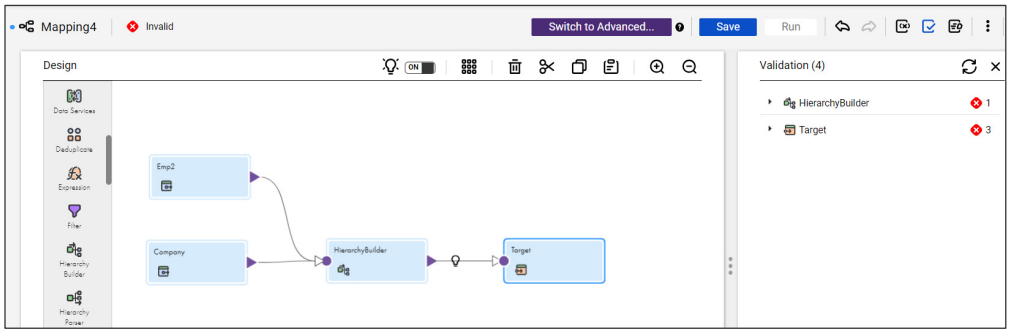
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- A pipeline is a source and all the transformations and targets that receive data from that source.
- When you configure the flow run order, you cannot configure the run order of the pipelines within a data flow.

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Validating a Mapping

- When you save a mapping, the Mapping Designer validates the mapping.
- The status of the mapping can be valid or invalid.
- Use the Validation panel to view the location and details of mapping errors.



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- When you save the mapping, the Mapping Designer automatically validates the mapping. The status of the mapping is displayed in the header of the Mapping Designer. The status can be either valid or invalid.
- If the mapping is invalid, you can use the Validation panel to view the location and details of the errors. The Validation panel displays a list of transformations in the mapping. An error icon is displayed next to the transformation that includes errors.
- As you can see in the image, the sample mapping has three errors in the Target Transformation.

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Running a Mapping

- When the mapping is valid, you can perform a test run to verify the results of the mapping.
- Test run creates a temporary mapping task.
- Data Integration service deletes the temporary mapping task after the test run is complete.

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- After you check the validity of a mapping, and ensure that it is free of errors, you can perform a test run to verify the results of the mapping.
- In the test run, you run a temporary mapping task. The task reads data from the source, writes data to the target, and performs all transformation logics in the data flow. The Data Integration service deletes the temporary task after the test run is complete.

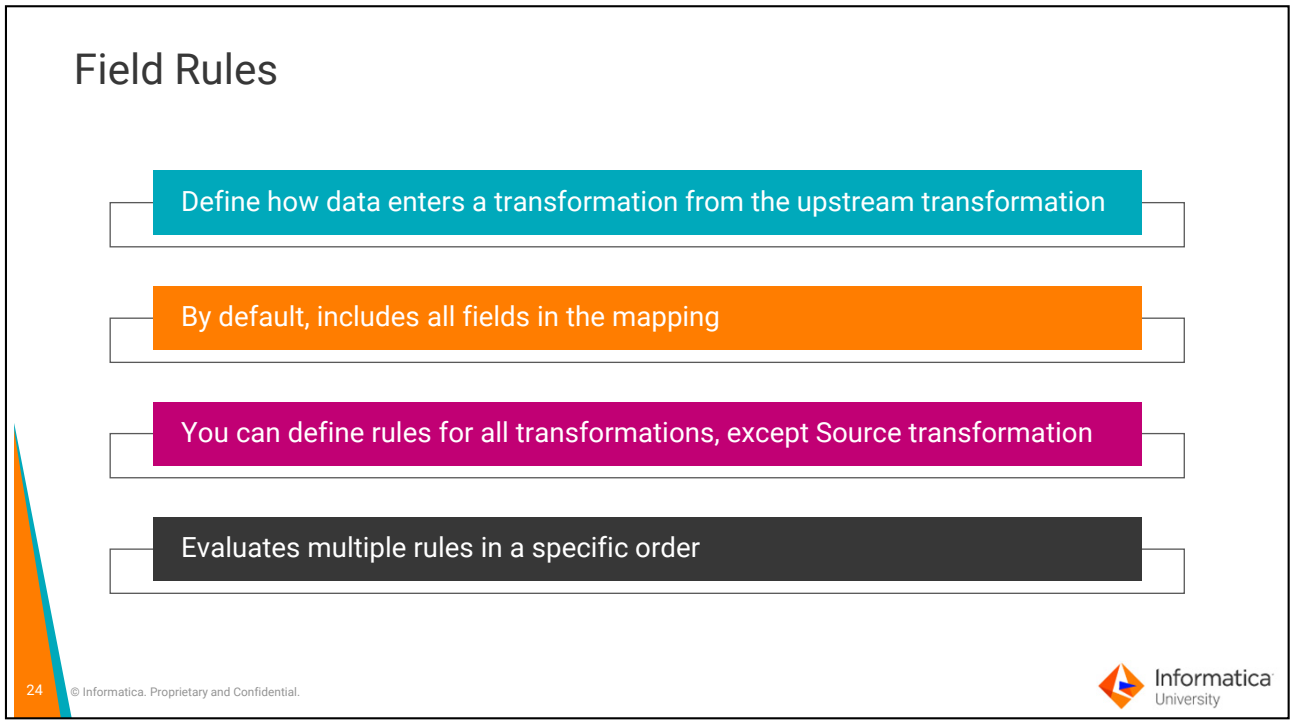
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Topic

Field Rules

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- Field rules define how data enters a transformation from the upstream transformation.
- If you do not define any field rules in the mapping, then by default, all fields are included.
- You can define field rules for all transformations, except the Source transformation, because it is always at the start of the mapping.
- If you define multiple field rules, then the rules are evaluated in the specified order. For example, if you want to include all incoming fields, except date fields, you can use the default field rule that includes all fields. You must then create a second field rule that excludes fields by data type, with the data type set to date-time.

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Field Rules – Selection Criteria

Criteria	Definition
All Fields	• Includes or excludes all fields • Can rename all fields in bulk
Named Fields	• Includes or excludes specific fields • Can create parameters to represent fields • Can rename fields individually or in bulk
Fields by Data Types	• Includes or excludes fields of a selected data type • Can rename fields in bulk
Fields by Text or Pattern	• Includes or excludes fields based on a prefix, suffix, or pattern • Can rename fields in bulk

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- Renaming fields avoids naming conflicts and helps in identifying the origin of the field. When you configure field rules, you can rename fields either individually or in bulk.
- Field naming conflicts propagate throughout the data flow. If you get a field naming conflict error, you must rename fields before the transformation where the error occurred.
- There are three options to rename fields in bulk. You can add a prefix, a suffix, or use a pattern.
- Renaming fields allows you to bring data from multiple sources into the mapping. For example, you can use “bulk rename in a prefix” to avoid naming conflicts and easily identify the original source of all fields.

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Bulk Renaming

You can rename fields individually or in bulk.

- You can rename all fields by adding a prefix, suffix, or pattern when you rename in bulk.
- Provide the text string to be used as a prefix or suffix when renaming fields.
  - For example, you can specify to rename all fields as FF_<field name>.
- Examples of how to rename fields in bulk using regular expressions are shown in the following table:

Goal	Expression
Replace all occurrences of Inc with LLC.	Inc/LLC
Replace occurrences of Inc that occur at the end of a field name with LLC.	Inc$/LLC
Replace occurrences of Loc that occur at the beginning of a field name with Branch.	^Loc/Branch
Remove all occurrences of A/C.	A\/C(.*)/$1
Add a prefix of FF and a suffix of _in to all fields.	FF_$0_in

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- When renaming fields using a pattern, you may either use a regular expression to represent the pattern or define the pattern using a task parameter.
- If the upstream transformation is a source where you cannot rename in bulk, you can add an Expression transformation to rename the fields.

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Best Practices for Creating Mappings

- Create a single data flow between the source and the target
- Connect all transformations to the data flow
- Use a Joiner transformation to join heterogeneous sources
- Resolve field naming conflicts before fields come into a transformation
- If you use a parameter for an object, you must use parameters for all conditions or field mappings in the data flow

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- You must create a single data flow between the source and the target. You must connect all the transformations to the data flow.
- To join heterogeneous sources, you must use a Joiner transformation.
- You must resolve field naming conflicts before fields come into a transformation.
- Finally, if you use a parameter for an object, you must use parameters for all conditions or field mappings in the data flow.

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Topic

Knowledge Check

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What does the field selection criteria - 'All Fields' do?

A. Does not include or exclude fields
B. Excludes all fields
C. Includes or Excludes all fields
D. Includes all fields

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Answer: d

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Instead of building a mapping from scratch, you can use a Mapping template.

A. True

B. False

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Answer: a

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Module Summary

This module showed you how to:

- Discuss Cloud Mapping Designer
- List the mapping designer terminologies
- Discuss Mapping Data Preview
- Define data flow run order
- Discuss Field rules
- Copy Transformations
- Customize Mapping Preview

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Module 6
IICS Transformations

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Module Objectives

After completing this module, you will be able to:

- Define IICS transformations
- Explain different types of transformations

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Topic

IICS Transformations

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What is a Transformation?

A mapping transformation represent the actions you perform on the data.

- It is a mapping object that modifies or passes data.
- Each transformation executes a specific function.

  Example: Source transformation reads data from a source, and an Expression transformation performs row-level calculations.
- Transformation can be active or passive.
- Active transformation changes the number of rows that pass through it.

  Example: Filter Transformation
- Passive transformation does not change the number of rows that pass through it.

  Example: Expression Transformation

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A transformation is a mapping object that modifies the data in the mapping or passes it on to the next step of the mapping. A transformation can either be active or passive.

An active transformation changes the number of rows that pass through it. For example, a Filter transformation is an active transformation because it passes data only when the criteria specified in the filter condition is fulfilled.

A passive transformation does not change the number of rows that pass through it. For example, an Expression transformation is a passive transformation because it transforms the data based on the expression that you specify. However, it does not change the number of rows that pass through it.

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Source Transformation

- It is an Active transformation
- It reads data from a source
- Defines connection and object
- Source transformation configure advanced options based on the connection type
- It add or remove fields from source
- You can read data from the following source types using a Source transformation:
  1. File
  2. Database
  3. Web service
  4. Informatica Cloud Data Integration connectors

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The Source transformation is an active transformation that reads data from a source. You can add one or more Source transformations to a mapping.

When you configure a Source transformation, you must specify a connection and an object. You can also configure advanced options based on the connection type. For example, when you select a Salesforce connection, you can use multiple related source objects and configure the Salesforce API advanced source option.

You can also add or remove fields from the Source transformation

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Target Transformation

- Defines the target connection and object for the mapping.
- Specify whether you want to use the Insert, Update, Upsert, or Delete operation.
- Configure advanced options based on the connection type.
- You can use file, database, and Informatica Cloud Data Integration connections in the Target transformation.

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The Target transformation defines the target connection and object for the mapping. When you configure the target, you must specify whether you want to use the Insert, Update, Upsert, or Delete operation. You should also map the incoming fields to the target fields.

In the Target transformation, you can configure advanced options based on the connection type. For example, when you select a Salesforce connection, you can configure options for success and error log details.

You can also create a flat file target at run time. To create dynamic files at runtime, you can use expressions to generate the file name.

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Topic

Aggregator Transformation

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Aggregator Transformation



- Allows you to perform aggregate calculations on group of data
- Active transformation
- Data Integration Service stores the data temporarily in an aggregate cache, until it completes the aggregation

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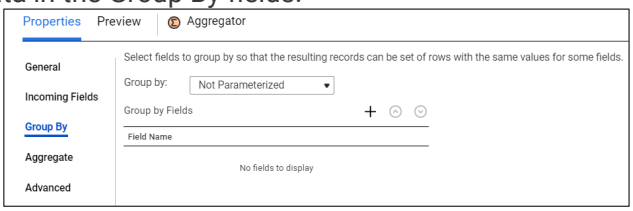
The Aggregator transformation allows you to perform aggregate calculations, such as average and sum on groups of data. The Aggregator transformation is an active transformation. This means, the number of rows that enter the transformation changes after you apply the transformation.

When you run a mapping that uses an Aggregator transformation, the Data Integration Service stores the data temporarily in an aggregate cache, until it completes the aggregation.

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Group by Fields

- Use Group by field to define how to group data for aggregate expressions.
- On the Group By tab of the properties panel configure group by fields.
- The mapping configuration task groups rows with the same data in the field when you configure a group by field.
- The task performs aggregate calculations on each group and writes the result to the last row in the group.
- When you select more than one Group By field, the task creates a group for each unique combination of data in the Group By fields.



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The Group By field groups data for aggregate expressions. When you configure a Group By field, the mapping task groups rows with the same data in the field. The task performs aggregate calculations on each group and writes the result to the last row in the group.

The result of an aggregate expression varies based on the Group By fields that you configure. When you select more than one Group By field, the task creates a group for each unique combination of data in the Group By fields. You can configure Group By fields in the Group By tab of the Properties panel.

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Aggregator Transformation – Source

- Distributed annual sales across four quarters for each store:

Store Name	Q1	Q2	Q3	Q4	Year
Walmart	30	50	48	80	2011
BestBuy	120	100	88	150	2011
Kellogs	80	108	123	134	2011
Walmart	40	60	88	100	2012
BestBuy	110	120	98	140	2012
Kellogs	100	98	133	234	2012
Walmart	40	56	68	80	2013
BestBuy	125	200	98	150	2013
Kellogs	90	128	123	134	2013
Walmart	90	86	88	180	2014
BestBuy	125	200	148	190	2014
Kellogs	90	138	126	154	2014

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Aggregator Transformation – Target

- To know the stores' annual sales and the quarterly average for each year, use the Aggregator transformation.

Store Name	Annual_Sales	Quaterly_Avg	Year
BestBuy	458	115	2011
BestBuy	468	117	2012
BestBuy	573	143	2013
BestBuy	573	143	2014
Kellogs	445	111	2011
Kellogs	565	141	2012
Kellogs	475	119	2013
Kellogs	475	119	2014
Walmart	208	52	2011
Walmart	288	72	2012
Walmart	244	61	2013
Walmart	244	61	2014

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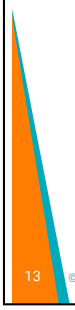
Filter Transformation

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Filter Transformation

- The Filter Transformation filters data based on the filter condition.
- Place the Filter transformation close to the mapping sources.
- It returns either a TRUE or FALSE value.
- The filter transformation can include multiple conditions.
- You can only link a single transformation to the Filter transformation.



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The Filter transformation filters data based on the filter condition that you define. To improve job performance, you must place the Filter transformation close to the mapping sources. This way, you can remove unnecessary data from the data flow.

A filter condition is an expression that returns either a TRUE or a FALSE value. When the filter condition returns a TRUE value for a row, the Filter transformation passes the row to the rest of the data flow. When the filter condition returns a FALSE value, the Filter transformation drops the row.

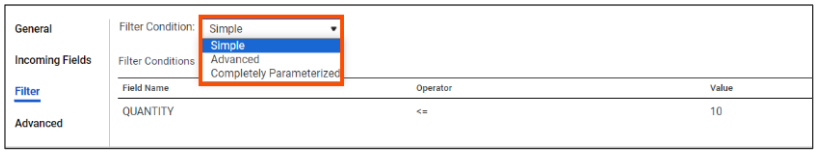
You can filter data based on one or more conditions. For example, to work with data within a specified date range, you can create conditions to remove data before and after the specified dates.

You must note that you can only link a single transformation to the Filter transformation. You cannot merge multiple transformations into it.

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Filter Condition

- Create one or more simple filter conditions.
- A simple filter condition includes a field name, an operator, and a value.
- Can use the following operators:
  - = (equals)
  - < (less than)
  - > (greater than)
  - < = (less than or equal to)
  - > = (greater than or equal to)
  - ! = (not equals)
- Use advanced filter condition to define complex expression.



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You have just seen that the filter condition is an expression that returns either a TRUE or a FALSE value. You can create one or more simple filter conditions. A simple filter condition includes a field name, an operator, and a value. You must note that filter conditions are case sensitive.

A simple filter includes operators such as – equals, less than, greater than, less than or equal to, greater than or equal to, and not equals.

When you define multiple simple filter conditions in a mapping task, the task evaluates the conditions in the order that you specify. The task evaluates the filter conditions using the AND logical operator and returns rows that match all the filter conditions.

You can also use an advanced filter condition to define a complex expression. When you configure a complex expression, you can incorporate multiple conditions using the AND or the OR logical operators.

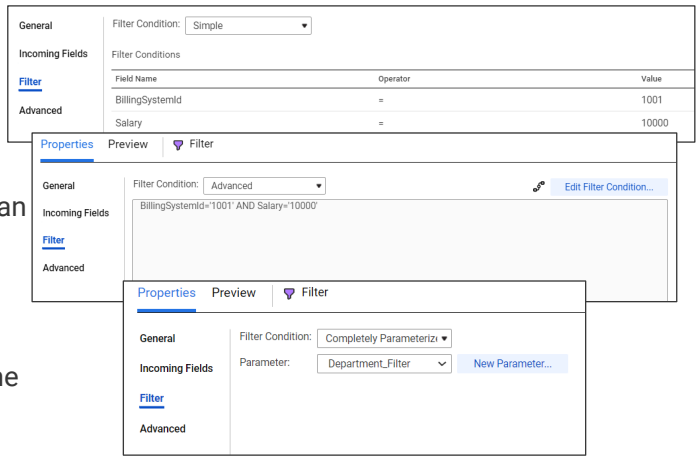
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Types of Filter Condition

- In the image of simple filter condition, filter condition will be:

  EMP_ID = '10' and SALARY = '10000'

- The next image is of Advanced filter condition, where multiple conditions can be included using the AND logical operator.
- By creating an input parameter of type expression, for which a value could be passed during the mapping runtime, the filter condition could be completely parameterized in filter transformation.



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Topic

Expression Transformation

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Expression Transformation

- Expression Transformation allows you to create new fields within the mapping.
- It allows you to create an expression field and a variable field.
- Expression field defines the calculations that you perform on an incoming field and acts as the output field for results.
- Variable field holds a variable in the expression or in other Expression transformations within the mapping
- Use an Expression transformation to:
  - perform non-aggregate calculations
  - concatenate or split incoming field values
  - insert a hard coded value into a field

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An Expression transformation allows you to create new fields within the mapping. When you create a new field, you must specify the field name, type, precision, and scale. The Expression transformation allows you to create an expression field and a variable field.

An expression field defines the calculations that you perform on an incoming field and also acts as the output field for results. You can use the field in the data flow. You can use multiple expression fields to perform calculations on incoming fields.

A variable field holds a variable in the expression or other Expression transformations within the mapping. A variable field is not available for use in downstream data flow.

You can use an Expression transformation to perform non-aggregate calculations, concatenate or split incoming field values, or insert a
hard-coded value into a field.

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Expression Transformation Field Types

- In Expression transformation new fields of below type can be created.
  1. Output Field
  2. Variable Field
  3. Input Macro Field
  4. Output Macro Field
- To perform a specific action on each record in the dataflow and pass it on to the downstream transformation, create an **Output field**.
- Create a **Variable Field** for calculations that you want to use within the transformation.

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Topic

Expression Macro

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Expression Macro Overview

- An Expression Macro is used to create repetitive or complex expressions in mappings.
- An expression macro is used to calculate values across a range of fields or constants.
  - For instance: To substitute null values in a group of fields or to label products according to a range of sales, utilize an expression macro.
- One or more input fields serve as the macro's source data in an expression macro.
- An expression represents the calculations that you want to perform.
- An output field represents the results of the calculations.
- At runtime, the task writes the results to the output fields after expanding the expression to include all the input fields and constants.
- You can create expression macros in Expression and Aggregator transformations.

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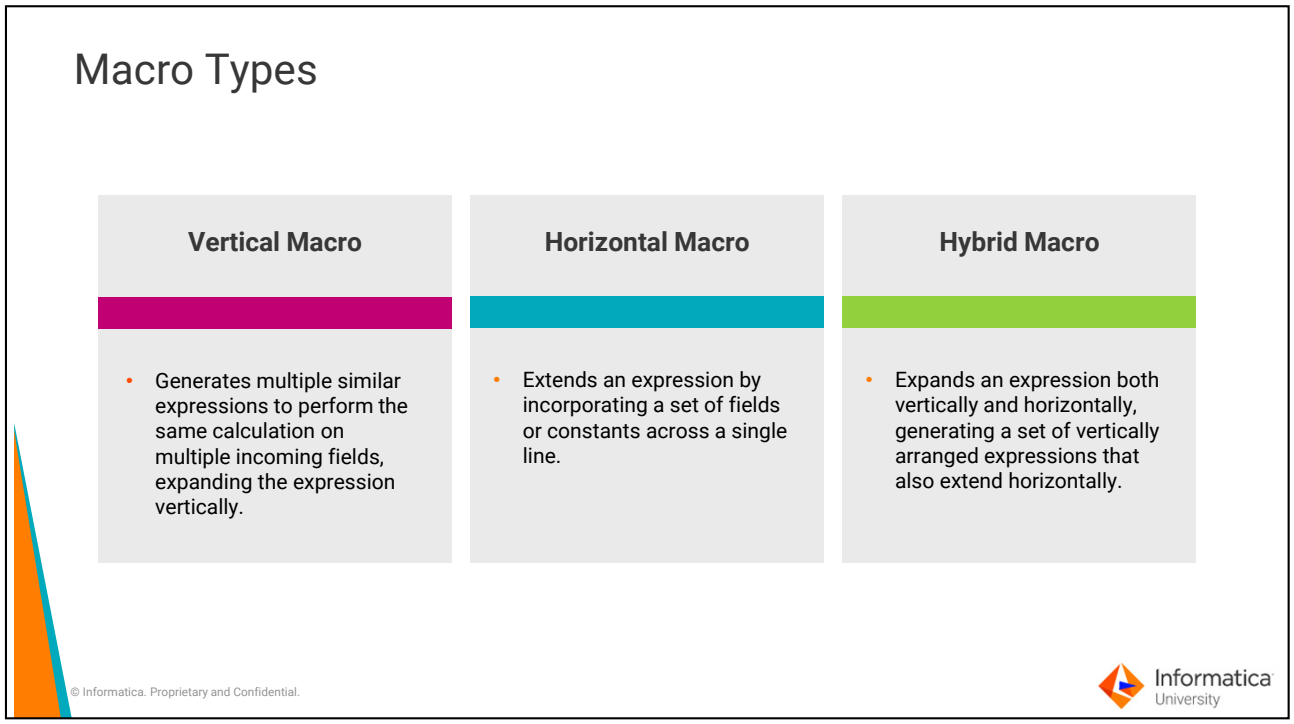
An expression macro allows you to create repetitive or complex expressions in mappings.

You can use an expression macro to perform calculations across a set of fields or constants. For example, you can use an expression macro to replace null values in a set of fields or to label items based on a set of sales ranges.

In an expression macro, one or more input fields represent source data for the macro. An expression represents the calculations that you want to perform, and an output field represents the results of the calculations. At run time, the task expands the expression to include all the input fields and constants, it writes the results to the output fields.

You can create expression macros in Expression and Aggregator transformations. However, you cannot combine an expression macro and an in-out parameter in an Expression transformation.

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Vertical Macro: A vertical macro expands an expression vertically. A vertical macro generates a set of similar expressions to perform the same calculation on multiple incoming fields.

Horizontal Macro: A horizontal macro expands an expression horizontally. A horizontal macro generates one extended expression that includes a set of fields or constants.

Hybrid Macro: A hybrid macro expands an expression both vertically and horizontally. A hybrid macro generates a set of vertical expressions that also expands horizontally.

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Macro Input Fields

- Represents input for expression macros, which can be fields or constants.
- Required for all expression macros.
- In a vertical macro, the macro input field represents a set of incoming fields.
- In a horizontal macro, the macro input field can represent incoming fields or a set of constants.
- Macro input fields follow the naming convention: **%<macro_field_name>%.**
  - Example: Create a macro input field, e.g., %AddressFields%, for applying an expression to a set of address fields.
  - Define a field rule to indicate the incoming fields to use.
  - When configuring the expression, use %AddressFields% to represent the incoming fields.

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Vertical Macro

- Macro input field represents the incoming fields
- Expression represents the calculations you want to perform on all the incoming fields
- Macro output field represents the results of the calculations
- The names of the output fields are not explicitly defined in the mapping
- Configure a field rule in the downstream transformation to include the output fields that the macro generates
- Link the output fields to target fields in the Target transformation
- When the task runs, it performs the following actions:
  - Generates multiple expressions based on the macro input fields
  - Replaces the macro output fields with the actual output fields
  - Uses the output fields to pass the results of the calculations to the rest of the mapping

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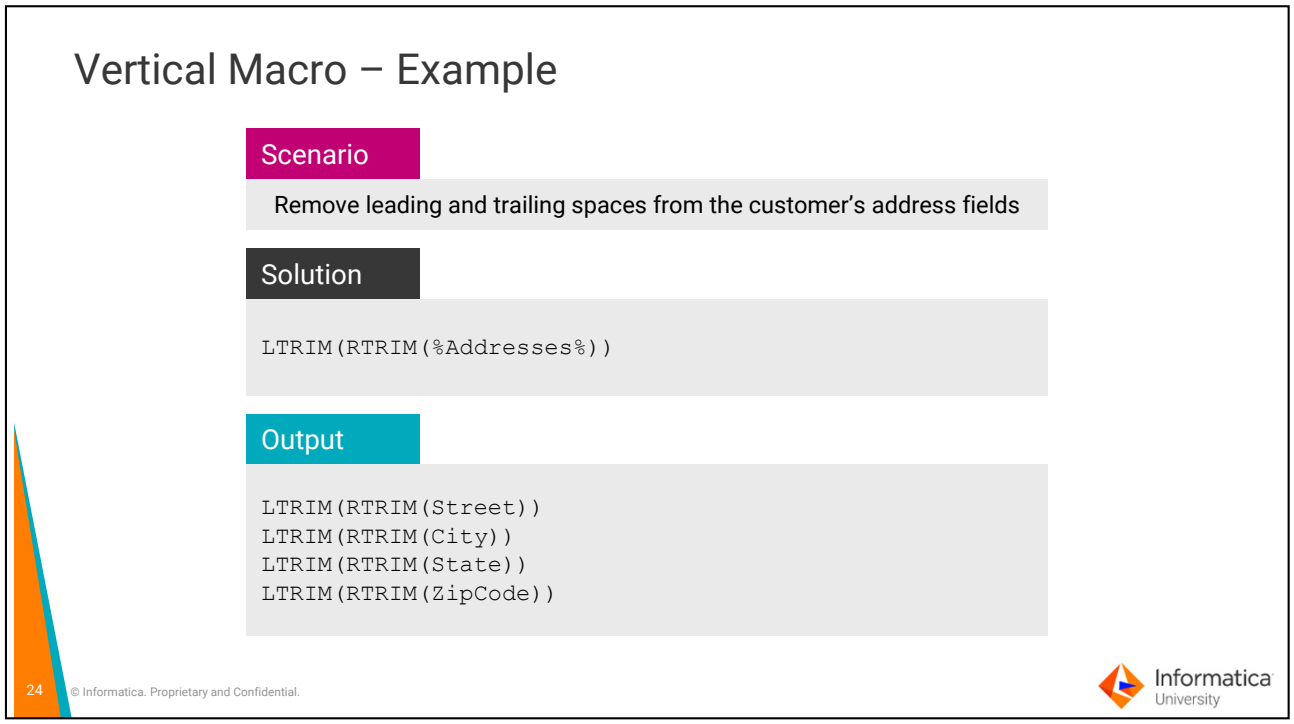
You can use a vertical macro to apply a macro expression to a set of incoming fields. The macro input field in a vertical macro represents the incoming fields. The expression represents the calculations that you want to perform on all incoming fields. The macro output field represents a set of output fields that passes the results of the calculations to the rest of the mapping. You can configure the macro expression in the macro output field.

The macro output field represents the output fields of the macro. However, the names of the output fields are not explicitly defined in the mapping. To include the results of a vertical macro in the mapping, you must first configure a field rule in the downstream transformation to include the output fields that the macro generates.

To write the results of a vertical macro to the target, you must link the output fields to target fields in the Target transformation. When the task runs, it generates multiple expressions to perform calculations on each field that the macro input field represents. The task also replaces the macro output fields with the actual output fields. The task then uses the output fields to pass the results of the calculations to the rest of the mapping.

Note: The macro output field does not pass any data.

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Look at an example of a vertical macro expression.

Consider that you want to remove leading and trailing spaces from the customer's address fields.

You can use the vertical macro expression to trim leading and trailing spaces from the address fields.

When the task runs, it generates a set of expressions to trim spaces from the address fields. As you can see, the expression removes the leading and trailing spaces from the Street, City, State, and Zip Code address fields.

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Horizontal Macro

- Use a horizontal macro to generate a single complex expression
- Macro input field represents a set of incoming fields or a set of constants
- Expression represents the calculations that you want to perform on the incoming fields or constants
- Horizontal macro expression produces one result, and the transformation output field passes the results to the rest of the mapping
- Results of the expression pass to the downstream transformation with the default field rule
- To write the results to the target, connect the transformation output field to a target field in the Target transformation

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You can use a horizontal macro to generate a single complex expression that includes a set of incoming fields or a set of constants.

The macro input field represents a set of incoming fields or a set of constants. The expression represents the calculations that you want to perform on the incoming fields or constants. You must include a horizontal expansion function in the expression.

The horizontal macro expression produces one result, and the transformation output field passes the results to the rest of the mapping. You can configure the horizontal macro expression in the transformation output field.

The results of the expression passes to the downstream transformation with the default field rule. Unlike a vertical macro, you don't have to configure additional field rules to include the results of a horizontal macro in the mapping. To write the results of a horizontal macro to the target, you must connect the transformation output field to a target field in the Target transformation.

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Horizontal Macro – Example

Scenario

Check for null values in all the fields of a customer record

Solution

```
%OPR_SUM{IIF(ISNULL(%AllFields%),1,0]%
```

Output

```
IIF(ISNULL (AccountID, 1,0)+IIF(ISNULL(AccountName, 1,
0)+IIF(ISNULL(ContactName, 1, 0)+IIF(ISNULL(Phone, 1,
0)+IIF(ISNULL(Email, 1, 0)...
```

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Assume that you want to check for null values in all the fields of a customer record.

You can use the horizontal macro expression to check for any null values in a field. The expression returns the value 1 when a field is null, and the horizontal expansion function **%OPR_SUM%** returns the total number of null fields.

When the task runs, it expands the expression horizontally to include all the incoming fields. As you can see, the expression checks for null values in the customer's AccountID, AccountName, ContactName, Phone, and Email fields.

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Hybrid Macro

- Expands an expression both vertically and horizontally
- Configure a hybrid macro based on your business requirements

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A hybrid macro expands an expression both vertically and horizontally. You can configure a hybrid macro based on your business requirements.

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Hybrid Macro – Example

Scenario

Format the date fields in a customer's record to the 'mm-dd-yyyy' format

Solution

```
%OPR_IIF[IsDate(%dateports%,%fromdateformat%),To_String(To_Date(%dateports%,%fromdateformat%),
'mm-dd-yyyy'),%dateports%]%
```

Output

```
IIF(IsDate(StartDate,'mm/dd/yy'),To_String(To_Date(StartDate,'mm/dd/yy'),'mm-dd-yyyy'),
IIF(IsDate(StartDate,'mm/dd/yyyy'),To_String(To_Date(StartDate,'mm/dd/yyyy'),'mm-dd-yyyy'),
StartDate))

IIF(IsDate(EndDate,'mm/dd/yy'),To_String(To_Date(EndDate,'mm/dd/yy'),'mm-dd-yyyy'),
IIF(IsDate(END _DT,'mm/dd/yyyy'),To_String(To_Date(EndDate,'mm/dd/yyyy'),'mm-dd-yyyy'),
EndDate))
```

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Assume that you want to format a date field in the customer's record to a specific format.

You can use the hybrid macro expression to convert the date fields to the required format. In the expression, the **%fromdateformat%** macro input field defines the different date formats used in the date fields.

When the task runs, it expands the expression vertically and horizontally. The expression expands vertically to create an expression for the **Start Date** and **End Date** fields that the **%dateports%** represents. The expression also expands horizontally to use the constants that the **%fromdateformat%** represents to evaluate the incoming fields.

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Topic

Union Transformation

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Union Transformation

- It is an active transformation.
- For Data Integration patterns, it is common to combine two or more data sources into a single stream that includes the union of all rows.
- Union transformation enables you to make the metadata of the streams alike.
- You can add, change, or remove specific fields when you merge data sources.
- At run time, the mapping task processes input groups in parallel.
- When the mapping runs, it merges data into a single output group based on the field mappings.

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The Union transformation is an active transformation that you can use to merge data from multiple pipelines into a single pipeline.

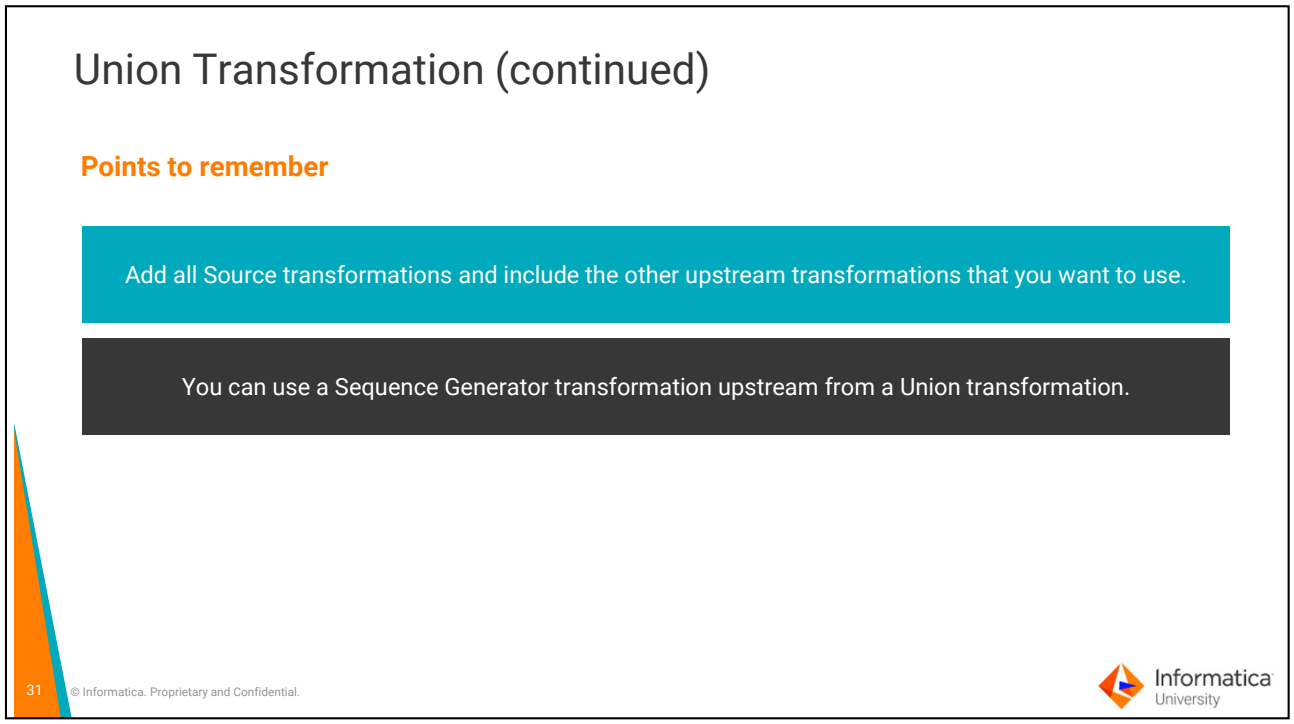
For Data Integration patterns, it is common to combine two or more data sources into a single stream that includes the union of all rows. The data sources often do not have the same structure, so you cannot freely join the data streams. The Union transformation enables you to make the metadata of the streams alike, so you can combine the data sources in a single target.

The Union transformation merges data from multiple sources similar to the UNION ALL SQL statement. For example, you can use the Union transformation to merge employee information from ADP Workforce Now with data from a Workday employee object.

With a Union transformation, you can add, change, or remove specific fields when you merge data sources.

At run time, the mapping task processes input groups in parallel. It concurrently reads the sources connected to the Union transformation and pushes blocks of data into the input groups of the transformation. When the mapping runs, it merges data into a single output group based on the field mappings.

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Remember a few important points when you use a Union transformation in a mapping.
Before you add a Union transformation to a mapping, add all Source transformations and include the other upstream transformations that you want to use.
You can use a Sequence Generator transformation upstream from a Union transformation, if you connect both the Sequence Generator and a Source transformation to one input group of the Union transformation

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Topic

Rank Transformation

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Rank Transformation

- Selects the top or bottom range of data
- Use the Rank Transformation to:
  - Return the largest or smallest numeric values in a group
  - Return strings at the top or bottom of the mapping sort order
- Rank transformation differs from the transformation functions MAX and MIN

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The Rank transformation selects the top or bottom range of data. You can use the Rank transformation to return the largest or smallest numeric values in a group. You can also use the Rank transformation to return strings at the top or bottom of the mapping sort order. For example, you can use a Rank transformation to select the top 10 customers by region, or you can identify the three departments with the lowest expenses in salaries and overhead.

The Rank transformation differs from the transformation functions MAX and MIN because the Rank transformation returns a group of values, not just one value. The SQL language provides many functions that are designed to handle groups of data. However, identifying top or bottom groups of values within a set of rows is not possible with the use of standard SQL functions.

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Rank Transformation (continued)

- An active transformation.
- Example:
  - Select the top 10 rows from a source that contains 100 rows
- When you run a mapping that contains a Rank transformation, Data Integration caches input data until it can perform the rank calculations.
- You can configure the Rank transformation to return the top or bottom values of a string field.

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The Rank transformation is an active transformation because, it can change the number of rows that pass through it. For example, you can configure the transformation to select the top 10 rows from a source that contains 100 rows. In this case, 100 rows pass into the transformation but only 10 rows pass from the Rank transformation to the downstream transformation or target.

When you run a mapping that contains a Rank transformation, Data Integration caches input data until it can perform the rank calculations.

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Topic

Sorter Transformation

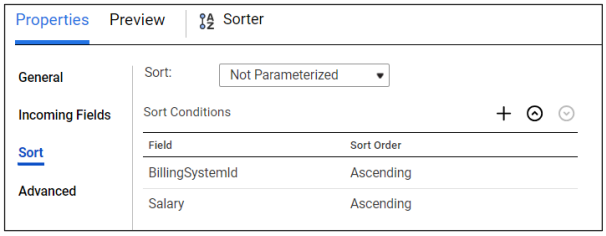
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Sorter Transformation

Data can be sorted using the Sorter transformation in either ascending or descending order, depending on the sort condition.

- Sorter transformation is an active and connected transformation.
- You can define the sorter properties on the following tabs of the Properties panel:
  - **Sort tab**: Define the Fields and Sort order on which data coming from upstream transformation should be sorted.
  - **Advanced tab**: Define advanced properties like Distinct of Sorter transformation.
- You can also modify the order of sort conditions based on the business requirement.



In the above example, the records are first sorted billing system Id wise and then the records inside each department are sorted according to salary.

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- The Sorter transformation can be used to speed up other transformations.
- For instance, you can sort data that goes through a Lookup or Aggregator transformation that is set up to use sorted incoming fields.
- Configure the sort condition to specify the sort fields and the sort order.
- When more than one sort condition is specified, the mapping job sorts each condition sequentially.

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Topic

Router Transformation

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Router Transformation

- Router Transformation is an active transformation.
- You can apply a condition to incoming data using the Router transformation.
- Data Integration uses Filter condition to evaluate each row of incoming condition.
- Uses output group conditions to route a data row.
- There can be one input group and more than one output group.
- There can be user defined output groups
- A given data row can pass through multiple user output groups if it satisfies the output group condition.

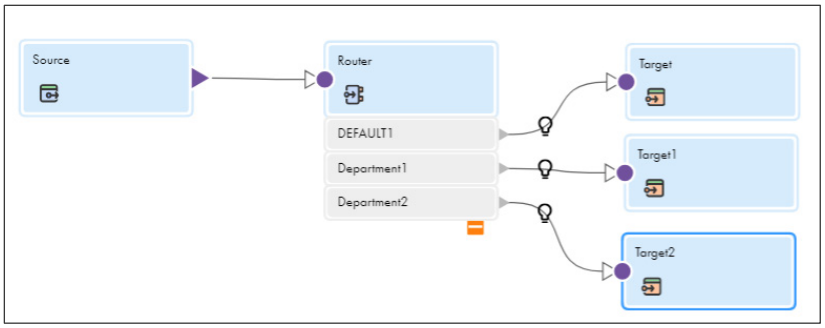
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Router Transformation (continued)

A unique data flow will be formed for each output group chosen and can be given to downstream transformations as indicated below.



Router transformation with multiple output groups, routing data to multiple targets

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Default Group in Router Transformation

- A DEFAULT group will be included with router transformation by default.
- The default group passes through any rows that do not match any filter requirement.
- In the example above, departments with IDs other than 10 and 20 will be passed through thee DEFAULT group.
- Conditions for router filters are not if and else.
- The router transformation passes data from all the output groups that satisfy the criteria if a row meets the filter condition in more than one group.

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- Comparable to what was recently covered in Filter transformation, Router transformation supports similar filter condition types.
- Depending on the demand, you can define a simple or Advanced filter condition as well as fully or partially parameterize the filter condition.

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Topic

Knowledge Check

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Which of the following transformations does not eliminate duplicates?

a. Source
b. Sorter
c. Union
d. Rank

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Answer: b

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Which of the following statements is False for Union transformation?

a. Allows defining of multiple input groups

b. Allows removal of duplicate rows

c. Allows merging of different data types

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Answer: b

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Lab Activity

Lab 6-1 Classifying the Product Shipping Details Using Various Transformations

In this lab, you will perform the following:

- Create a Mapping
- Add Joiner, Expression, and Filter transformations and complete the logic

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Lab Activity

Lab 6-2 Using Union, Sorter, and Rank Transformations in a Mapping

In this lab, you will perform the following:

- Create a Mapping using the Union, Rank, and Sorter Transformations

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Lab Activity

Lab 6-3 Using Expression Macro in a Mapping

In this lab, you will perform the following:

- Use expression macro in a mapping using Informatica Cloud

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Lab Activity

Lab 6-4 Using Vertical Macro in a Mapping

In this lab, you will perform the following:

- Create a mapping using Vertical macro

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Lab Activity

Lab 6-5 Using Horizontal Macro in a Mapping

In this lab, you will perform the following:

- Create a mapping using Horizontal macro

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Module Summary

This module showed you how to:

- Define IICS transformations
- Explain different types of transformations

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Module 7

Joiner and Lookup Transformations

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Module Objectives

After completing this module, you will be able to:

- Discuss Joiner Transformation
- Describe Join Condition
- Explain Join Type
- Describe Lookup Transformation

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Topic

Joiner Transformation

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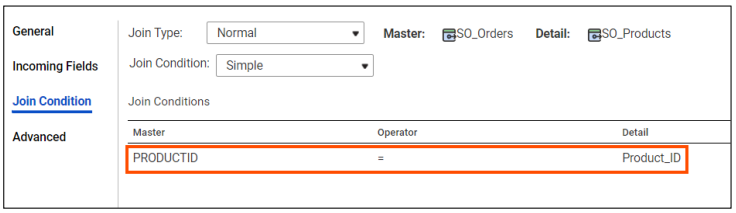
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Joiner Transformation

Joins two related heterogenous sources

Joins data based on a join condition

Performs inner join operation



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A Joiner transformation joins two related heterogeneous sources. The sources can reside in different source systems. For example, you can join data from the Salesforce Account object with the data in the customer's database table.

The Joiner transformation joins data based on a join condition that you define. You can also create multiple join conditions.

The Joiner transformation performs an inner join, which results in rows from both sources that match all the conditions. Source rows that do not match the join conditions are dropped from the data flow. You can change the Join Type from the Join Condition tab of Joiner transformation.

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Joiner Transformation (continued)

- Select Master or Detail group to connect a source to Joiner transformation
- Connect source with smaller data set to Master group
- Use multiple Joiner transformations to join more than two sources in a mapping
- To avoid field name conflicts:
  - rename matching fields in upstream transformation
  - rename matching fields in Source transformation
  - pass data through an Expression transformation and rename fields

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In a Joiner transformation, there are two sources used for joins. These two sources are called, Master Source and Detail Source.

In the properties of a Joiner transformation, you can select which data source can be Master and which source can be Detail source. When the task runs, the Master Source is cached into the memory for joining purpose. So, it is recommended to select the source with a smaller number of records as the Master Source.

You can join only two sources at a time to a Joiner transformation. If you want to connect more than two sources in the mapping, you must use multiple Joiner transformations.

Sometimes, when you join sources with matching field names, there can be a field name conflict. To avoid these conflicts, you must:

- Rename matching fields in an upstream transformation
- Rename matching fields in the Source transformation, and
- Pass data through an Expression transformation and rename fields

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Join Condition

- The Join Condition in Joiner transformation represents the condition upon which the rows coming from two sources must be matched.
- You can create one or more conditions which define the relationship between both the sources.

  Example - If there are two sets of employee data, the following condition will match any rows that have the same employee ID number in both sets of data:

  EMP_ID1 = EMP_ID2

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Join Type

The result set passed to the remaining parts of the mapping depends on the join type.

- Joiner transformation supports four different types of joins

Normal Join	Master Outer	Detail Outer	Full Outer
• Includes only rows with matching join conditions. • Discards rows that do not match the join conditions.	• Includes all rows from the Detail pipeline and the matching rows from the Master pipeline. • It discards the unmatched rows from the Master pipeline.	• Includes all rows from the Master pipeline and the matching rows from the Detail pipeline. • Discards the unmatched rows from the Detail pipeline.	• Includes rows with matching join conditions and all incoming data from the Master pipeline and Detail pipeline.

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- The Joiner transformation does not match null values.
- To join rows with null values, you can replace null values with default values, and then join on the default values.

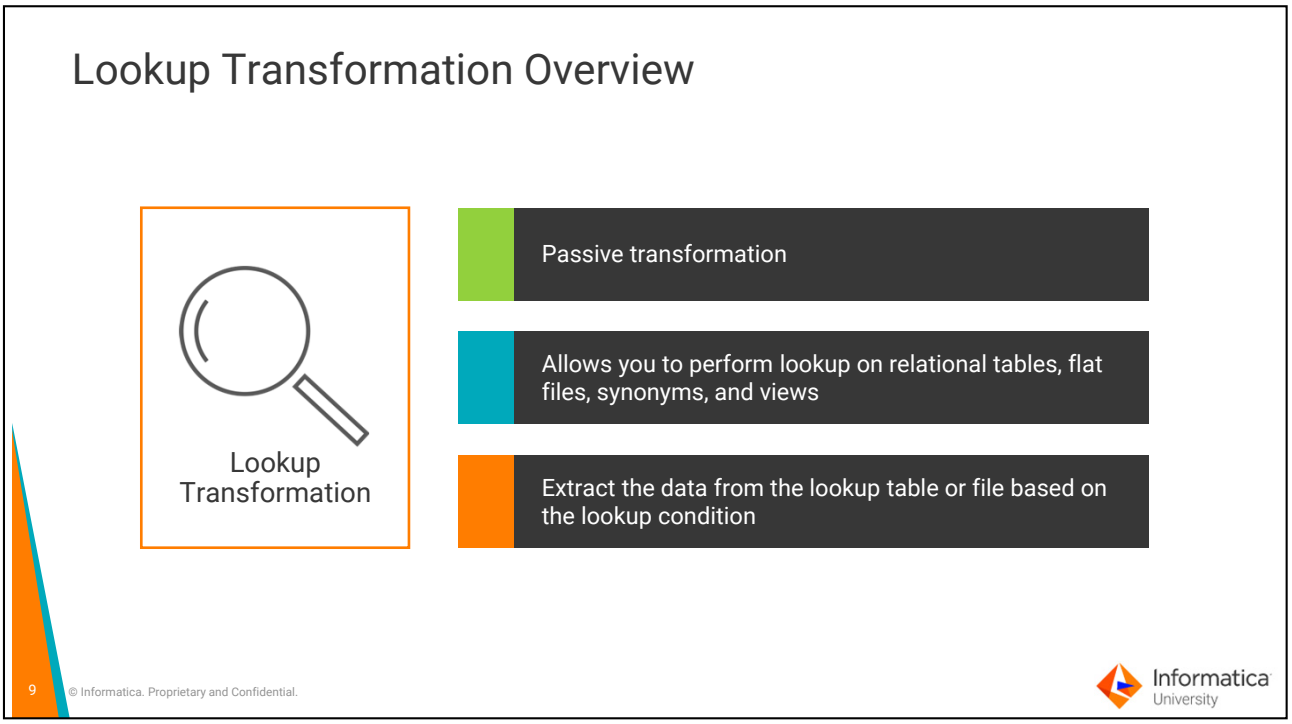
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Topic

Lookup Transformation

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The Lookup transformation is a passive transformation. It allows you to perform lookup on relational tables, flat files, synonyms, and views. You must use a lookup condition to extract data from lookup tables or flat files.

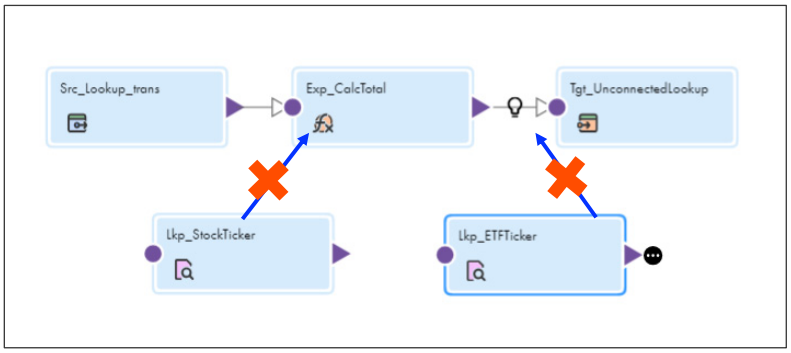
When you configure a Lookup transformation, there are several options that you must configure. You must select the lookup connection and object. After that, you must specify how the lookup must behave if multiple matches are found. For example, you can choose to return any row that matches, return the first row, return the last row, or report an error, and so on.

You must also configure the lookup condition and specify how incoming rows must match with rows in the lookup object. Finally, you must also specify the fields that you want to return.

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Unconnected Lookup

- Neither connected to the source nor to the target
- Use when you want the lookup to return only one value
- Reuse the lookup multiple times in a mapping



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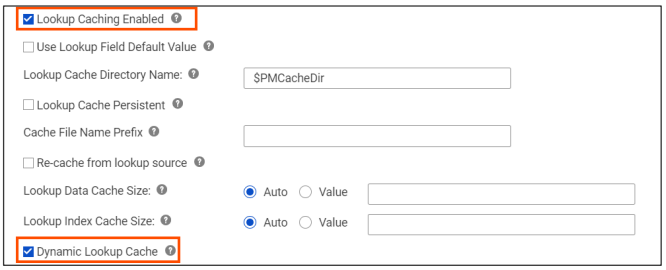
An Unconnected Lookup transformation is neither connected to the source nor to the target. You can use the Unconnected Lookup transformation when you want the lookup to return only one value. You can reuse the lookup multiple times in a mapping.

You can use an unconnected lookup only with a database or flat file data source.

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Dynamic Lookup

- When you enable lookup caching, a mapping task builds the lookup cache when it processes the first lookup request.
- Cache can be static or dynamic.
- If the cache is static, the data in the lookup cache does not change as the mapping task runs.
- If the cache is dynamic, the task updates the cache based on the actions defined in the task.



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When you enable lookup caching, a mapping task builds the lookup cache while it processes the first lookup request. The cache can be static or dynamic.

If the cache is static, the data in the lookup cache does not change as the mapping task runs. If the task uses the cache multiple times, the task uses the same data.

If the cache is dynamic, the task updates the cache based on the actions defined in the task. So, if the task uses the lookup multiple times, the downstream transformations uses the updated data.

You can use a dynamic lookup cache to keep the lookup cache synchronized with the target. You can also use a dynamic lookup cache if the source data contains duplicate primary keys.

When you use a dynamic lookup cache, you can configure to ignore some lookup fields when Data Integration compares the values in the lookup fields with the values in the associated incoming fields.

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Dynamic Lookup (continued)

- The mapping task performs one of the following actions on the dynamic lookup cache when it reads a row from the source:
  - Inserts the row into the cache
  - Updates the row in the cache
  - Makes no change to the cache
- The dynamic Lookup transformation includes the return field and the New Lookup Row.

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Now, based on the results of the lookup query, the row type, and the Lookup transformation properties, the mapping task performs one of the following actions on the dynamic lookup cache when it reads a row from the source:
Inserts the row into the cache: The mapping task inserts the row when it is not in the cache. The task flags the row as insert.
Updates the row in the cache: The mapping task updates the row when it exists in the cache. It updates the row in the cache based on the input fields, and flags the row as an update row.
Makes no change to the cache: The mapping task makes no change when the row is in the cache, and nothing changes. The task flags the row as unchanged.

The dynamic Lookup transformation includes the return field and the New Lookup Row, which describes the changes that the task makes to each row in the cache.

Note that you cannot use a parameterized source, target, or lookup with a Lookup transformation that uses a dynamic cache.

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Dynamic Lookup Example

- To configure a Lookup transformation to be dynamic, use a dynamic lookup cache.
- Example of using a dynamic cache:
  1. You want to update your payroll table with data from your Human Resources department.
  2. The payroll table includes the following data, where ID is the primary key:
  3. Create a mapping with a Lookup transformation and use the payroll table for the target and the lookup cache.
  4. Configure the cache to be dynamic since the Human Resources department's table contain duplicate keys.
  5. Specify the Human Resources department's table to be the source.
  6. The source table includes the following:

ID	Name	Location
1	Abhi	USA
2	Alice	UK

ID	Name	Location
1	Abhi	USA
2	Alice	UK
3	James	Japan
3	James	USA

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A dynamic cache is helpful when the source table contains a large amount of data or it contains duplicate primary keys.

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Dynamic Lookup Example (continued)

- You create a mapping task to update the payroll table.
- When the mapping task begins, it creates the cache file that contains the rows in the target table.
- As the task processes the rows, it flags the first row as an update, and it updates the cache.
- It flags the third row as an insert and inserts the row in the cache.
- It flags the fourth row as an update because the row exists in the cache.

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Lab Activity

Lab 7-1 Using Lookup Transformation in a Mapping

In this lab, you will perform the following:

- Use Lookup transformation in a mapping

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Lab Activity

Lab 7-2 Merging Current and Previous Data Loads Using Full Outer Join

In this lab, you will perform the following:

- Create a mapping in IICS
- Use Full Outer Join in the mapping

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Topic

Knowledge Check

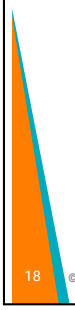
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You cannot use an unconnected lookup with a flat file data source.

a. True

b. False



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Answer: b

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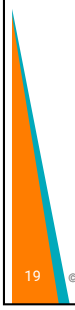
To join 4 different sources, how many joiner transformation are required

a. 4

b. 3

c. 5

d. 1



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Answer: b

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Which of the following statements is FALSE?

a. Unconnected look up returns only one output field.

b. If the lookup cache file is still present after the successful completion of the task, the lookup in the mapping is configured to have Dynamic cache.

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Answer: b

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Module Summary

This module showed you how to:

- Discuss Joiner Transformation
- Describe Join Condition
- Explain Join Type
- Describe Lookup Transformation

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Module Objectives

After completing this module, you will be able to:

- Mapplets Overview
- Discuss Types of IICS mapplets
- Explain Parameters in mapplets
- Understand Mapplet Transformation
- Explain PowerCenter mapplet
- Create mapplet in IICS

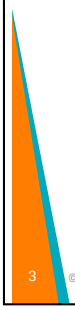
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Mapplets Overview

- A mapplet is reusable transformation logic used to transform source data before it is loaded into the target.
- A mapplet can be made in one of the following ways:
  - Create a mapplet in Data Integration.
  - Import a mapplet from PowerCenter.
  - Generate an SAP BAPI or IDoc mapplet.
- Create mapplets in Data Integration in the Mapplet Designer.



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- You can import a PowerCenter mapplet to use in Data Integration.
- You can add a mapplet to a Mapplet transformation to use its transformation logic.
- A mapplet can be used in another mapplet.
- However, you cannot cyclically reference a mapplet. For example, if Mapplet A uses Mapplet B, Mapplet B cannot also use Mapplet A.

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Topic

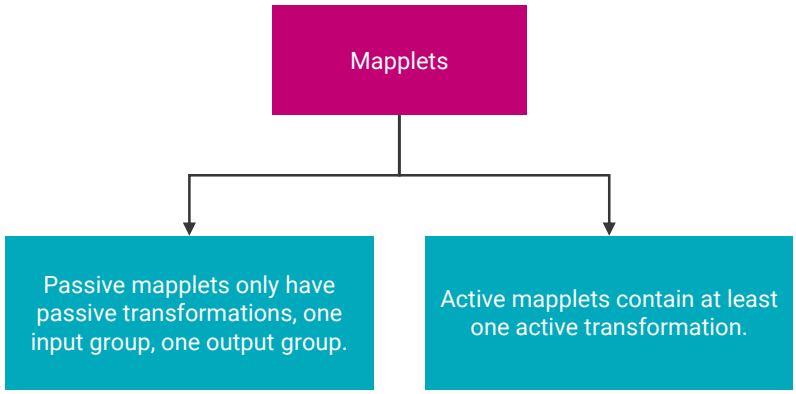
Types of IICS Mapplets

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Active and Passive Mapplets

- Two types of mapplets are possible: Active and Passive



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The mapplet type appears on the Mapplet tab when you use a mapplet in a mapplet transformation.

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Mapplet Input and Output

- You can use a mapplet in mapplet transformation, configure the mapplet input and output.
- Mapplet input can be an Input transformation, a Source transformation, or both.
- You can use an Input transformation when the mapplet receives input data from one or more upstream transformations.
- A mapplet can have multiple input transformations.
- Mapplet output can be to an Output transformation, a Target transformation, or both.
- When a mapplet needs to provide information to one or more downstream transformations, use an output transformation.
- For the mapplet to write data to a target, use Target Transformation.

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- A mapplet can receive input from a source transformation, an upstream transformation, both, or neither through an input transformation.
- A mapplet sends output through an output transformation, a target transformation, or both to a downstream transformation.
- At least one input transformation and one output transformation must be present in a mapplet.
- To give source data, one or more Source transformations are included in a mapplet.
- A mapplet must contain at least one Input transformation or Source transformation.
- **Note**: Not all source types can be used in a Source transformation in a mapplet.
- A mapplet must contain at least one Output transformation or Target transformation.

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Parameters in Mapplets

- Use input parameters and in-out parameters in a mapplet.
- You can define the value of the parameters while configuring the mapping task.
- The name of the Mapplet transformation is prefixed to each parameter name.
  - Example: Select Completely Parameterized as the filter condition when using the Filter transformation in a mapplet.
  - "Filter Param" is the name of the string parameter you construct.
  - You apply the mapplet to a transformation in Mapplet called "MyMappletTx."
  - The filter parameter is now known as "MyMappletTx Filter Param" in the Mapplet transformation.
- If in-out parameters are defined in a mapplet, even if they are not used, the mapplet is not valid for elastic mappings.

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- The same principles apply to configuring parameters in mapplets as in mappings.
- When parameters are added to a mapplet, when the mapplet is used in a mapplet transformation, Data Integration renames the parameters.

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Topic

Create Mapplet in IICS

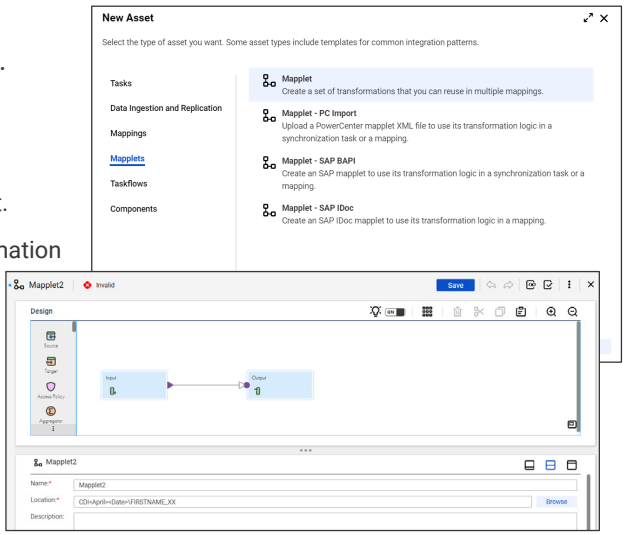
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Creating a Mapplet

- Create a mapplet in Mapplet Designer.

1. Click **New** > **Mapplets** > **Mapplet**.
2. Enter the mapplet name and location.
3. Enter the mapplet description, if you want.
4. Filter the transformations in the transformation palette.
5. Configure the mapplet input.
6. Add transformations to the mapplet.
7. Configure the mapplet output.
8. Validate the mapplet
9. Save the mapplet.



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The same design sections and features found in the Mapping Designer are also present in the Mapplet Designer.
The Mapplet Designer appears with an Input transformation and an Output transformation on the canvas.
After saving the mapping, you can edit the name or location in the Explore page.

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Editing a Mapplet

- You can use a Mapplet Designer to edit the mapplet.
- To edit a mapplet, open the mapplet from the Explore page.
- The Mapplet Designer verifies the transformation logic when you edit a mapplet.
- Few changes you can make to a mapplet without impacting its validity or session:
  1. Add input or output ports
  2. Change input or output comments or transformation names
  3. Change comments or port names
  4. Change transformation properties, comments or names
  5. Change port default values for transformations

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Mapplet Transformation

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Mapplet Transformation

- The Mapplet transformation inserts a mapplet built in Data Integration or imported from PowerCenter into a mapping.
- Each Mapplet transformation can contain one mapplet.
- Mapplet transformation can be active, or passive based on the transformation logic within the mapplet.
- An active mapplet includes one active transformation.
- A passive mapplet includes only passive transformations.

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- A mapping or mapplet may contain many Mapplet transforms.
- At least one active transformation is present in an active mapplet.
- The number of source rows passed to an active mapplet may differ from the number of rows it returns.
- A passive mapplet includes only passive transformations.
- A passive mapplet returns the same number of rows that are passed from the source.

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Mapplet Transformation (continued)

- Mapplet transformation accomplish the following goals:

Extend the data transformation capabilities of Data Integration.

Reuse transformation logic in different mappings.

Hide complex transformation logic.

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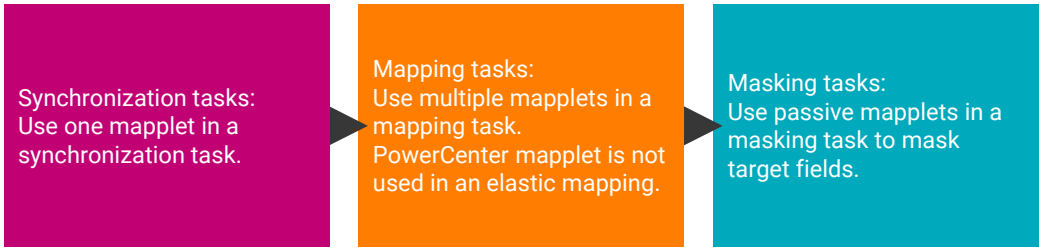
PowerCenter Mapplets

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PowerCenter Mapplets

- If you use Informatica PowerCenter, you can create a mapplet for Data Integration using a PowerCenter mapplet.
- You must construct a PowerCenter mapplet and export it to an XML file in order to use it.
- Upload the XML file into Data Integration.
- Use the PowerCenter mapplets in the following Data Integration tasks:



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- A mapplet contains a set of transformations.
- A PowerCenter mapplet may contain one or more Source transformations, but cannot contain Target transformation.

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Active and Passive PowerCenter Mapplets

- You can define whether a PowerCenter mapplet is active or passive when you upload it to Data Integration.
- An active mapplet includes at least one active PowerCenter transformation.
- An active mapplet will return a different number of rows than the source rows that were supplied.
  - For example, an active mapplet might aggregate five source rows and return one row.
- A passive mapplet includes only passive PowerCenter transformations.
- It returns the same number of rows that are passed from the source.

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PowerCenter Mapplets in Data Integration Tasks

For using the PowerCenter mapplets in Data Integration tasks, use the following rules and guidelines:

- Add Lookup transformations between the source and a mapplet.
- Add Expression transformations between the source and a mapplet, and between a mapplet and the target.
- The synchronization task wizard deletes any existing field mappings when you add a mapplet to the task.
- When you use an active mapplet in a synchronization task that uses a saved query, the task's configured target option is ignored, and data is instead inserted into the target.
- Data Integration retains PowerCenter session-level overrides to the mapping.

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The synchronization task wizard doesn't remove existing field mappings if you add a passive mapplet between the source and target.

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Topic

Knowledge Check

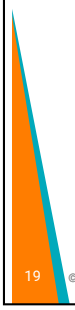
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Mapplets are reusable objects that contain a set of transformations that you can use in multiple mappings.

a. True

b. False



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Answer: a

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Which of the following statements is FALSE about Mapplet Transformation?

a. You can add multiple Mapplet transformations to a mapping or mapplet
b. The Mapplet transformation can be active or passive.
c. Each Mapplet transformation can contain more than one mapplet.
d. Hide complex transformation logic.

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Answer: c

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Lab Activity

Lab 8-1 Using Mapplet Transformation in a Mapping

In this lab, you will perform the following:

- Create a Mapplet
- Use Mapplet Transformation in a mapping

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Lab Activity

Lab 8-2 Create Mapplet and Use It in the Existing Mapping

In this lab, you will perform the following:

- Create a Mapplet
- Replace the existing mapping objects with the mapplet
- Replace Filter transformation with Router in the mapping

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Lab Activity

Lab 8-3: Creating a Mapping Using Existing Mapplet and Aggregator Transformations

In this lab, you will perform the following:

- Use the existing Mapplet
- Use the Aggregator transformation

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Module Summary

This module showed you how to:

- Mapplets Overview
- Discuss Types of IICS mapplets
- Explain Parameters in mapplets
- Understand Mapplet Transformation
- Explain PowerCenter mapplet
- Create mapplet in IICS

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Module 9
Mapping Parameterization

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Module Objectives

After completing this module, you will be able to:

- Explain Parameters
- List the Types of Parameters
- Describe a Parameter File
- Discuss some Best Practices for Parameters

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What are Parameters?

- Parameters are placeholders that represent values in a mapping.
- You can use parameters to hold values that:
  - you want to define at runtime
  - change between task runs
- You can override complete source queries in relational database connections.

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Parameters are placeholders that represent values in a mapping. You can use parameters to hold values to define at runtime, such as a source connection, a target object, or the join condition for a Joiner transformation. You can also use parameters to hold the values that change between task runs, such as a time stamp that increments each time you run a mapping.

You can also use parameters to override complete source queries in the relational database connections.

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Uses of Parameters

- Used to parameterize source and target connections.
- Reuse parameterized mapping in Development, Test, and Production environments.
- You can also parameterize filter conditions and expressions.

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There are various use cases where you add a parameter to a mapping.

You can use parameters when you want to parameterize source and target connections for a mapping and reuse it in your Development, Test, and Production environments. You can create multiple mapping tasks using the same mapping.

You can also parameterize filter conditions and expressions to allow business users to easily specify values without editing the mapping.

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Adding Parameters to a Mapping

- Allows you to create flexible mapping templates
- Any part of a mapping can be parameterized:
  - Source connection and source object
  - Target connection, target object, and field mapping
  - Join condition
  - Filter criteria
  - Lookup connection, lookup object, and lookup condition
  - Expression

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You can add parameters to mappings to create flexible mapping templates that business users can use to create multiple mapping tasks.

In IICS, you can parameterize any part of the mapping. Some example includes the following:

- Source and Target connections and objects.
  Note: The target connection supports full and partial parameterization of the field mapping.
- Field mapping, Join conditions, and filter criteria.
- Lookup connection, lookup object, lookup condition, as well as Expressions.

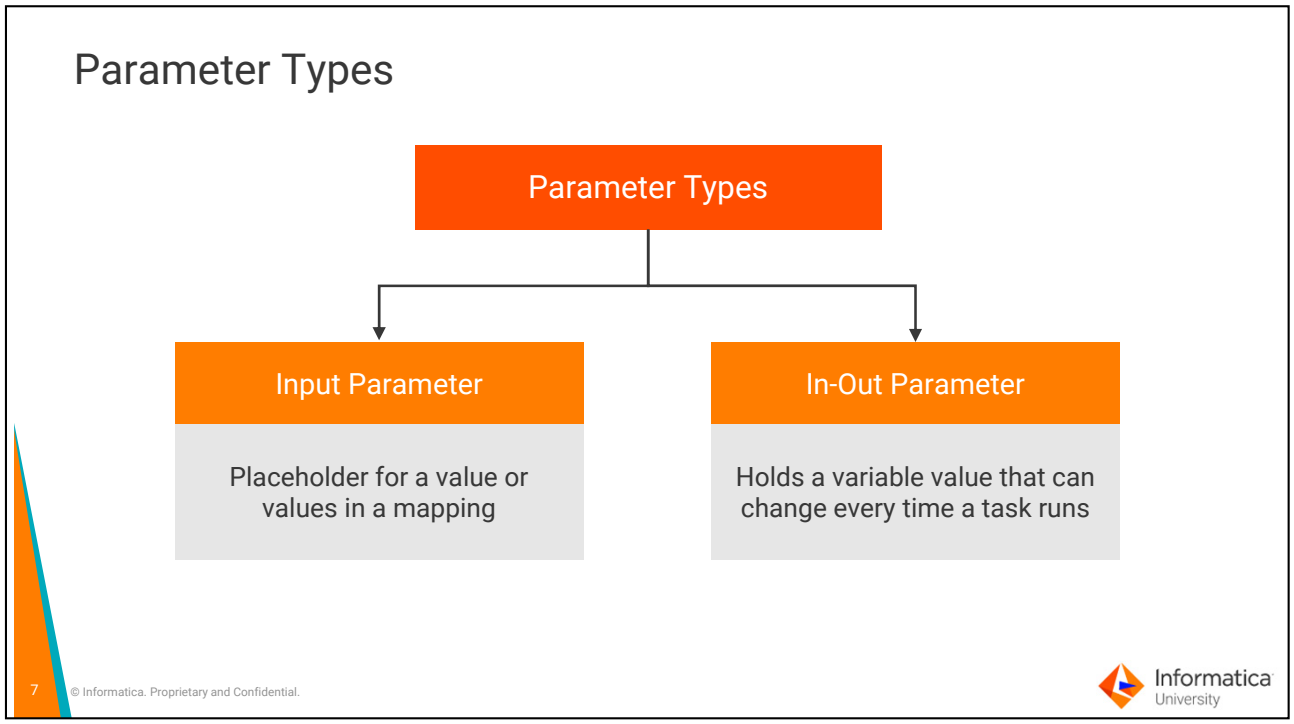
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Topic

Parameter Types

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You can create the following types of parameters in a mapping – Input Parameters and In-Out Parameters.

An input parameter is a placeholder for a value or values in a mapping. Input parameters help you to control the logical aspects of a data flow or to set other variables that you can use to manage different targets. You can define an input parameter in a mapping and set the value of the parameter when you configure a mapping task.

An In-Out parameter holds a variable value that can change every time a task runs. When you define an In-Out parameter, you can set a default value in the mapping. However, you would typically set the value at run time using an Expression transformation. You can also change the value in the mapping task.

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Input Parameter

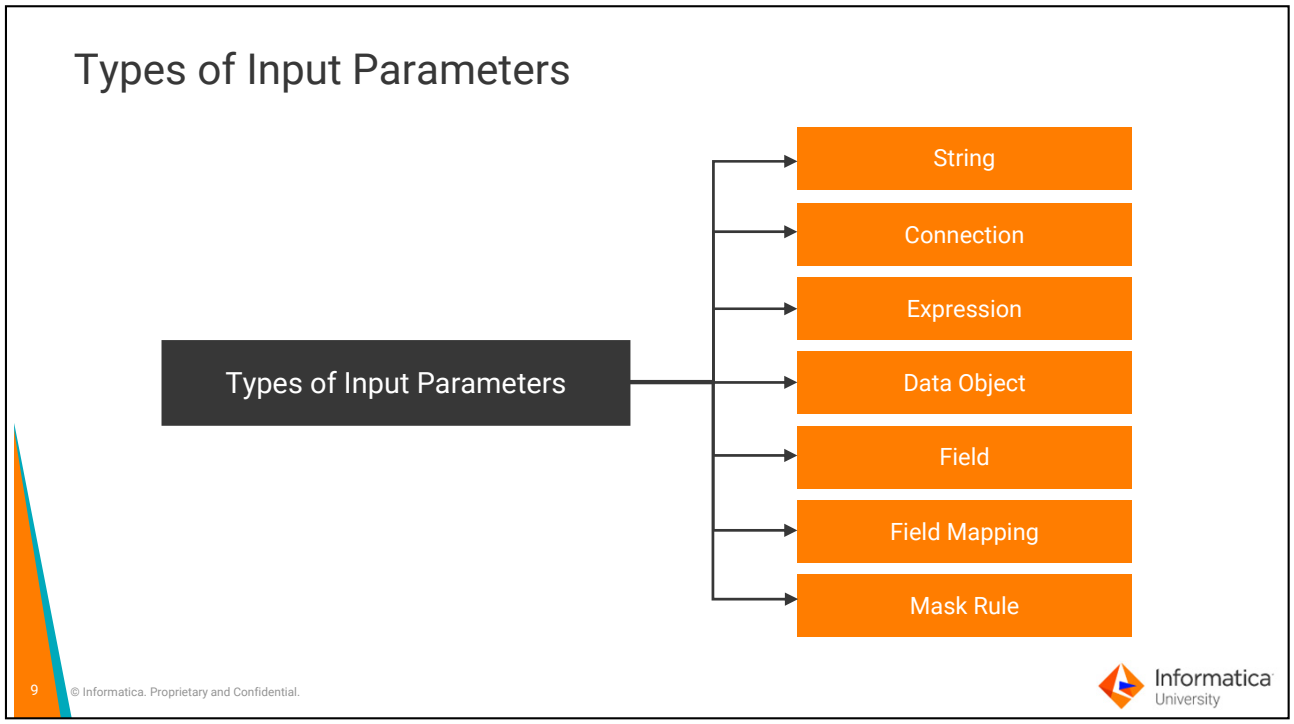
- You can use input parameters in the following transformations:

Source	Lookup
Target	Mapplet
All transformations with incoming fields	Rank
Aggregator	Router
Data Masking	Sorter
Expression	SQL
Filter	Structure Parser
Joiner	Union

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- **String**: A String parameter represents a string value. In the task, the string parameter displays as a textbox in most instances.
- **Connection**: A Connection parameter represents a connection. You can specify the connection type for the parameter or allow any connection type. In the task, the connection parameter displays a list of connections.
- **Expression**: An Expression parameter represents an expression. In the task, the expression parameter displays the Field Expression dialog box to configure an expression.
- **Data Object**: A Data Object parameter represents a data object, such as a source table or a source file. In the task, the data object parameter appears as a list of available objects from the selected connection.
- **Field**: A Field parameter represents a field. In the task, the field parameter displays a list of available fields from the selected object.
- **Field Mapping**: A Field Mapping parameter represents field mappings for the task. You can create a full or partial field mapping. A full field mapping parameter displays all fields for configuration, and a partial field mapping parameter displays the unmapped fields.
- **Mask Rule**: A mask rule parameter represents a masking technique. In the task, the mask rule parameter displays a list of masking techniques.

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In-Out Parameters

- Act as persistent task variables.
- Can store a date value for the last record that loads from a data warehouse.
- Can help you manage the update process for a slowly changing dimension table.
- For example, you can:
  - update values after each task execution
  - handle incremental data loading for a data warehouse

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In-Out parameters act as persistent task variables. The parameter values update during task execution. The parameter can store a date value for the last record that loads from a data warehouse, or it can help you manage the update process for a slowly changing dimension table.

Some examples where you can use an In-Out parameter:

- You can use an In-Out parameter to update values after each task execution. So, you can use the SetVariable, SetMaxVariable, SetMinVariable, or SetCountVariable functions in an Expression transformation to update parameter values each time you run a task.
- You can also use an In-Out parameter to handle incremental data loading for a data warehouse. In this case, you set a filter condition to select records from the source that meets the load criteria. When the task runs, you include an expression to increment the load process.

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In-Out Parameters (continued)

- Use In-Out parameters in the following transformations:
  - Source
  - Target
  - Aggregator
  - Expression
- You cannot use In-Out parameters in expression macros and in an at-scale mapping.
- Unlike Input parameters, an In-Out parameter can change each time a task runs.
- You can reset the In-Out parameters in a mapping task.

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You cannot use In-Out parameters in expression macros and in an at-scale mapping.

For each In-Out parameter, you need to configure the variable name, datatype, default value, aggregation type, and retention policy. You can also use a parameter file that contains the parameters to apply at runtime. For a specific task run, you can change the value in the mapping task.

Unlike input parameters, an In-Out parameter can change each time a task runs. The latest value of the parameter displays in the job details when the task completes successfully. Next time the task runs, the mapping task compares the In-Out parameter to the saved value.

You can also reset the In-Out parameters in a mapping task, and view the saved values in the job details.

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Topic

Parameter File

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Parameter File

- A list of user-defined parameters and their associated values.
- Defines update values without editing the task.
- Parameter values are treated as String values.
- You cannot use a parameter file if the mapping task is based on an elastic mapping.
- Access parameter file from local directory or from cloud platform.

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A parameter file is a list of user-defined parameters and their associated values. You can use user-defined parameters in data filters, expressions, and lookup expressions in a Synchronization task or a Mapping task.

You can use a parameter file to define values that you want to update without having to edit the task. For example, you can use a parameter file for a Sales quota that changes quarterly.

Parameter values are treated as String values. So, when you use a parameter in an expression, you must use the appropriate function to convert the value to the necessary data type. The parameter values are applied when the task runs.

Note: You cannot use a parameter file if the mapping task is based on an elastic mapping.

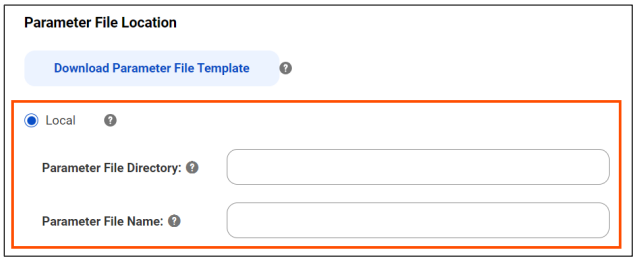
IICS allows use of parameter file that is saved on local machine or on cloud platform.

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Parameter File (continued)

Local

- Select Parameter File Location as Local.
- Provide absolute path for the directory that contains the parameter file.
  - exclude the parameter file name
- Default directory:
  - `<Secure Agent installation directory>/apps/Data_Integration_Server/data/userparameters`



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When you want to use parameter values from a parameter file saved on a local machine, you must select the Parameter File Location as Local.

You must also provide the absolute path for the directory that contains the parameter file, excluding the parameter file name.

The directory storing the parameter file must be accessible by the Secure Agent.

If you do not enter a location, the following directory is used:
<Secure Agent installation directory>/apps/Data_Integration_Server/data/userparameters

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Parameter File (continued)

Local

- You can also specify a path relative to a $PM system variable in the Parameter File Directory field.
- Following system variables are supported:
  - $PMRootDir
  - $PMTargetFileDir
  - $PMSourceFileDir
  - $PMLookupFileDir
  - $PMCacheDir
  - $PMSessionLogDir
  - $PMExtProcDir
  - $PMTempDir

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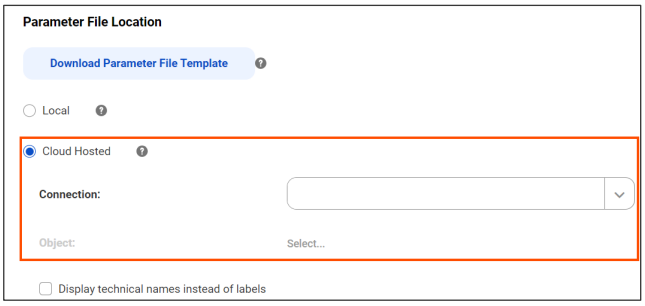
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Parameter File (continued)

Cloud Hosted

- Select Parameter File Location as Cloud Hosted.
- Select Connection containing the parameter file.
- Supported Connection types:
  - Amazon S3
  - Google Storage V2
  - Azure Data Lake Store Gen2
- In Object field, select the parameter file.



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When you want to use parameter values from a parameter file saved on a cloud-hosted platform, you must select the Parameter File Location as Cloud Hosted.

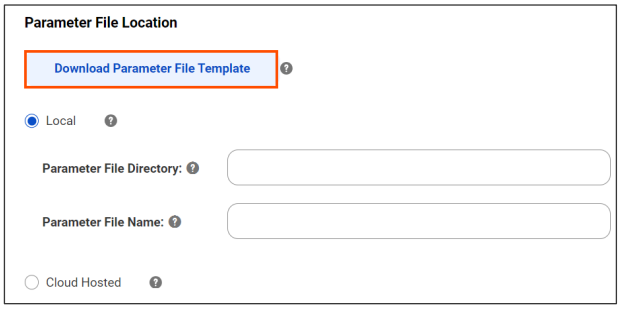
You can use any one of the supported connection types to save the parameter file.
- Amazon S3
- Google Storage V2
- Azure Data Lake Store Gen2

You must also provide the parameter file name in the object field that contains the definitions and values of user-defined parameters used in the task.

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Parameter File Template

- Contains mapping parameters and their default values.
- Includes input and in-out parameters that can be overridden at runtime.
- Parameter file template does not contain:
  - Runtime values
  - Partial parameters
  - Advanced session properties
  - System defined variables



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A parameter file template contains mapping parameters and their default values. It also includes input and in-out parameters that you can override at runtime. When you generate a parameter file template, the file contains the default parameter values from the mapping. If you do not specify a default value during parameter creation, the value for the parameter in the template is blank.

The parameter file template does not contain information about runtime values specified in the mapping task, Partial parameters, Advanced session properties, and System defined variables.

To download a parameter file template, perform the following steps:

1. In the mapping task, on the **Schedule** tab, click **Download Parameter File Template**.
2. The file name is <mapping task name>.param.
3. If you want to use the file in subsequent task runs, save the parameter file in a location that is accessible by the Secure Agent.
4. Enter the file name and directory on the **Schedule** tab when you configure the task.

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Parameter Best Practices

- You must first configure the mapping with a specific connection to select a source, target, or lookup object.
- Use parameters for all conditions or field mappings in the data flow that use fields from the object.
- Enter the appropriate label for the parameter.
- Provide a description for the parameter.

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You must first configure the mapping with a specific connection to select the source, target, or lookup object. When the mapping is complete, you can replace the connection with a parameter.

When you use a parameter for an object, use parameters for all conditions or field mappings in the data flow that use fields from the object.

When you create a parameter, you must enter the appropriate label for the parameter. The label displays in the mapping task and helps users to enter the right data.

You must also provide a description for the parameter. The value that you enter in the description field displays as a tooltip in the mapping task wizard.

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Topic

Knowledge Check

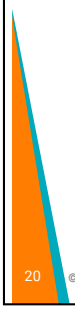
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You can use parameters to hold values that change between task runs. True or False?

a. True

b. False



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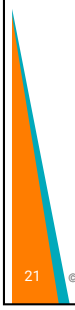
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Answer: a

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Which type of parameter is used to hold a variable value that can change every time a task runs?

a. Input parameter
b. Output parameter
c. In-out parameter



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Answer: c

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Lab Activity

Lab 9-1 Performing Complete Parameterization

In this lab, you will perform the following:

- Create a completely parameterized mapping

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Lab Activity

Lab 9-2 Using In-Out Parameters for Incremental Data Loading

In this lab, you will perform the following:

- Create a mapping using Input-Output Parameters

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Module Summary

This module showed you how to:

- Explain Parameters
- List the Types of Parameters
- Describe a Parameter File
- Discuss some Best Practices for Parameters

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Module 10
Mapping Task

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Module Objectives

After completing this module, you will be able to:

- Describe Mapping Task
- Discuss mapping updates and deployment
- Describe Simultaneous task runs

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Topic

Mapping Task

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Mapping Task

- Mapping task allows you to process data based on the data flow logic defined in a mapping.
- You can define parameters that associate with the mapping
- You can configure the task to run on a schedule.
- You can add pre and post-processing commands to the task.

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A Mapping task allows you to process data based on the data flow logic defined in a mapping. When you create a Mapping Task, you must select a mapping to use in the task. You can define parameters that associate with the mapping. You can also configure the Mapping Task to run on a schedule. You can also add pre and post-processing commands to a Mapping Task.

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Mapping Task Features

- Download a Mapping Task and import it into PowerCenter.
- Invoke a Mapping Task via the REST API or the Salesforce outbound message.
- Simultaneous mapping task runs.

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A Mapping task has the following important features:
You can download a Mapping Task and import it into PowerCenter.
You can invoke a Mapping Task via the REST API or the Salesforce outbound message.
You can enable simultaneously runs of the mapping task on the Schedule tab. This feature allows you to run the instances in a Parallel Paths step of a taskflow or in two taskflows that run concurrently.

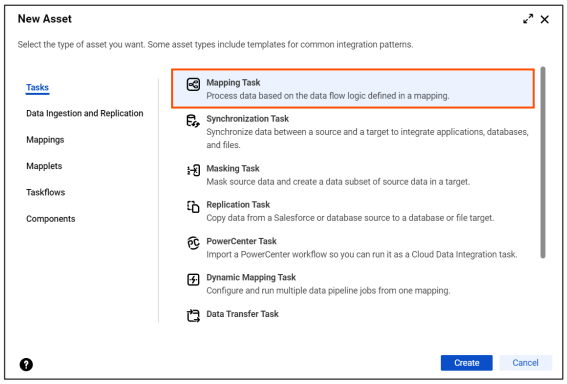
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Create Mapping Task

To execute a mapping job without first constructing a mapping, use a mapping task template.

To create a mapping task, click **New** > **Tasks**, and then complete one of the following steps:

- To create a mapping task, select Mapping Task and then click Create.
- To create a mapping task using a template, expand the appropriate template category, select the template you want to use, and then click Create.



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- To create a mapping task, select the mapping.
- A task can also be scheduled.
- Mapping task can be downloaded and imported into the PowerCenter.
- It can be invoked via REST API

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Topic

Advanced Settings for Mapping Task

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Advanced Settings for Mapping Task

- When you create an advanced relationship, the wizard converts any relationships that you defined to an SQL statement that you can edit.
- Multiple instances of a mapping task are run at the same time.
- To enable simultaneous task runs, select Allow the mapping task to be executed simultaneously on the Schedule tab when you configure the task.

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To create an advanced relationship, you add the primary source object in the Objects and Relationships table.
Then you select fields and write the SQL statement that you want to use. Use an SQL statement that is valid for the source database.
You can also add additional objects from the source.

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Advanced Session Properties in Mapping Tasks

Advanced connection attributes are defined in the mapping for tasks based on mappings.

- To add advanced session property, click **Add** in the **Advanced Session Properties** tab on the Schedule page.
- The following table describes the advanced session property values to add:

Advanced Property	Required Value
Commit on End of File	**No.** This setting avoids duplicate data from occurring on the target after a mapping task job is restarted. The default is No.
Recovery Strategy	Resume from last checkpoint. With this setting, the mapping task can resume processing change data from the point of interruption in the change data stream.

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- Depending on the connection type and use, the advanced attributes will appear.
- Advanced properties, such as page size, flush interval, or row limit, can be displayed on connections for Informatica Intelligent Cloud Services Connectors.

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Deploy Mapping Updates

- If you want the task to use the new mapping, you must deploy the mapping to a Mapping Task.
- If you make changes to a mapping, you can:
  - Save changes as a new mapping
  - Save changes and deploy to Mapping Task's
  - May require editing of task

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Simultaneous Task Runs

A mapping task can be run in several instances simultaneously.

Multiple task instances may be used in two parallel taskflows or in the Parallel Paths stage of a taskflow.

Using the REST API version 2 job resource, you can also perform many instances of a mapping task at once.

When you run the task instances simultaneously, mapping features that alter each time the task is executed, such as in-out parameters and sequence generator values, could result in unexpected outcomes.

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Knowledge Check

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Which of the following can be used to add advanced session properties?

A. Mapping
B. Mapping Task
C. Task flow
D. Masking task

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Answer: b

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IICS validates the SQL during the design time.

A. True

B. False

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Answer: b

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Lab Activity

Lab 10-1: Creating a Mapping Task to Perform Basic History Tracking and Configure Advanced Settings

In this lab, you will perform the following:

- Create a Mapping to perform basic history tracking
- Create a Mapping Task
- Configure Email Notifications

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Module Summary

This module showed you how to:

- Describe Mapping Task
- Discuss mapping updates and deployment
- Describe Simultaneous task runs

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Module 11
Replication Task

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Module Objectives

After completing this module, you will be able to:

- Explain the purpose of a Replication task
- List the features of a Replication task
- Discuss load type options in a Replication task
- Describe source and target options in a Replication task

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Topic

Replication Task

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Replication Task Overview

- Replicates data from a cloud-based application or a relational database table to a target
  - Replicates all rows
  - Replicates only the changed rows
- Use cases:



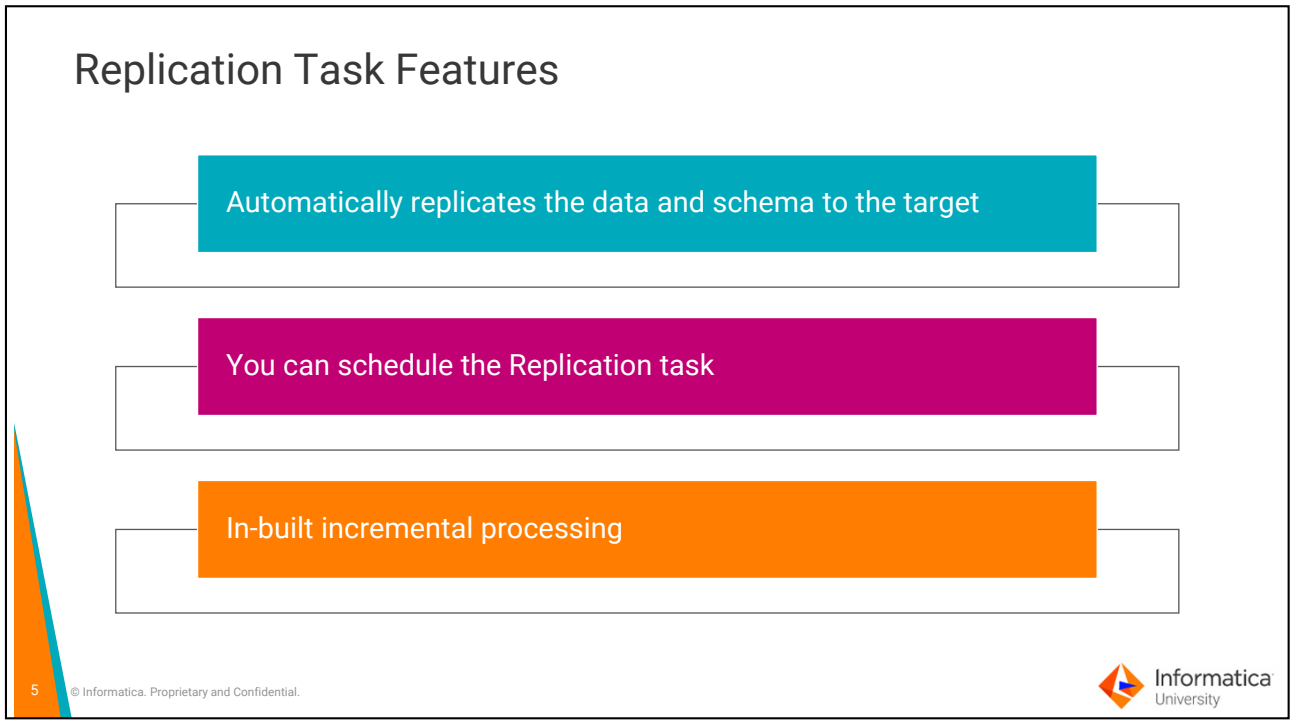
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A Replication task allows you to replicate data from a cloud-based application or a database table, to a target. You can replicate all rows of a source object each time the task runs, or replicate only those rows that changed since the last time the task ran. You can also use a Replication task to reset target tables and create target tables.

You can create and use a Replication task when you need to perform a regular back-up of data, when you need to archive data for compliance purposes, or when you need to move data to a data warehouse to perform offline reporting.

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A Replication task automatically replicates data and schema to the target. One of the major differences between a Synchronization task and a Replication task is that, in a synchronization task, you can transform the data before loading it to the target. However, in a Replication task, you can replicate the data from source to target without transforming the data.

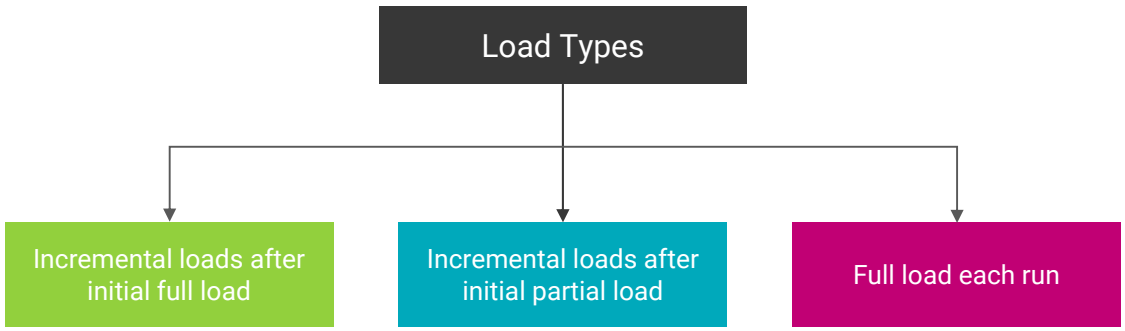
You can configure a Replication task to run on a schedule.

The Replication task has in-built incremental processing. It means that when you replicate data from a cloud-based application to a database table, you can choose to capture only the changed data.

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Load Types

- Determines the type of operation that you can use when you replicate data from the source to the target.



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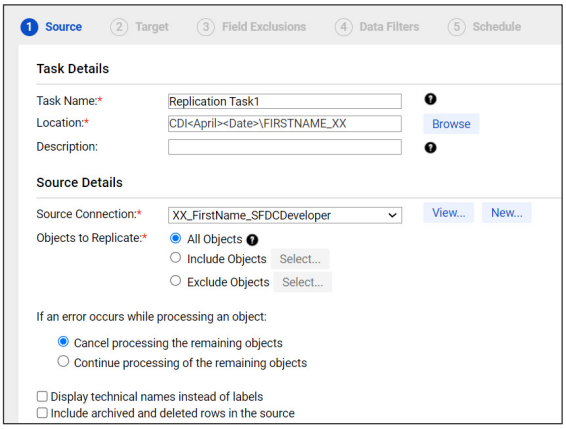
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The load type determines the type of operation you can use when you replicate data from a source to the target. When you replicate data, you can use one of three load types:
Incremental loads after initial full load: The first time the Replication task runs, it performs a full load, replicating all rows of the source. For each subsequent run, the Replication task performs an incremental load. In an incremental load, the Replication task uses an Upsert operation to replicate rows that changed since the last time the task ran.
Incremental loads after initial partial load: The Replication task always performs an incremental load with this load type. The first time the Replication task runs, the Replication task processes rows created or modified after a specified point in time. For each subsequent run, the Replication task replicates rows that change since the last time the task ran.
Full load each run: The Replication task replicates all rows of the source objects in the task during each run. You can specify this load type when the task uses a Salesforce source and a relational target.

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Replication Task – Source Options

- You can replicate a single object or multiple objects.
- If an error occurs while replicating multiple objects, you can choose to either cancel or continue with the processing of the remaining objects.
- For Salesforce sources only, you can include archived and deleted rows.



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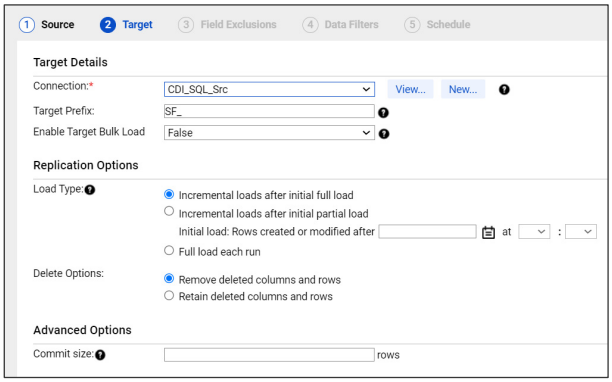
When you configure a Replication task, you have the following source options:

You can replicate a single object or multiple objects.
If an error occurs while replicating multiple objects, you can choose to either cancel or continue with the processing of the remaining objects.
For Salesforce sources only, you can include archived and deleted rows.

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Replication Task – Target Options

- A prefix is added to name the resulting table or flat file.
- You can change the default prefix and assign a unique prefix.
- Specify the load type to be incremental or full.
- In the target, retain, or delete rows that are deleted from the source.
- Specify the number of rows to commit.



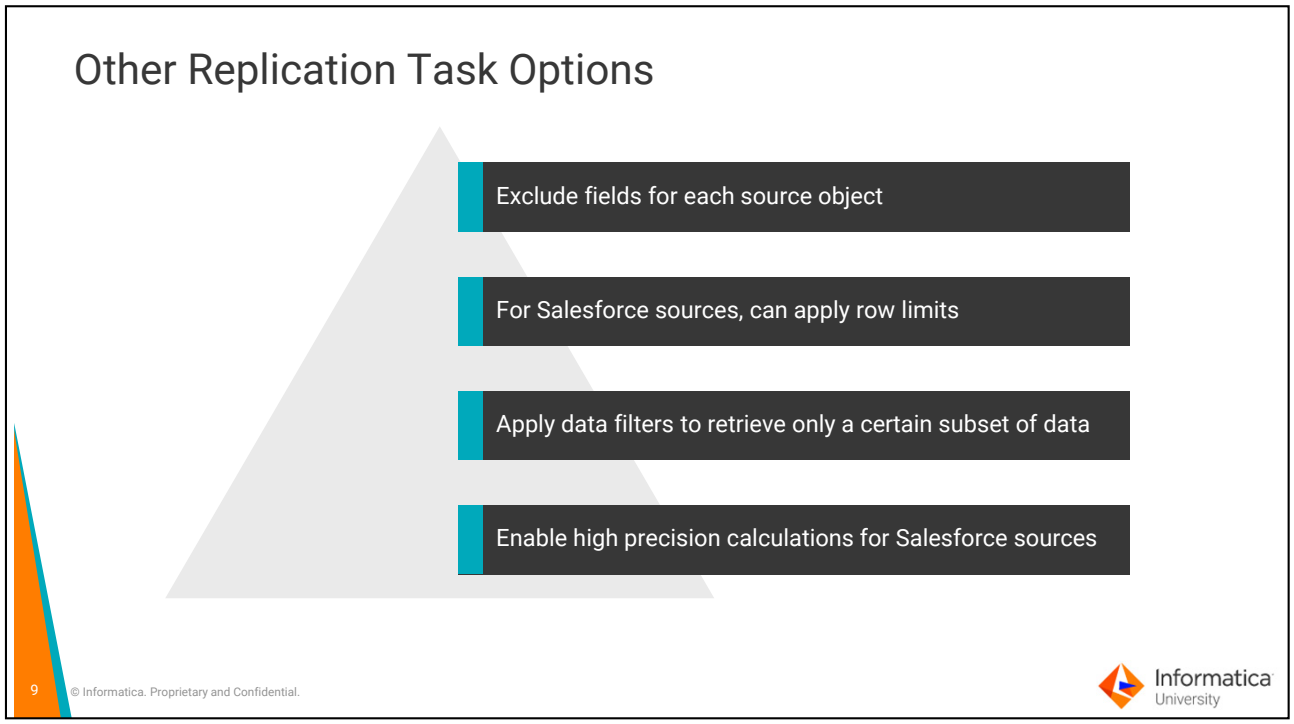
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When you replicate the data from a Salesforce source to a database table, you can specify the load type to be incremental or full.
In the target, you can also define if you want to retain or delete the rows that are deleted from the source during replication.
By default, a prefix is added to the name of the resulting table or flat file. For example, if you use the Replication task to replicate data from the Salesforce Account object to a database table, the default name of the resulting table would be SF_ACCOUNT.

You can change the default prefix and assign a unique prefix. Assigning a unique prefix allows users to replicate data using the same database connection, without the risk of overwriting data.

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You can exclude fields for each source object. For example, if you have fields in Salesforce that the business is not using, then you can exclude those fields.

If your source is a Salesforce object, you can apply row limits. For example, if you want to perform a test run of a task, you can limit the number of rows you export. For non-Salesforce sources, this option is disabled.

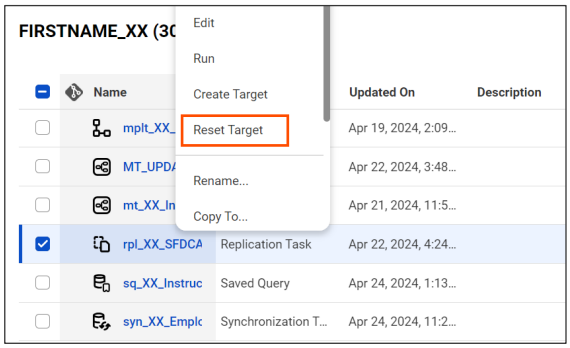
You can also apply data filters to retrieve only a certain subset of data. For example, if you want to replicate data to a data warehouse for reporting, you can filter and replicate only records of a certain type.

Finally, you can enable high precision calculations for Salesforce sources. When you can enable high precision calculations, the Replication task reads data with a precision of up to 28 in Salesforce calculated fields and writes the data to the target.

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Resetting the Target Table

- While using the Replication task, IICS provides 'Reset Target' option to drop the target table
- Use this option:
  - when the task load type is incremental, and
  - if any one of the following changes are made to a source field – datatype, precision, and scale



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How can you reset a target table using a Replication task?

Usually, to drop a target table, you have to go to the database that stores the table, and then drop the table there. When you use a Replication task, IICS provides a Reset Target option that enables you to drop the target table within the IICS interface.

You can use this option when the task type is incremental and if you change the data type, precision, or scale of a source field.

Assume that you have a Replication task scheduled to run every day, with an incremental load type. If you change the data type, precision, or scale of a source field, you will get an error when the task runs again because the data type, precision, or scale for the source field and target field are inconsistent. In this scenario, you can use the Reset Target option to drop the target table after you make changes to the source fields. The next time the task runs, it will run a full load and create the target table.

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Generating Non-Unique Index

When you replicate salesforce sources to the target database tables:

The Replication application generates Non unique index for each table.

The index is generated based on the Salesforce ID field.

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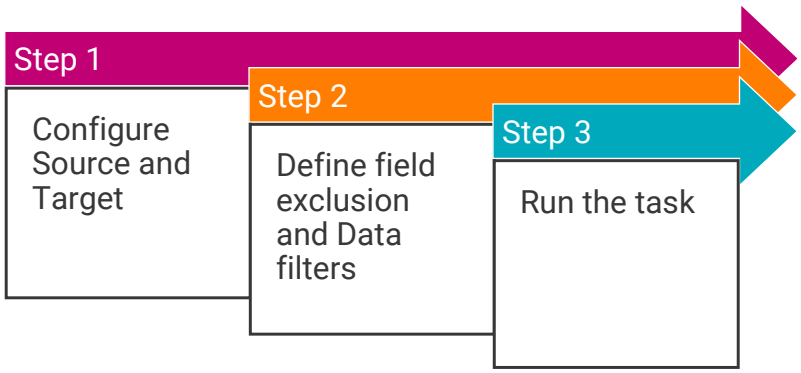
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The data replication application generates no unique index for each target table.
The index is generated based on the Salesforce ID field.

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Demonstration

Replication Task



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How can you reset a target table using a Replication task?

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Topic

Knowledge Check

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Replication task provides built-in incremental processing.

A. True

B. False

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Answer: a

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Which of the following load types always perform an incremental load in Replication Task?

A. Full load each run

B. Incremental loads after initial full load

C. Incremental loads after initial partial load

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Answer: b

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Lab Activity

Lab 11-1 Replicating Data to a Flat File

In this lab, you will perform the following:

- Create a replication task to replicate data to a CSV file

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Module Summary

This module showed you how to:

- Explain the purpose of a Replication task
- List the features of a Replication task
- Discuss load type options in a Replication task
- Describe source and target options in a Replication task

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Module 12
Masking Task

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Module Objectives

After completing this module, you will be able to:

- Explain Masking Task
- Discuss Masking task source and target options
- Refresh Masking task metadata
- Define Masking Rule Types
- Reset a Masking Task
- Discuss guidelines for Masking data
- Describe Data Masking Transformation

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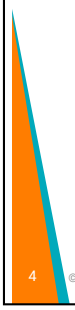
Masking Task

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Masking Task Overview

- Allows you to mask sensitive fields in the source data with realistic test data for non-production environments.
- Allows you to migrate data from one Salesforce Org to another Salesforce Org.
- Allows you to choose the source and the target and select a masking rule.
- Allows you to mask data “inplace” to overwrite existing data.



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You can use a masking task to mask the sensitive fields in source data with realistic test data for non-production environments. A masking task allows you to migrate data from one Salesforce Org to another. When you migrate data, you can mask sensitive fields with realistic test data.

When you configure a masking task, you must choose the source and the target and select a masking rule for each field in the source that you want to mask.

You can also use the “inplace” masking option to mask the data in the same system from which the masking task reads the data. In simpler terms, the “inplace” masking option allows you to overwrite the existing data in a Salesforce Org.

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Masking Task Steps

STEP 1	Definition
STEP 2	Configure source
STEP 3	Configure target
STEP 4	Configure data filters
STEP 5	Define masking rules
STEP 6	Schedule

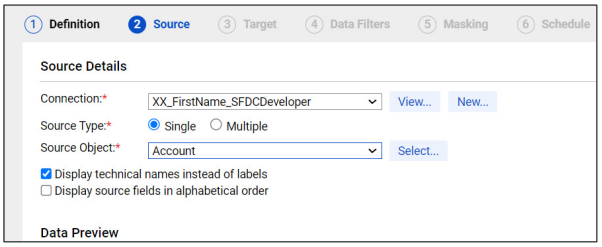
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Masking Source Options

- You can add a single object or multiple related objects.
- A single object does not contain any related objects.
- Multiple objects have an explicit relationship defined in Salesforce.
  - Salesforce Opportunity object is related to the Campaign object.
- If you select multiple source objects, you can choose an object and add the related parent, child, and self-reference objects manually.



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You can add a single object or multiple related objects in a masking task.

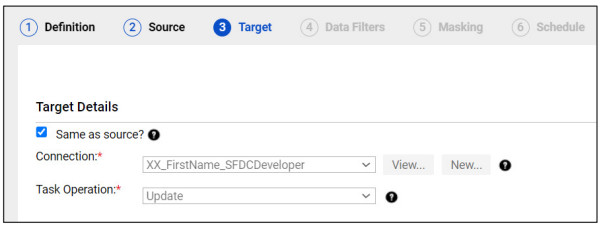
You can add a single object that does not contain any related objects. You can also add multiple objects that have an explicit relationship defined in Salesforce. For example, if you use the Opportunity object in Salesforce as a source, you can add the related Campaign object as well. All Salesforce objects in a multiple-object source must have a predefined relationship in Salesforce.

If you select multiple source objects, you can choose an object and add the related parent, child, and self-reference objects manually. A self-reference relationship is the one in which a source object references to itself within a task.

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Masking Target Options

- For a Salesforce source, you can select another Salesforce connection for your target.
- Select the task operation that you want to perform in the target.
- Select the 'Same as Source' option in the target step if you want to mask data "inplace".
- When the target is different from the source, select one the following task operations:
  - Insert
  - Update
  - Upsert



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If you have a Salesforce source, you can select another Salesforce connection for your target. This means that the source and target type must be the same in a masking task.

When you configure the target options, you can select the task operation that you want to perform in the target. If you want to mask the data "in place", you must select the 'Same as Source' option in the target step. When the target is same as the source, you can perform only an Update operation.

When the target is different from the source, you can select one of the following task operations:
Insert: This operation ignores the existing target data and inserts all the source data.
Update: This operation updates data in the target location based on the source data.
Upsert: This operation updates existing target data. If data does not exist in the target, the masking task inserts the data.

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Data Subset

- Allows you to specify row limit or data filter options.
- Available only if the source and target are not from the same connection.
- For Salesforce sources, create a filter on a single Salesforce object.

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A Data Subset allows you to specify row limit or data filter options in a masking task. You must note that these options are available only if the source and target connections are different. You cannot use the filter if the source and target connections are the same or from the same account. Therefore, you cannot use these options if you want to mask the data inplace.

For Salesforce sources, you can create a filter on a single Salesforce object.

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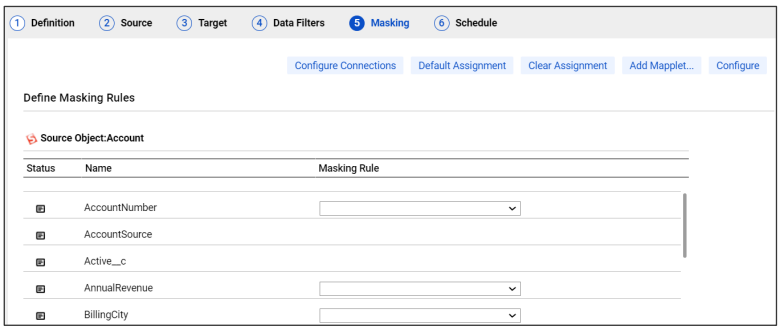
Field Masking Rules



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Defining Field Masking Rules

- Specify masking rules for columns that you want to mask.
- Column data type determines the available masking rules.
- If you do not specify a masking rule for a column, the column is copied unmasked.



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In the Masking step of the wizard, you can select masking rules for columns that you want to mask.

The column data type determines the available masking rules. For example, if the field data type is numeric, you can define a masked value that is within a fixed or percent variance from the original value.

If you do not specify a masking rule for a column, the column is copied 'unmasked' to the target.

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Masking Rule Types

Masking Technique	Description
Credit Card Masking	Masks a credit card number with a number that has a valid checksum.
Email Masking	Masks an email address with a random email address.
Email Advanced Masking	Masks an email address with a realistic email address from a first name, a last name, and a domain name.
IP Address Masking	Applies a built-in mask format to disguise an IP address.
Nullification Masking	Changes input values in a field to null.
Phone Masking	Masks a phone number with random numbers in the same format as the original number.

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A masking rule defines the logic that masks the data. As we saw earlier, the type of masking rule that you can apply depends on the data type of the field that you want to mask.

Credit card masking: A credit card masking rule applies a built-in mask format to mask credit card numbers.
Date masking: A date masking rule applies a date mask format to the columns of string data type that contain dates.
Email masking: An email masking rule applies an email mask format to the columns of the string data type that contains email addresses.
IP address masking: An IP address masking rule applies an IP address mask format to columns of string data type that contain IP addresses.
Key masking: A key masking rule produces repeatable results for the same source data. You can apply key masking to datetime, string, and numeric data types.
Nullification masking: A nullification masking rule transfers a null value from the source to the target.

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Masking Rule Types (continued)

Masking Technique	Description
SIN Masking	Masks a Social Insurance number.
SSN Masking	Masks a Social Security number with a valid number.
Substitution Masking	Replaces a field with similar but unrelated data from a dictionary file.
URL Masking	Masks a URL by searching for the '://' string and masking the substring to the right of it.
Custom Masking	Applies an expression to mask the target field.
Mapplet Masking	Applies masking rules from a PowerCenter mapplet. The mapplet contains the logic to mask the input fields and return data to the target.

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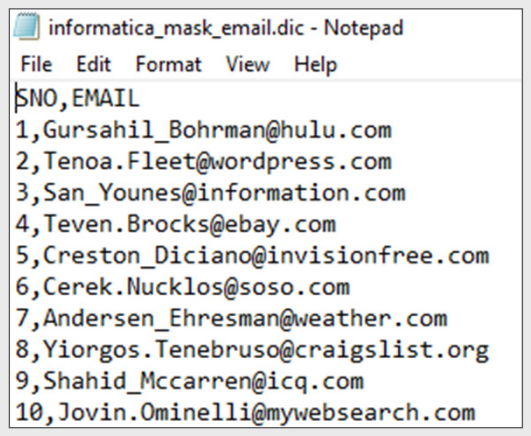
Phone masking: This rule applies a phone number mask format to columns of string data type that contain phone numbers.
Random masking: This rule produces random, non-repeatable results for the same source data and masking rules. You can apply random masking to datetime, string, and numeric data types.
Social Insurance Number (SIN) masking: This rule applies a built-in mask format to modify the Social Insurance Numbers according to the specified format.
Social Security Number (SSN) masking: This rule applies a built-in mask format to modify the Social Security numbers.
Substitution masking: This rule replaces a column of data with similar but unrelated data from a default dictionary. You can apply substitution masking to columns with string data type.
URL masking: This rule applies a URL mask format to columns of string data type that contain URLs.

Remember, for some of the masking rules, you can configure additional options.

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Dictionary Files

- Masking task uses a set of built-in dictionary files or the custom dictionary files.
- Masking task performs a lookup on the selected dictionary and replaces the source data with data from the dictionary.
- You can find dictionary files in the following directory:
- C:\Program Files\Informatica Cloud Secure Agent\apps\Data_Integration_Server\data.
- You cannot edit or rename dictionary files.



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The masking task uses a set of built-in dictionary files or custom dictionary files that you create. When you configure a substitution masking operation, you can select a dictionary that contains substitute values.

The masking task performs a lookup on the selected dictionary and replaces the source data with data from the dictionary.

The dictionary files are located in the secure agent installation directory. When you cannot edit or rename the dictionary files, you can change the content within the specified file structure.

The image shows a sample dictionary file for email masking.

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Refresh Metadata

- Masking task imports the source and target metadata.
- If you make changes to the Salesforce objects or objects in the task, the metadata imported when you created the task can get outdated.
- Masking task requires the latest metadata to define relationships between objects and to determine fields that you can mask.
- Refresh the metadata before you run a masking task to ensure that the source and target metadata in the task is up to date.

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When you create a masking task, the task imports the source and target metadata. Over the time, you might update the Salesforce objects and add or delete objects.
You might also add or delete objects in the masking task.

When there are changes to the Salesforce objects or objects in the task, the metadata imported when you created the task can become outdated. If you run the same masking task at regular intervals, the metadata imported in the task will not be the latest. The masking task requires the latest metadata to define relationships between the objects and to determine the fields that you can mask.

A masking task can fail if it does not use the updated metadata in the Salesforce source and target. You can refresh the metadata before you run a masking task, to ensure that the source and target metadata in the task is up to date.

There are two ways in which you can refresh the metadata in a masking task. We will discuss metadata refresh in the upcoming slides.

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Best Practices For Refreshing Metadata

- If you want to update many objects, refresh the metadata without editing the task.
- If you want to update fewer objects, refresh the metadata from within the task.

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You can choose how you want to refresh the metadata based on the number of objects to refresh. As a best practice, if you want to update many objects, it is recommended that you refresh the metadata without editing the task. To update fewer objects or less metadata, you can refresh the metadata from within the task.

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Reset Task

- You can reset a masking task that has a different source and target and contains data filters.
- Masking task with data filters performs different steps.
- If the task fails at any of the steps, it continues from the point of failure when you restart the task.
- Reset returns the task status to Start.
- When you restart the task, the task starts from the first step.
- Tasks that use the same source and target or do not include data filters do not require subset computation or staging tables.

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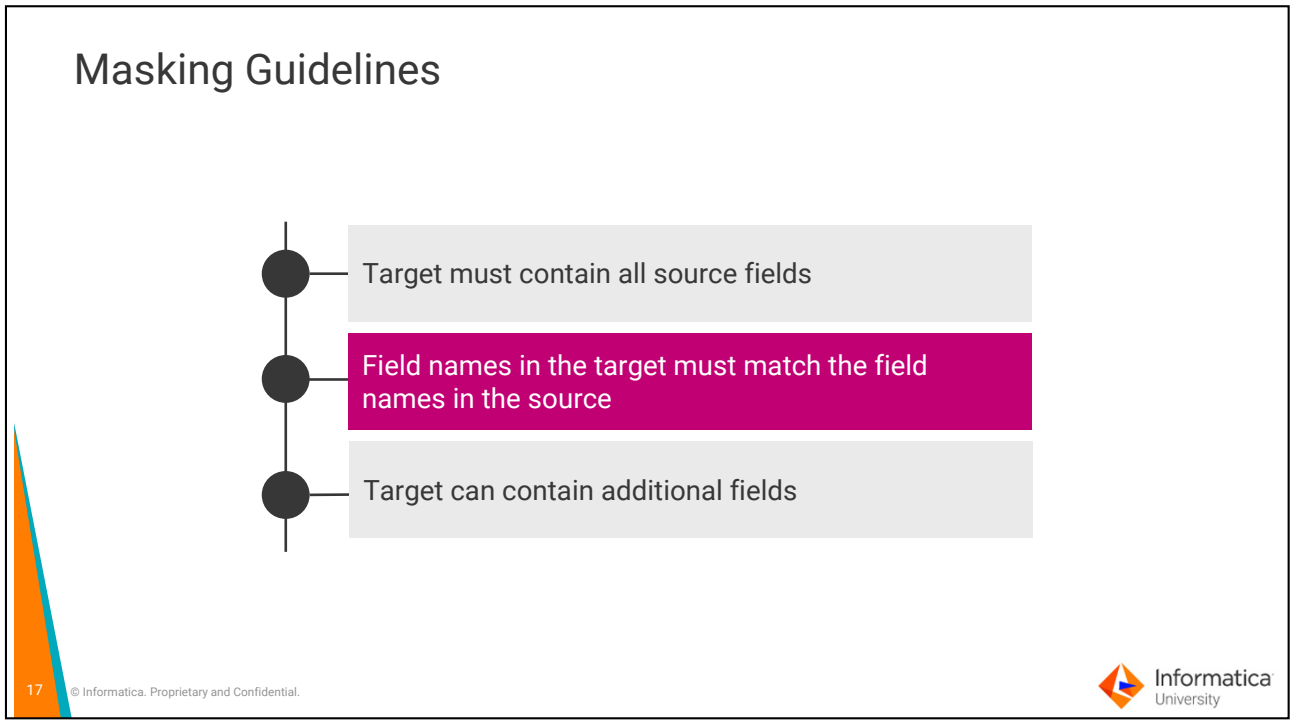
You can reset a masking task that has a different source and target and contains data filters.

A masking task with data filters performs different steps including staging data, subset computation, load to target, and drop staging tables. If a task fails at any of the steps, it continues from the point of failure after the restart.

You can choose to reset the task before you restart the task. The reset returns the task status to Start. When you restart the task, it starts from the first step. It performs all steps of staging, subset computation, load to target, and drop staging tables, based on how you configure the task.

Tasks with the same source and target or without data filters, do not require subset computation or staging tables.

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When you mask the data from a source Org to a target Org, you must ensure that the target Org contains all the source fields. Additionally, the field names in the target Org must match the field names in the source Org.

The target Org can contain additional fields. However, the additional fields will not have masked data.

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Topic

Data Masking Transformation

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Data Masking Transformation

To transform sensitive production data into accurate test data for non-production situations, use the Data Masking transformation.

- Data Masking transforms source data according to masking rules that you provide for each column.
- For software development, testing, training, and data mining, create masked data.
- Data masking is a passive transformation.

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- Based on the source data type and masking type you provide for a port, the Data Masking transformation offers masking rules.
- For strings, you can specify which characters should be replaced and which should be applied to the mask.
- You can give the masked data a range of numbers for dates and numbers.

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Data Masking Transformation Example

You cannot use the data without the risk of compromising sensitive data.

- You can use the Data Masking transformation to mask sensitive data.
- The production data includes a table Personnel_Information with the following data:

First Name	Surname	DOB	Address	State
Uma	Hilel	03/12/1985	24, Atkins Avenue	CA
John	Sen	07/15/1963	49, Wheeler Road	MN
Keiko	Burns	11/24/1989	13, Draker Drive	PA
Isadora	Buckley	08/16/1990	20, Fountain Cente	CA

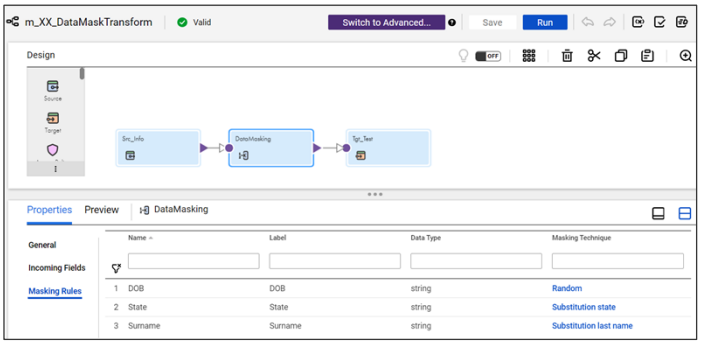
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Data Masking Transformation Example (continued)

- Add the Personnel as a source transformation in the Mapping Designer.
- Add a target transformation Personnel_test.
- To ensure sensitive data is masked, you need to mask the Surname, DOB, and the State columns.



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Data Masking Transformation Example (continued)

When the mapping run completes successfully, view the masked data in the Personnel_test output table:

First Name	Surname	DOB	Address	State
Uma	Acothley	05/12/1985	24, Atkins Avenue	Michigan
John	Mcgovern	04/15/1963	49, Wheeler Road	Oklahoma
Keiko	Garsia	03/24/1990	13, Draker Drive	Montana
Isadora	Sonnier	01/16/1991	20, Fountain Cente	South Carolina

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Topic

Knowledge Check

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What determines which data masking rules are available?

A. Data Type, Data Content and Data Length
B. Data length
C. Data content
D. Data Type

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Answer: d

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Lab Activity

Lab 12-1 Creating a Masking Task

In this lab, you will perform the following:

- Create a data masking task
- Mask the phone number fields values

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Lab Activity

Lab 12-2 Using Data Masking Transformation in a Mapping

In this lab, you will perform the following:

- Configure a mapping using Data Masking transformation
- Mask the Surname, DOB, and State field values in the source data

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Module Summary

This module showed you how to:

- Explain Masking Task
- Discuss Masking task source and target options
- Refresh Masking task metadata
- Define Masking Rule Types
- Reset a Masking Task
- Discuss guidelines for Masking data
- Describe Data Masking Transformation

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Module 13
Taskflows

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Module Objectives

After completing this module, you will be able to:

- Define a taskflow
- List the steps in a taskflow
- Define Linear taskflow
- List the task types in a Linear taskflow
- Discuss the taskflow templates
- Explain the use of parameters in a taskflow

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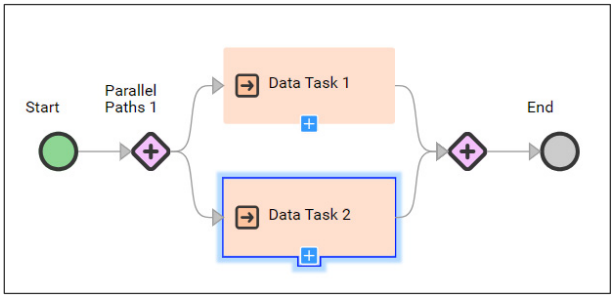
Taskflow Overview

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Taskflow Overview

- A taskflow controls the execution sequence of the tasks
- Create tasks and add them to a taskflow
- Can configure email notifications for a taskflow
- Can configure a taskflow to run on a schedule
- Taskflow allows you to:
  - run parallel tasks
  - use advanced decision making
  - perform other advanced orchestrations



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A taskflow controls the execution sequence of a mapping task or a synchronization task based on the output of the previous task. You must first create tasks and then add them to a taskflow. You can configure email notifications for a taskflow. Email notifications allows users to receive notifications about the status of the taskflow. You can also configure a taskflow to run on a schedule. A sample taskflow is shown in the image.
With the various tasks and taskflow templates available in IICS, you can run tasks in parallel, use advanced decision making criteria, and perform other advanced orchestrations.

You will see more about taskflow templates later in the module.

Note that the taskflow name must always start with an alphabet. This is because, the tasks in a taskflow are called using an API call internally, and the API calls cannot be starting with a number.

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Taskflow Steps

- Use taskflow steps to add and orchestrate Data Integration tasks

Assignment Step	Sets a value to a field
Data Task Step	Adds a task to a taskflow
Notification Task Step	Sends an email notification to specified recipients
Command Task Step	Runs shell scripts or batch commands from a file on the Secure Agent machine
Subtaskflow Step	Embeds and reuses an existing taskflow
Decision Step	Enables a taskflow to take different paths based on the value of a specific field
Parallel Paths Step	Enables a taskflow to run multiple tasks at the same time
Jump Step	Jumps from one part of the taskflow to another

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Assignment Step: An Assignment step allows you to set a value to a field. With this step, you can use input, output, and temporary fields.
Data Task Step: A Data Task step allows you to add a task to a taskflow. You can configure how the taskflow handles errors and warnings, perform actions based on a schedule, and override runtime parameters.
Notification Task Step: A Notification Task step allows you to send an email notification to specified recipients.
Command Task Step: A Command Task step allows you to run shell scripts or batch commands from a file on the Secure Agent machine. You can use the Command Task outputs to orchestrate subsequent tasks in the taskflow.
Subtaskflow Step: A Subtaskflow step allows you to embed and reuse an existing taskflow. You can configure the input fields to provide the input when you run the taskflow.
Decision Step: You can use a Decision step when you want a taskflow to take different paths based on the value of a specific field.
Parallel Paths Step: You can use a Parallel Paths step when you want a taskflow to run multiple tasks at the same time.
Jump Step: A Jump step allows you to jump from one part of the taskflow to another.

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Taskflow Steps (continued)

- Use taskflow steps to add and orchestrate Data Integration tasks

File Watch Step	Listens to files in a defined location and monitor file events
Ingestion Task Step	Adds an existing file Ingestion and replication task to a taskflow
Throw Step	Catches fault and returns the fault details
End Step	Defines the HTTP status code that must be used when a taskflow completes
Wait Step	Pauses the taskflow execution for a specific duration

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File Watch Step: You can use a File Watch Task step to listen to files in a defined location and monitor file events. In the File Watch Task step, you can select an existing file listener with the connector source type. You can use file events to orchestrate taskflow execution.

Ingestion Task Step: You can use the Ingestion Task step to call a file ingestion and replication task. For example, you can use the Ingestion Task step to read a large number of files from a source location, perform data integration step on the files using the Data Task step, and use another Ingestion Task step to write the updated files to the final target location.

Throw Step: You can use a Throw step to catch faults and return its details. The fault step prevents the execution of the subsequent steps in a taskflow and sets the taskflow status to failed.

End Step: You can use an End step to define the HTTP status code shown when a taskflow completes.

Wait Step: A Wait step allows you to pause the taskflow execution for a specific duration.

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Taskflow Features

Properties	Description
Taskflow Binding	Defines how a taskflow is invoked and run. You can invoke a taskflow using one of the following binding types: • REST/SOAP • Event
Input fields	Used to add fields that a taskflow uses at the beginning of a step.
Temporary fields	Used to create temporary fields for use in a taskflow step. Temporary fields do not appear in the input or output of a taskflow.
Advanced properties	Used to configure a taskflow to suspend on a fault that occurs at the taskflow level and to configure email notification after suspension of the taskflow.

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When you create a taskflow, you can define the following taskflow properties:
1. Taskflow binding: It defines how a taskflow is invoked and run. You can invoke a taskflow using REST/SOAP binding type or as an Event binding type.
- When the binding type is REST/SOAP, you can run the taskflow by using a REST or SOAP endpoint. You can also define the Allowed Groups and Allowed Users fields to provide access to the user groups and users who can run a published taskflow as an API.
- When the binding type is Event, the taskflow is invoked when the specified event occurs. You must also select the file listener that will monitor the event.

2. Input fields: These are used to add fields that a taskflow uses at the beginning of a step.
3. Temporary fields: These are used to create internal temporary fields for use in a taskflow step.
4. Advanced properties: You can configure a taskflow to suspend on a fault that occurs at the taskflow level and to configure the taskflow to send an email notification to specified recipients when it is suspended on a fault.

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Topic

Linear Taskflow



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Linear Taskflow Overview

- Simplified version of the Data Integration taskflow.
- Cannot control the execution sequence of the tasks based on the previous task in the taskflow.
- Groups multiple Data Integration tasks and runs them serially.
- Can configure email notifications for a linear taskflow.
- Can also configure a linear taskflow to run on a schedule.

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A linear taskflow is a simplified version of the Data Integration taskflow. It cannot control the execution sequence of tasks based on the previous task in the taskflow.

A linear taskflow groups multiple Data Integration tasks and runs them serially in the order that you specify.

You can configure email notifications for a linear taskflow. You can also configure the taskflow to run on a schedule.

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Linear Taskflow Example

Scenario

- Update a list of contacts on a monthly basis

Solution

- Create a linear taskflow with a Synchronization task to Upsert accounts followed by a Synchronization task to Upsert contacts for the accounts
- Schedule the linear taskflow to run each month

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Assume that you need to update a list of contacts on a monthly basis. To do this, you can upsert recent account information and then upsert contact information for each account. So, you can create a linear taskflow with a Synchronization task to upsert accounts followed by a Synchronization task to upsert contacts for the accounts. After that, you can schedule the linear taskflow to run each month.

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Tasks in a Linear Taskflow

- Linear taskflow can include the following task types:
  - Synchronization Task
  - Replication Task
  - Mapping Task
  - Masking Task
  - PowerCenter Task
- You can edit a linear taskflow
- You cannot delete a taskflow that was published, previously run from the taskflow designer, or associated with a schedule

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A linear taskflow can include Synchronization Task, Replication Task, Mapping Task, Masking Task, and PowerCenter Task.

You can also edit a linear taskflow. If you add a task to a linear taskflow that is currently running, Data Integration does not run the new task until the next time the linear taskflow runs.

Note: You cannot delete a taskflow that is published, previously run from the taskflow designer, or associated with a schedule. You must first unpublish the taskflow, and delete it.

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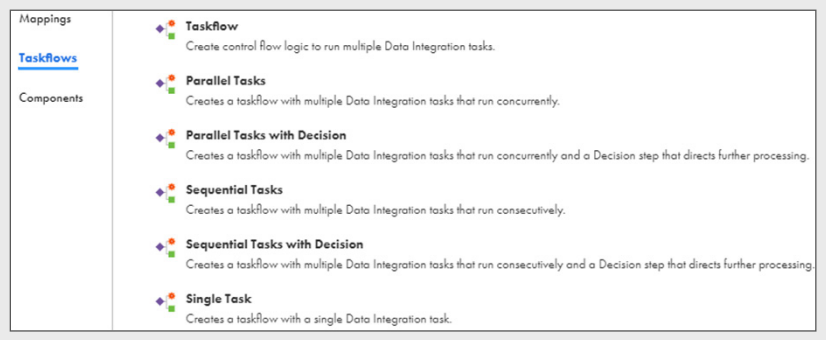
Taskflow Templates

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Taskflow Templates

- Basic
- Parallel Tasks
- Parallel Tasks with Decision
- Sequential Tasks
- Sequential Tasks with Decision
- Single Tasks



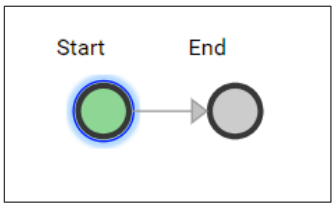
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You can use a taskflow template instead of creating a taskflow from scratch. IICS provides pre-created templates such as Basic, Parallel Tasks, Parallel Tasks with Decision, Sequential Tasks, Sequential Tasks with Decision, and Single Task, that you can use as per your business requirements. You can find these templates when you create a new asset from the Data Integration Service home page.

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Basic Template

- Provides a canvas with a Start and an End step.



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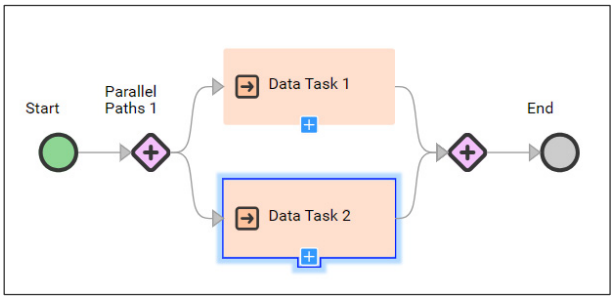
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Basic Template: The basic template provides a canvas with a Start and an End step. You can create and configure a taskflow based on your business requirements.

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Parallel Tasks Template

- You can use it to run two or more Data Integration tasks in parallel.



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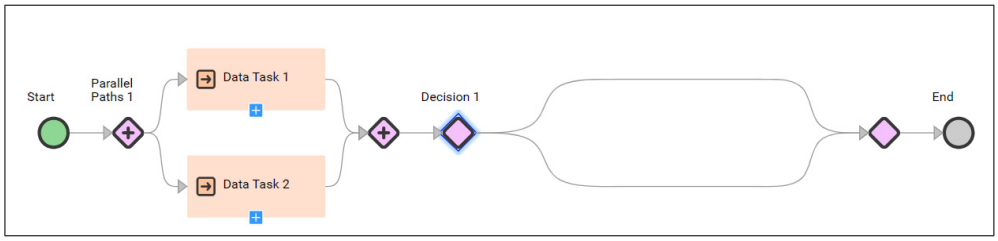
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Parallel Tasks Template: You can use this template when you want to run two or more Data Integration tasks in parallel.

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Parallel Tasks with Decision Template

- You can use it to run two or more Data Integration tasks in parallel, and make a decision based on the outcome of any task.



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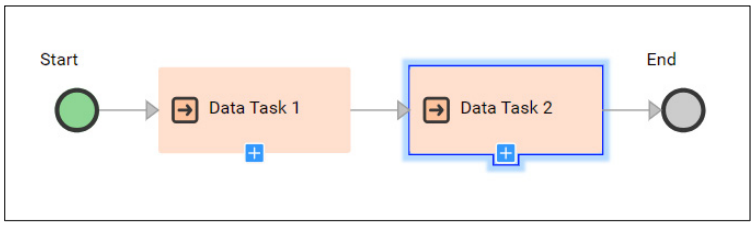
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Parallel Tasks with Decision Template: You can use this template when you want to run two or more Data Integration tasks in parallel, and then make a decision based on the outcome of any task.

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Sequential Tasks Template

- You can use it to run two Data Integration tasks consecutively.



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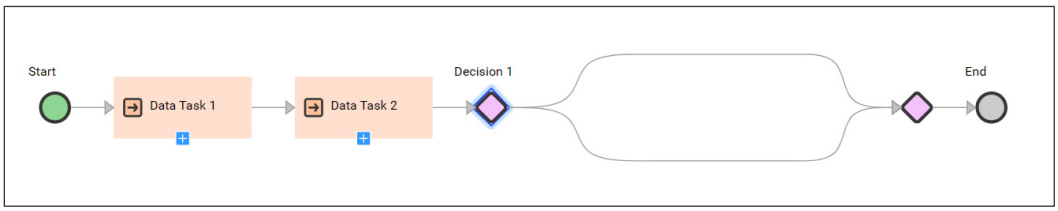
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Sequential Tasks Template: You can use this template when you want to run two Data Integration tasks consecutively.

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Sequential Tasks with Decision Template

- You can use it to run two Data Integration tasks consecutively, and make a decision based on the output of either of the two tasks.



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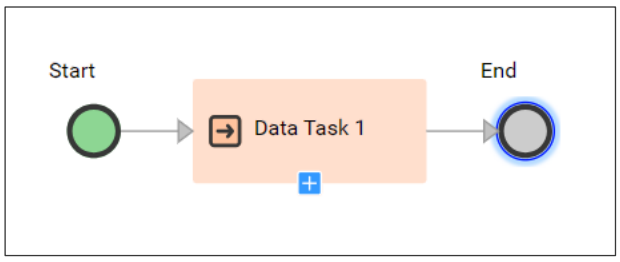
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Sequential Tasks with Decision Template: You can use this template when you want to run two Data Integration tasks consecutively, and then make a decision based on the output of either of the two tasks.

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Single Task Template

- You can use it to run a Data Integration task on a daily or weekly basis.



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Single Task Template: You can use this template when you want to run a Data Integration task on a daily or weekly basis.

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Topic

Parameters in a Taskflow

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Parameters in a Taskflow

- Use a taskflow to pass Input and In-Out parameters to a task.
- An Input parameter is a placeholder for a value or values in a mapping.
- An In-Out parameter is a placeholder for a value that you can pass in to or out of a mapping.
- Override the parameter file directory and parameter file name of a mapping task at the taskflow level using Data Task step.
- When you add a mapping task to a taskflow, you can override parameter values with the Data Task step or with the Assignment step.

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You can use a taskflow to pass Input and In-Out parameters to a task. An Input parameter is a placeholder for a value or values in a mapping. An In-Out parameter is a placeholder for a value that you can pass in to or out of a mapping.

You can design a mapping with Input or In-Out parameters.

When you add a mapping task to a taskflow, you can override the parameter file directory and parameter file name of a mapping task at the taskflow level using Data Task step. To override parameter values use the Data Task step or with the Assignment step. The mapping task passes the parameters to the mapping.

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Input Parameters

- You can define the value of the input parameter when you configure the mapping task.
- You can override the following subset of mapping input parameters:
  - **Source object**: You can change the object from which the mapping task reads the data.
  - **Source connection**: You can change the connection that the mapping task uses to read data from the source.
  - **Target connection**: You can change the connection that the mapping task uses to write data to the target.
  - **Target object**: You can change the object to which the mapping task writes the data.

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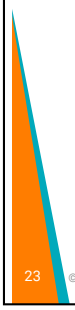
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You can define the value of the input parameter when you configure the mapping task. When you use the mapping task in a taskflow, you can override the parameter values for the source object, source connection, target connection, and target object.
When you override the parameter value for the source object, you can change the object from which the mapping task reads the data.
When you override the parameter value for the source connection, you can change the connection that the mapping task uses to read the data from the source.
When you override the parameter value for the target connection, you can change the connection that the mapping task uses to write the data to the target.
When you override the parameter value for the target object, you can change the object to which the mapping task writes the data.

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In-Out Parameters

- You can define the value of the in-out parameter when you configure the mapping task.
- An in-out parameter can change each time a task runs.



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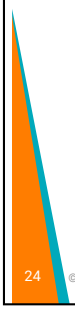
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You can also define the value of the in-out parameter when you configure the mapping task. Unlike input parameters, an in-out parameter can change each time a task runs. You can use the taskflow to override any type of in-out parameters that the mapping task supports.

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Guidelines for Using Parameters in Taskflows

- Use the Data Task step to override input or in-out parameters.
- For advanced use cases, use the Assignment step to override the parameters.
- If you override the same parameter in both the Assignment step and the Data Task step, the taskflow uses the value assigned in the Data Task step.
- You can get the current value of the in-out parameter only after the taskflow executes the Data Task step.



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You can use the Data Task step to override input or in-out parameters. However, for advanced use cases, you can override the input or in-out parameters with an Assignment step.

If you define a field to override the same parameter in both the Assignment step and the Data Task step, the taskflow uses the value assigned in the Data Task step.

You can get the current value of the in-out parameter only after the taskflow executes the Data Task step.

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Error Handling

- Use the Error handling tab to specify how the taskflow behave when a Command Task step encounters an error.
- When the Command Task step fails the taskflow behaviour can also be configured.
- After you select a task, configure the following error handling properties:

On Error	The path that a taskflow takes when it encounters an error in a Command Task step.
Fail taskflow on completion	Taskflow behavior when the Command Task step fails or does not run.

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When a Command Task step fails, an error is produced.
You can the choose from the following options:
Select Ignore to ignore the error and move to the next step.
Choose suspended taskflow to move the taskflow to the suspended state when it encounters an error.
Select Custom error handling to handle the error in a manner you choose.
If the Command Task step does not run, you can arrange a taskflow to terminate when it is finished.

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Topic

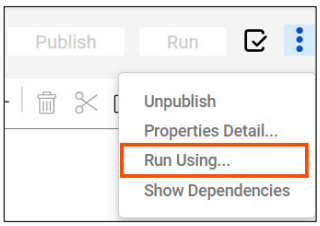
Running a Taskflow

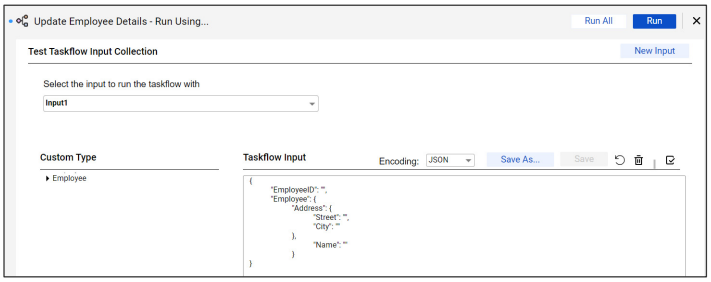
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Running a Taskflow Using Taskflow Input

- Create one or more taskflow inputs and run the taskflow using taskflow inputs to test it.
- After you run the taskflow, you can view details of the successful and unsuccessful executions of the taskflow instances.
- To run a taskflow with a specific taskflow input or all the taskflow inputs, from the Actions menu, select Run Using.





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After you publish a taskflow, you can create one or more taskflow inputs and run the taskflow using taskflow inputs to test it. After you run the taskflow, you can view details of the successful and unsuccessful executions of the taskflow instances.
To run a taskflow using taskflow inputs, you must include Allowed Users and Allowed Groups in the taskflow designer. To run a taskflow with a specific taskflow input or all the taskflow inputs, from the Actions menu, select Run Using.

Note:
- To run a taskflow using taskflow inputs, you must include Allowed Users and Allowed Groups in the taskflow designer.
- The Run Using option is disabled if the taskflow contains unsaved or unpublished changes. To run the taskflow, you must save and publish the taskflow.

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Running a Taskflow Using REST API

- You can invoke a taskflow as an API by publishing the taskflow as a service.
- Data Integration generates the service URL and the SOAP service URL.
- When you invoke a taskflow as an API, you can dynamically provide input parameters for the tasks in the taskflow.

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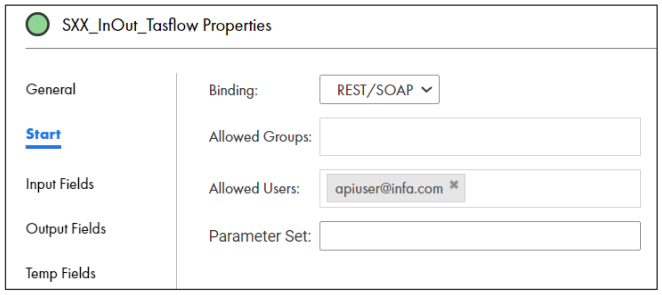
You can invoke a taskflow as an API by publishing the taskflow as a service. When you publish a taskflow, Data Integration generates the service URL and the SOAP service URL. You can use these endpoint URLs to invoke the taskflow as an API.

When you invoke a taskflow as an API, you can dynamically provide input parameters for the tasks in the taskflow.

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Generating the Service URL

1. In the Start tab, enter the user details in the Allowed Users field.



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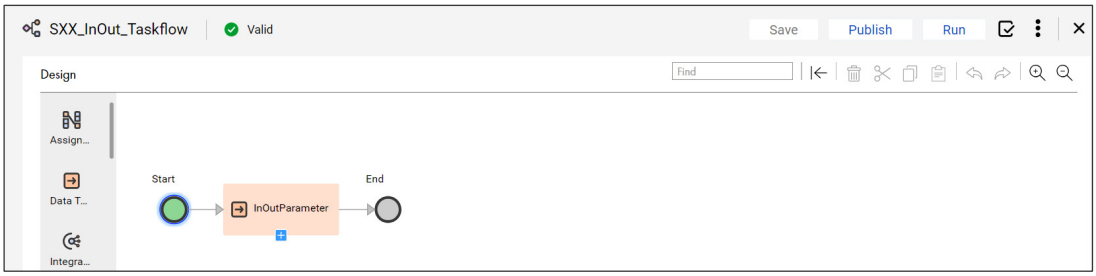
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After you create a taskflow, navigate to the Start tab under the taskflow properties. In the Allowed Users field, enter the user details that can start the taskflow.

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Generating the Service URL (continued)

2. Publish the taskflow.



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After saving the taskflow, you must publish it so that the Data Integration can generate the service API for the taskflow.

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Generating the Service URL (continued)

3. Navigate to the Properties Detail to get the service URL.



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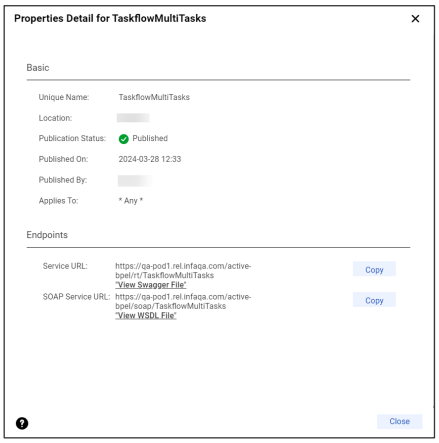
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After the taskflow is published, navigate to the Properties Detail to get the service URL.

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Generating the Service URL (continued)

4. Copy the Service URL. Use the service URL to start the taskflow.



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The Service URL is specified under the Endpoints section on the Properties Detail page.

You can use the Service URL to start the taskflow from a REST client such as the Postman.

On the Explore page, you can view the publish status and publish date of taskflows by adding Published and Published On columns for taskflows.

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Topic

Knowledge Check

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If you need to transform data from a source format to a target format in a linear taskflow, which task type would you use?

a. Mapping task
b. Replication task
c. Synchronization task
d. Masking task

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Answer: b

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Which of the following is incapable to be a Taskflow step?

a. Data Task Step
b. Jump Step
c. Pause Step
d. End Step

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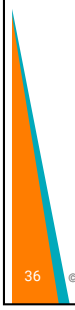
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Answer: c

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You can use a Taskflow to pass parameters to a task.

a. True
b. False



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Answer: a

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Lab Activity

Lab 13-1 Creating a Linear Taskflow

In this lab, you will perform the following:

- Configure a Linear Taskflow
- Configure email notifications for Linear Taskflow

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Lab Activity

Lab 13-2 Creating a Taskflow with Conditional Logic

In this lab, you will perform the following:

- Create a Taskflow and run the task
- Use conditional logic and add decision-based logic based on the output of the previous task
- Configure email notifications

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Module Summary

This module showed you how to:

- Define a taskflow
- List the steps in a taskflow
- Define Linear taskflow
- List the task types in a Linear taskflow
- Discuss the taskflow templates
- Explain the use of parameters in a taskflow

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Module 14
IICS REST APIs

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Module Objectives

After completing this module, you will be able to:

- Explain REST API
- Describe IICS REST API
- Discuss IICS REST API versions
- Understand request header and request body configurations
- Use REST APIs
- Create REST V2 Connector
- Explain Web Service Transformation

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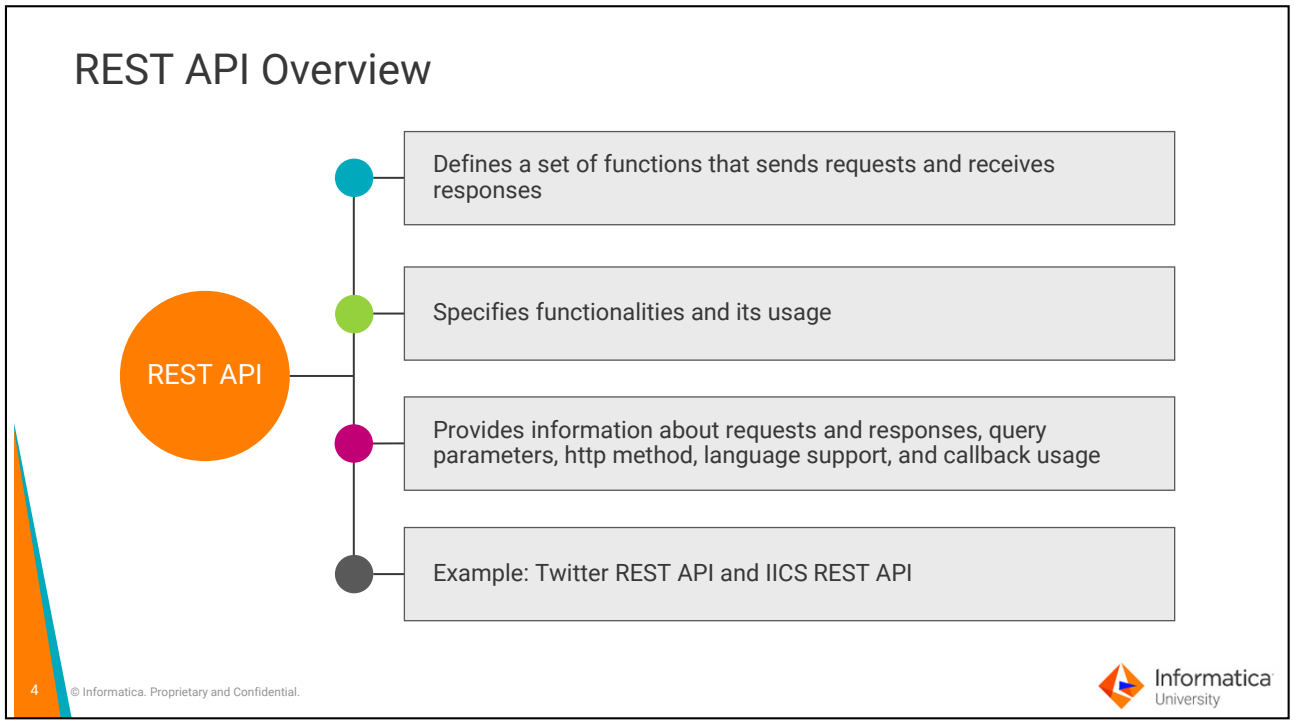
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Topic

REST API

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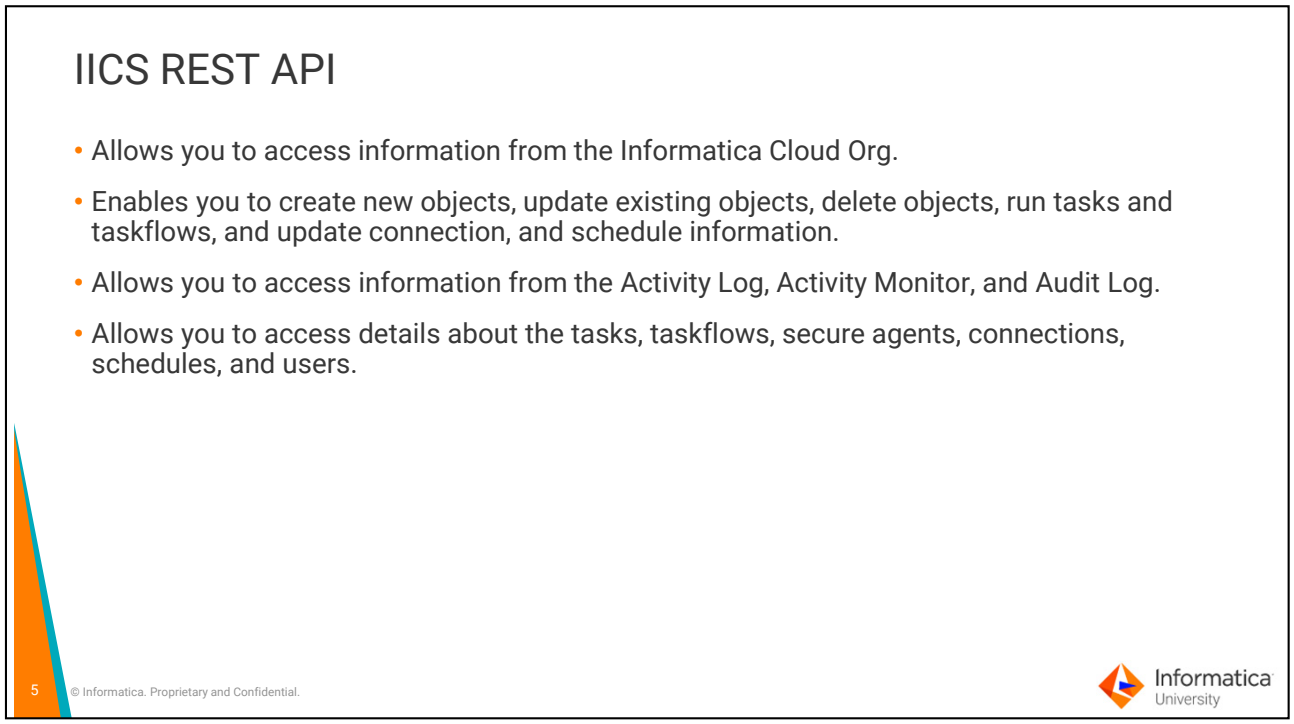


A REST API defines a set of functions that allows you to send requests and receive responses using the HTTP protocol.

The REST API specifies the functionalities it can provide, and also specifies how to use those functionalities. It also provides information about request and response formats, query parameters, HTTP method to be used – for example, GET, POST, PUT, or DELETE, language support, callback usage, and so on.

Some common examples of REST APIs include, Twitter REST API, and IICS REST API.

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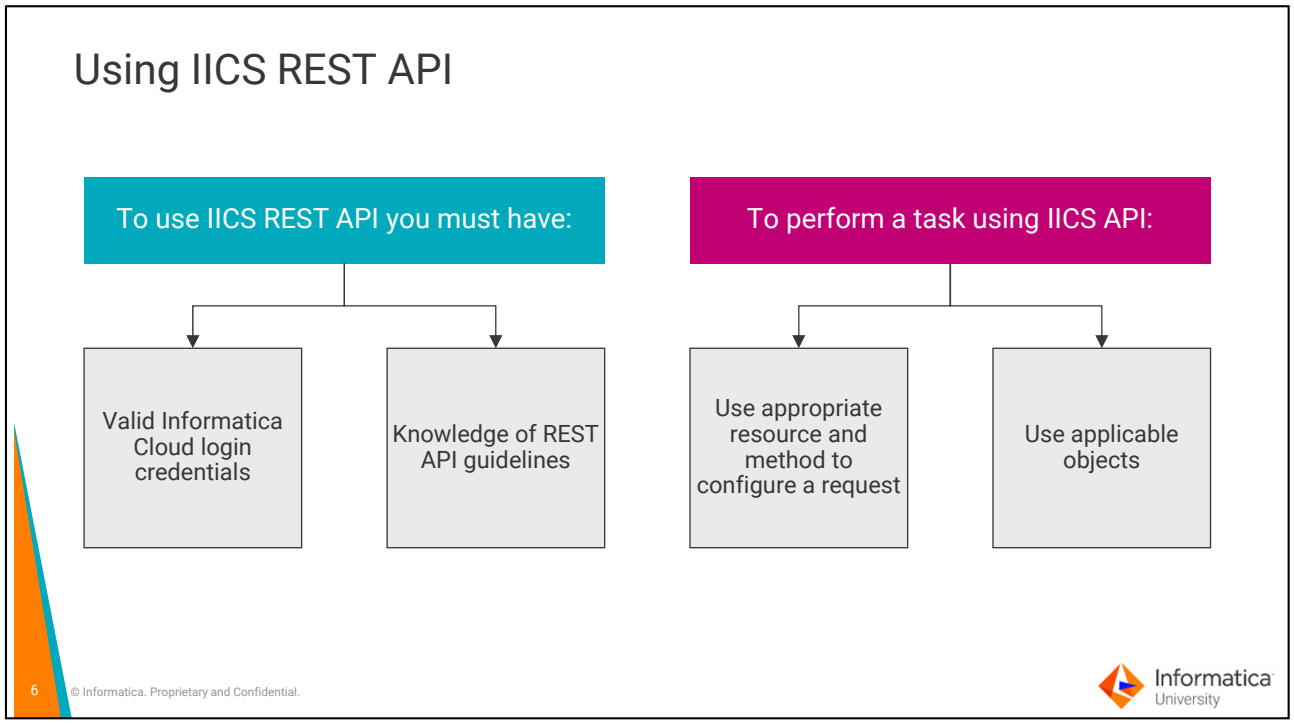


IICS REST API allows you to access information from the Informatica Cloud Org using a third-party application or service. You can use the IICS REST API to create new objects, update existing objects, delete objects, run tasks and taskflows, update connection, and schedule information for the Org.

The REST API includes a complete set of resources that allows you to access information from the Activity Log, Activity Monitor, and Audit Log. You can also access details about tasks, taskflows, secure agents, connections, schedules, users, and so on.

When you use the Informatica Cloud REST API, you do not have to manually log in to the Org to perform the tasks.

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To use IICS REST API, you must have valid Informatica Cloud login credentials and knowledge of REST API guidelines.

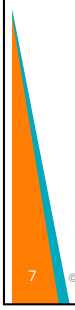
To perform a task, you must first configure a request using the IICS REST API. To configure a request, you must use the appropriate resource and method. You must also use all the applicable objects.

Informatica Cloud returns the requested information, performs the requested task, or returns an error object, and related messages.

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IICS Platform REST APIs

- IICS includes common platform functionality that is applicable to all services.
  - Platform Resource task enables you to list all the tasks in your organization
- Some functionalities are specific to a service.
  - Mapping task is applicable only to the Data Integration service



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IICS includes common platform functionality applicable to all the services. For example, you can use the platform resource task to list all the tasks in your organization.

It is also important to know that apart from the platform functionalities, some functionalities are specific to a service. For example, a mapping task is applicable only to the Data Integration service.

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Topic

IICS REST API Versions

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IICS REST API Versions

- Supports the platform REST API version 2 and version 3 resources and service-specific resources.
- You can log in to IICS using the platform REST API version 2 or version 3 login resource.

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IICS supports the platform REST API versions 2 and 3 resources and service-specific resources.

You can log in using the REST API versions 2 or 3 platforms.

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IICS REST API Versions – A Comparison

- Format:
  - version 2 supports XML and JSON calls
  - version 3 supports JSON calls
- Login URL:
  - for version 2, use https://dm-<POD region>.informaticacloud.com/ma/api/v2/user/login
  - for version 3, use https://dm-<POD region>.informaticacloud.com/saas/public/core/v3/login

North America	Europe	Asia
us	eu	ap

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The format that you can use depends on the API version. Version 2 supports XML and JSON calls. Version 3 supports only JSON calls.

The login URLs that you can use also depends on the API version. In the login URL, the Point of Deployment or POD region is based on the location of the Informatica Cloud data center.

- For the North America data center, the POD region is us.
- For Europe, the POD region is eu.
- For Asia, the POD region is ap.

The URL that you receive when you register with IICS includes the POD region.

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IICS REST API Versions – A Comparison (continued)

- Base URL:
  - for version 2 resources, use <serverUrl>/api/v2/<API name>
  - for version 3 resources, use <baseApiUrl>/public/core/v3/<API name>
- Request URL:
  - for version 2 resources, use <serverUrl>/api/v2/<API name>
  - for version 3 resources, use <baseApiUrl>/public/core/v3/<API name>
- Session ID:
  - for version 2 resources, use icSessionId in the header
  - for version 3 resources, use INFA-SESSION-ID in the header

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The login response includes the base URL that you must include in subsequent calls. The name of the base URL attribute and the URL that you use after login depends on the API version. The URL that you use in requests differs between the version 2 and version 3 resources.

The login response includes a session ID that you must include in headers during the session. You can use the same session ID for version 2 and version 3 resources. The name of the attribute for session ID also depends on the API version.

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Request Header – Version 2

For version 2 calls, use the following format in the REST API request header:

```
<METHOD> <serverUrl>/<URI> HTTP/<HTTP version>
Content-Type: application/<json | xml>
Accept: application/<json | xml>
icSessionId: <SessionId>
```

Note: If you use the Postman tool, requests automatically include the HTTP version.

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Request Header – Version 3

For version 3 calls, use the following format in the REST API request header:

```
<METHOD> <baseApiUrl>/<URI> HTTP/<HTTP version>
Content-Type: application/json
Accept: application/json
INFA-SESSION-ID: <SessionId>
```

Note: If you use the Postman tool, requests automatically include the HTTP version.

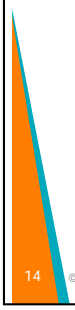
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Request Body

- Request body passes additional attributes for the resource.
- Passes attributes as part of an object.
- If a request includes sub-objects for attributes, you must declare the sub-objects before listing the related attributes.



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You can use the request body to pass additional attributes for the resource. When you pass attributes in a request body, you pass the attributes as part of an object.

For example, to log in with the login resource, you pass the required username and password attributes in a login object.

Some requests include sub-objects for attributes. You must declare the sub-objects before listing the related attributes.

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Request Body – JSON Format

When you use JSON format for version 2 REST API calls, you can optionally define a request object with the @type attribute.

```
{
  "@type": "<request object>",
  "<attribute1>": "<value1>",
  "<attribute2>": "<value2>",
}
```

When an attribute includes an object, state the attribute and use the object name.

```
{
  "@type": "<request object>",
  "<attribute1>": "<value1>",
  "<attribute2>": {
     "@type": "<attribute object>",
     "<attributeA>": "<valueA>",
     "<attributeB>": "<valueB>",}
     "@type": "<attribute object>",
     "<attributeD>": "<valueD>",
     "<attributeE>": "<valueE>",}
  "<attribute3>": "<value3>",
}
```

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When you use the JSON format for version 2 REST API calls, you can optionally define a request object with the @type attribute, as shown in the example.

When an attribute includes an object, you must state the attribute and use the object name.

For version 3 REST API calls, do not use the @type attribute.

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Request Body – XML Format

When you use XML format, define a request object as an enclosing set of tags.

```
<request object>
   <attribute1>value1</attribute1>
   <attribute2>value2</attribute2>
</request object>
```

When an attribute includes an object, enclose the attribute object within the attribute tags.

```
<request object>
   <attribute1>value1</attribute1>
   <attribute2>
     <attribute object>
       <attributeA>valueA</attributeA>
     </attribute object>
     <attribute object>
       <attributeB>valueB</attributeB>
     </attribute object>
   </attribute2>
   <attribute3>value3</attribute3>
</request object>
```

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When you use the XML format, you must define a request object as an enclosing set of tags.

When an attribute includes an object, you must enclose the attribute object within the attribute tags.

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Return Lists – JSON Format

JSON does not use additional attributes. The REST API encloses the list in square brackets ([ ]).

```
[
 {
    "<attribute1>": "<value1>",
    "<attribute2>": "<value2>",
 }{
    "<attribute1>": "<value1>",
    "<attribute2>": "<value2>",
 }
]
```

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Return Lists – XML Format

When the REST API returns a series of objects in XML, it encloses the list in the root tag.

```
<root>
  <return object 1>
    <attribute1>value1</attribute1>
    <attribute2>value2</attribute2>
  </return object 1>
  <return object 2>
    <attribute1>value1</attribute1>
    <attribute2>value2</attribute2>
  </return object 2>
</root>
```


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Topic

Create REST V2 Connector

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REST V2 Connection

- You can use REST V2 Connectors:
  - To interact with web service applications that support REST API.
  - In a Source transformation, Target transformation, or midstream in a Web Services transformation.
  - Midstream in a mapping to pass a single or multiple requests to a web service application and process the response data.
- In a source, target, and midstream transformation, you can use the REST methods such as GET, PUT, POST, or DELETE.
- Configure one of the REST authentication types such as BASIC, DIGEST, or OAUTH Version 1.
- Use REST V2 Connector to process XML and JSON data.

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REST V2 Connection (continued)

Create a REST V2 connection to make calls to a web service application.

- The swagger specification file and the authentication method must be specified when creating a connection.
- Specify the following authentication methods:

Standard | OAuth 2.0 client credentials | OAuth 2.0 authorization code | JWT bearer token

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Create REST V2 Connection

- Create REST V2 connections on **Connections** page.
- Click **Administrator** > **Swagger Files** to generate a swagger specification file.
- Configure the connection properties while creating a REST V2 connection.

Connection property	Description
Runtime Environment	Name of the runtime environment where you want to run the tasks.
Authentication Type	Select the authentication method that the connector must use to login to the web service application. Default is none.
Auth User ID	The user name to login to the web service application when you select the Basic authentication.
Auth Password	The password associated with the user name when you select the Basic authentication.
OAuth Consumer Key	The client key associated with the web service application.
OAuth Token	The access token to connect to the web service application.

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Create REST V2 Connection (continued)

Connection property	Description
OAuth Token Secret	The password associated with the OAuth token.
Swagger File Path	The absolute path along with the file name or the hosted URL of the swagger specification file.
TrustStore File Path	The absolute path of the truststore file that contains the TLS certificate to establish a one-way or two-way secure connection with the REST API.
TrustStore Password	The password for the truststore file that contains the SSL certificate.
KeyStore File Name	The absolute path of the keystore file that contains the keys and certificates required to establish a two-way secure communication with the REST API.
KeyStore Password	The password for the keystore file required for secure communication.
Proxy Type	Type of proxy.
Proxy Configuration	The proxy configuration format: <host>:<port>
Advanced Fields	Enter the arguments that the Secure Agent uses when connecting to a REST endpoint.

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Topic

Web Service Transformation

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Web Service Transformation

- Web Services transformation connects to a web service as a web service client to access, transform, or deliver data.
- Web service client request and web service response are Simple Object Access Protocol messages.
- Mapping task processes SOAP messages with document or literal encoding.
- Web Services transformation does not support Remote Procedure Calls (RPC)encoded or document encoded Web services description language (WSDL) files.
- SOAP request and response messages can contain hierarchical data.

Example:

- In order to execute the web service action getCityWeatherByZIP, the Web Services transformation sends a SOAP request to a web service.
- The Web Services transformation passes zip codes in the request.
- The weather data is retrieved by the web service, which then returns it as a SOAP response.

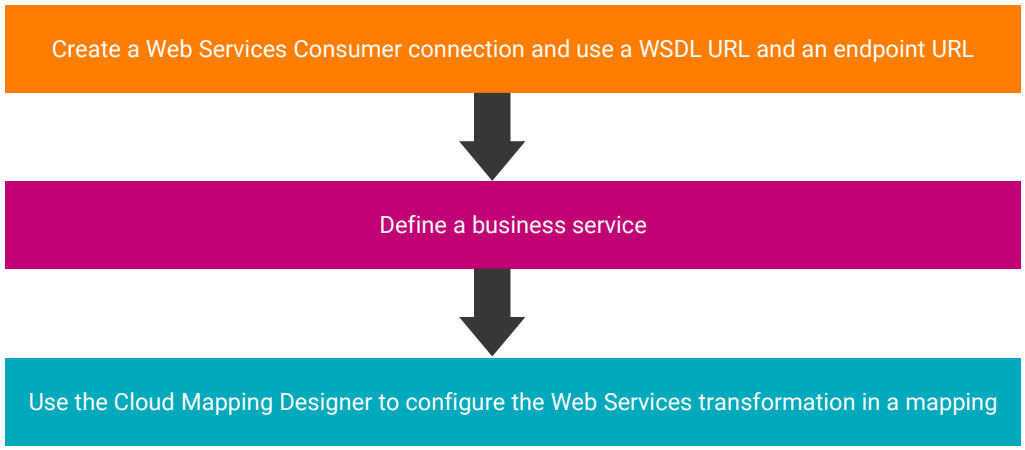
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- A Web Services transformation connects to a web service as a web-service-client, to access, transform, or deliver data. The web-service client-request and the web-service response are SOAP messages. The Mapping task processes SOAP messages with document or literal encoding. The Web Services transformation does not support RPC encoded or document encoded WSDL files.
- SOAP request messages and response messages can contain hierarchical data, such as data that follows an XML schema.

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Using Web Services Transformation



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Topic

Knowledge Check

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Which of the following tasks can you perform using the IICS REST API?

a. Update and delete schedules

b. Configure permissions

c. Run tasks and taskflows

d. All the above

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Answer: d

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Parameterized mapping tasks can be modified using the IICS REST API.

a. True

b. False

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Answer: a

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Lab Activity

Lab 14-1 Creating a Mapping Using a REST V2 Connector

In this lab, you will perform the following:

- Create mapping using REST V2 connector

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Lab Activity

Lab 14-2 Using Web Services Transformation in a Mapping

In this lab, you will perform the following:

- Use Web services transformation in mapping using Mapping Designer.

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Lab Activity

Lab 14-3 Running a Mapping Task Using REST API

In this lab, you will perform the following:

- Use a REST client application to call the login resource and obtain a session ID
- Start a Mapping task
- Log out of the Informatica Cloud API session

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Lab Activity

Lab 14-4 Monitoring Tasks Using REST API

In this lab, you will perform the following:

- Use a REST client application to call the login resource and obtain a session ID
- Get the log information about the running jobs
- Use API to logout of IICS

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Module Summary

This module showed you how to:

- Explain REST API
- Describe IICS REST API
- Discuss IICS REST API versions
- Understand request header and request body configurations
- Use REST APIs
- Create REST V2 Connector
- Explain Web Service Transformation

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